SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume I – Gameplay Foundation

Chapter 1 – Vision & Design Philosophy

Document Version: 1.0

Status: Approved Foundation

Priority: Highest

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1. Purpose of this Document

This chapter establishes the philosophical foundation of SPIKE LIFE™. Every future design decision, gameplay mechanic, user interface, financial algorithm, and software component must align with the principles defined here.

This chapter serves as the constitutional document of the game. If future design decisions conflict with these principles, this chapter takes precedence.

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2. Product Vision

SPIKE LIFE™ is a multiplayer financial decision simulation game that allows players to experience the financial consequences of life decisions in a realistic, engaging, and enjoyable environment.

The game combines life simulation, financial planning, strategy, and decision-making into an experience where learning emerges naturally through play.

Rather than teaching finance through lectures, calculations, or static lessons, SPIKE LIFE teaches through repeated experience.

Players learn because they experience the consequences of their decisions.

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3. Product Mission

To improve financial decision-making by allowing players to safely experience the opportunities, trade-offs, risks, and rewards of real life before encountering them in reality.

---

4. Long-Term Vision

SPIKE LIFE is envisioned as a global Financial Decision Platform.

The initial release focuses exclusively on the Philippine Financial World.

Future releases may introduce additional Financial Worlds representing different countries and financial environments without changing the Core Financial Decision Engine.

The platform should eventually support:

- Personal Financial Literacy
- University Education
- Corporate Financial Wellness
- Financial Advisor Training
- Family Financial Planning
- Cooperative Education
- Banking and Financial Institution Programs

The platform is designed for decades of expansion.

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5. Design Philosophy

SPIKE LIFE is built upon six core principles.

Principle 1

Life First.

Finance Second.

The game is not about money.

The game is about life.

Money exists to support life goals.

Every financial decision should begin with a life objective rather than an investment objective.

Players should constantly ask:

"What kind of life am I trying to build?"

instead of

"How much money can I make?"

Principle 2

Decisions Create Consequences

Every meaningful decision produces both benefits and costs.

There are no perfect financial decisions.

Every good decision delays or sacrifices something else.

Examples:

Buying health protection may delay buying a house.

Funding retirement may postpone an overseas vacation.

Supporting parents may delay starting a business.

Starting a business increases potential wealth but also increases financial risk.

These trade-offs are intentional and represent the essence of adult financial life.

Principle 3

Learning Through Experience

The game should never feel like a lecture.

Players should discover financial principles naturally through repeated gameplay.

The Financial Decision Engine performs sophisticated analysis behind the scenes.

Players experience only:

Situation

↓

Decision

↓

Consequence

Repeated exposure builds intuition before introducing financial terminology.

Principle 4

Simplicity on the Surface

Complexity underneath.

Players should never:

- compute interest
- calculate inflation
- build amortization schedules
- manually allocate percentages
- perform financial analysis

Players simply make life decisions.

The engine performs all calculations automatically.

Principle 5

Every Turn Must Be Fun

Educational value alone is insufficient.

Each Planning Cycle should create:

Anticipation

Surprise

Discussion

Emotion

Learning

Players should look forward to the next Planning Cycle.

If the game is enjoyable, learning becomes effortless.

Principle 6

Authentic Philippine Experience

Version 1.0 represents life in the Philippines.

Players should immediately recognize familiar situations.

Examples include:

Supporting parents

Typhoon damage

13th Month Pay

Christmas expenses

OFW opportunities

Family obligations

Business opportunities

Government benefits

Medical emergencies

Community responsibilities

The experience should feel familiar rather than imported.

---

6. Educational Philosophy

SPIKE LIFE teaches financial planning indirectly.

The game does not teach by telling.

It teaches by allowing players to experience:

Cash flow

Inflation

Opportunity cost

Risk

Protection

Investment growth

Emergency funds

Debt

Business ownership

Goal planning

Retirement planning

Players learn because they repeatedly experience outcomes.

---

7. The Role of the Financial Advisor

The Financial Advisor is not an instructor.

The Advisor is a guide.

The Advisor explains consequences and trade-offs but never forces decisions.

Players remain responsible for every decision.

This preserves agency while introducing professional financial thinking.

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8. Core Learning Outcomes

Every completed game should improve the player's understanding of:

Managing cash flow

Building emergency funds

Long-term planning

Goal prioritization

Protection planning

Risk management

Investment principles

Business risk

Delayed gratification

Financial resilience

Players should finish the game thinking differently about money.

---

9. The Player Experience

The emotional journey of the player should follow this progression.

Curiosity

↓

Excitement

↓

Uncertainty

↓

Decision

↓

Consequence

↓

Reflection

↓

Growth

↓

Confidence

The player gradually develops better financial instincts.

---

10. The Core Gameplay Formula

The entire game revolves around one repeated cycle.

Life Happens

↓

Player Decides

↓

Financial Engine Calculates

↓

Life Changes

↓

Repeat

Every feature in the game should reinforce this loop.

---

11. The Financial Decision Engine

The Financial Decision Engine is the heart of SPIKE LIFE.

Its responsibilities include:

Cash Flow

Living Expenses

Savings

Emergency Fund

Protection

Investment Growth

Goal Funding

Inflation

Debt

Business Performance

Retirement

Education Planning

Life Score

The engine performs all calculations invisibly.

Players interact only with decisions and outcomes.

---

12. The Meaning of Winning

Winning is intentionally different from traditional financial games.

The objective is **not** to become the richest player.

The objective is to build the most financially balanced life.

The winner is determined by Life Score, which rewards:

Financial Security

Protection

Goal Achievement

Wealth Creation

Life Stability

Balanced planning consistently outperforms reckless wealth accumulation.

---

13. MVP Design Constraints

Version 1.0 deliberately limits complexity.

The MVP will include:

- Philippine Financial World
- Semi-Annual Planning Cycles
- 2–6 Players
- Simultaneous Multiplayer
- One Major Decision per Planning Cycle
- Automatic Financial Engine
- Fixed Life Domains
- Hidden Financial Calculations

- Annual 13th Month Planning
- Life Score

The MVP intentionally excludes:

- Multiple countries
- Online multiplayer
- AI-generated scenarios
- Marketplace
- Campaign editor
- User-generated content
- Advanced investment products
- Multiplayer trading
- Real financial product integration

These belong in future expansions.

---

14. Design Rules

Every future feature proposal must answer the following questions.

Does it make the game more enjoyable?

Does it simplify the player experience?

Does it teach through experience rather than explanation?

Does it reinforce meaningful financial decision-making?

Does it fit naturally into the existing gameplay loop?

If the answer is "No," the feature should be redesigned or rejected.

---

15. Success Criteria

SPIKE LIFE succeeds when players:

Enjoy the game.

Replay the game.

Discuss their decisions.

Debate financial trade-offs.

Recognize mistakes.

Change future decisions.

Apply lessons to real life.

The game should leave players saying:

"I never realized how one financial decision could affect so many parts of my life."

rather than:

"I learned about budgeting."

---

16. Guiding Principle

SPIKE LIFE is not trying to teach players how to become rich.

It is teaching players how to make wiser financial decisions throughout life.

Every mechanic, every animation, every scenario, every algorithm, and every line of code should ultimately support one simple belief:

A better financial life is built one good decision at a time.

End of Chapter 1

This chapter defines the philosophical and educational foundation of SPIKE LIFE. All subsequent chapters—including gameplay mechanics, financial algorithms, UI/UX, and technical architecture—must remain consistent with these principles. Future enhancements should expand the experience without compromising the core vision established here.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume I – Gameplay Foundation

Chapter 2 – Core Gameplay Loop

Document Version: 1.0

Status: Approved Foundation

Priority: Highest

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1. Purpose

This chapter defines the complete gameplay flow of SPIKE LIFE.

It establishes:

- The player's journey
- The game sequence
- Planning Cycles
- Decision flow
- Simultaneous multiplayer
- Win condition
- User interaction rules

Every gameplay screen, animation, and system must conform to this chapter.

---

2. Gameplay Philosophy

SPIKE LIFE is built around one simple idea:

Life presents situations. Players make decisions. Decisions shape life.

Unlike traditional board games where players move around a track, SPIKE LIFE simulates the progression of life itself.

The player never controls the events that occur.

The player controls only how they respond.

---

3. Complete Player Journey

The complete gameplay sequence is:

Launch Game

↓

Sign In

↓

Join or Create Game

↓

Choose Avatar

↓

Create Player Profile

↓

Select Starting Career

↓

Build Dream Board

↓

Financial Engine Initialization

↓

Game Begins

↓

20 Planning Cycles

↓

Game Ends

↓

Life Summary

↓

Winner Announced

---

4. Game Lobby

Players may choose:

Host Game

Join Game

Single Player (Future)

AI Players (Future)

Current MVP supports:

2–6 Players

Recommended:

4–6 Players

The host configures:

- Number of Players
- Game Difficulty
- Decision Timer
- Starting Age (Default: 22)
- Campaign Length (Default: 10 Years)

---

5. Character Creation

Each player creates a simple character.

Required:

- Name
- Avatar
- Gender (optional)
- Starting Career

- Starting Monthly Income

Future versions may include:

Education

Province

Personality Traits

Financial Habits

---

6. Starting Careers

Each career has:

- Monthly Income
- Growth Potential
- Promotion Probability
- Job Security
- Business Compatibility

Examples:

Graduate

Teacher

Nurse

Engineer

Government Employee

BPO Professional

Sales Professional

Entrepreneur

Freelancer

OFW (future)

The selected career influences future situations.

---

7. Dream Board (Life Blueprint)

Before gameplay begins, players define the life they wish to build.

Goals include:

🏠 Own a Home

✈️ Travel

🚀 Start a Business

👶 Child Education (optional)

🏖️ Retirement

Emergency Fund (automatic)

Each goal includes:

Target Age

Target Amount

Priority

The game automatically calculates:

Future Value

Required Savings

Progress Tracking

using the Financial Engine.

---

8. Game Initialization

Before the first Planning Cycle, the engine calculates:

Current Age

Monthly Income

Living Expenses

Cash Flow

Financial Capacity

Emergency Fund Target

Dream Board Costs

Future Values

Protection Status

Life Score (baseline)

This creates each player's unique financial profile.

---

9. Game Timeline

One game covers:

10 Years

Each year consists of:

Planning Cycle 1

January–June

Planning Cycle 2

July–December

↓

13th Month Pay

↓

Annual Planning

↓

Age +1

Total:

20 Planning Cycles

---

10. Planning Cycle Structure

Every Planning Cycle follows exactly the same sequence.

Planning Cycle Begins

↓

Life Domain Selection

↓

Situation Selection

↓

Decision Timer Starts

↓

Players Decide Simultaneously

↓

Financial Engine Computes

↓

Consequences Revealed

↓

Planning Cycle Ends

The rhythm should become familiar and satisfying.

---

11. Simultaneous Multiplayer

SPIKE LIFE does **not** use sequential turns.

Every player participates simultaneously.

During each Planning Cycle:

Each player receives:

- One Life Domain
- One Situation
- One Decision

All players decide at the same time.

When all players finish (or timers expire), outcomes are revealed simultaneously.

Benefits:

- No waiting
- Continuous engagement
- Discussion among players
- Faster gameplay

---

12. Life Domain Selection

There are twelve Life Domains.

Career

Opportunity

Family

Health

Business

Lifestyle

Housing

Education

Government

Community

Investment / Finance

Milestone

At the beginning of each Planning Cycle:

The twelve domains animate.

Squares pulse and flash.

One domain is randomly selected for each player.

Selection is weighted based on:

Age

Career

Income

Previous Decisions

Life Stage

Goals

The animation creates anticipation before the situation is revealed.

---

13. Situation Reveal

After the Life Domain is selected:

A situation is randomly drawn.

Example:

Career

↓

Promotion

or

Housing

↓

Typhoon Damaged Your Home

Each domain contains a large pool of situations to maximize replayability.

---

14. Decision Phase

The player is presented with:

Situation

↓

Three simple choices

Typical decision styles include:

Secure Future

Balanced

Enjoy Today

The player selects one strategy.

The player never performs calculations or manual financial allocation.

---

15. Decision Timer

Every Planning Cycle starts a countdown timer.

Default:

15 Seconds

Available modes:

No Timer

20 Seconds

15 Seconds

10 Seconds

5 Seconds (Blitz)

The timer encourages instinctive decision-making and keeps gameplay moving.

---

16. Auto Advisor

If the timer expires:

The game does **not** choose randomly.

Instead,

the Financial Advisor automatically makes a balanced recommendation.

This reinforces an important lesson:

Indecision has consequences.

Doing nothing is also a decision.

---

17. Advisor Assistance

Players may request advice before making a decision.

The Advisor:

- Explains trade-offs
- Highlights possible future effects
- Does not guarantee outcomes
- Never forces a decision

Advice is educational, not prescriptive.

---

18. Financial Engine Processing

Once all players have finalized their decisions:

The Financial Decision Engine computes:

Income

Living Expenses

Savings

Emergency Fund

Protection

Debt

Investment Growth

Goal Funding

Business Performance

Random Events

Risk Exposure

Life Score

All calculations occur behind the scenes.

---

19. Consequence Reveal

After calculations:

Each player's results appear.

Typical feedback includes:

Cash Flow

Net Worth

Emergency Fund Progress

Goal Progress

Protection

Life Score Change

Some effects are immediate.

Others are stored as delayed consequences.

Example:

Buying health protection may produce no immediate benefit but substantially reduce losses from a future hospitalization.

---

20. Hidden Delayed Consequences

Many decisions create future effects.

Examples:

Health Insurance

↓

Reduced medical losses

Emergency Fund

↓

Absorbs future emergencies

Business Investment

↓

Possible future expansion

Luxury Spending

↓

Goal delays

Delayed consequences are invisible until triggered.

This teaches long-term planning.

---

21. End of Year Sequence

After Planning Cycle 2:

Players receive:

Mandatory 13th Month Pay

Players then complete Year-End Planning.

Typical uses:

Emergency Fund

Retirement

House

Business

Travel

Education

Protection

Debt Reduction

Lifestyle

The bonus creates an exciting annual planning opportunity.

---

22. Annual Financial Review

At the end of every year:

Players see a concise dashboard showing:

Age

Cash Flow

Net Worth

Emergency Fund

Protection Readiness

Goal Progress

Financial Health

Advisor Insight

The review lasts only a few seconds before the next year begins.

---

23. Game End

After ten years:

The simulation ends.

The Financial Engine performs a final evaluation.

---

24. Final Results

Players receive:

Life Score

Financial Health

Net Worth

Goals Achieved

Emergency Fund Completion

Protection Readiness

Investment Growth

Retirement Progress

Business Success

Major Decisions Timeline

Every player also receives a personalized life story summarizing their financial journey.

---

25. Winning

The winner is the player with the highest **Life Score**.

Life Score rewards:

Financial Security

Protection

Goal Achievement

Wealth Creation

Life Stability

The richest player may not necessarily win.

The player who created the most balanced financial life is declared the winner.

---

26. Replayability

Every new game should feel different.

Replayability is achieved through:

Random Life Domains

Random Situations

Weighted Events

Different Careers

Different Dream Boards

Business Outcomes

Market Conditions

Life Stage Progression

Player Decisions

No two games should produce the same story.

---

27. Gameplay Principles

Every Planning Cycle must:

Take less than 30 seconds.

Require only one meaningful decision.

Reveal immediate consequences.

Teach one financial lesson.

Encourage discussion.

Create anticipation for the next cycle.

The player should never feel overwhelmed.

---

28. Cursor Implementation Rules

The following gameplay rules are mandatory:

- Never require manual financial calculations.
- Never interrupt gameplay with lengthy reports.
- Never require spreadsheet-style budgeting.
- Never present more than one major decision per Planning Cycle.
- Always resolve all players simultaneously.
- Keep animations under three seconds.
- Prioritize responsiveness over visual complexity.
- Ensure the entire game is playable within approximately one hour with six players.

---

29. Chapter Summary

Chapter 2 defines the operational heartbeat of SPIKE LIFE.

The player experience should feel effortless:

Life presents opportunities and challenges.

Players make one important decision.

The Financial Engine silently performs sophisticated planning.

Consequences shape the player's future.

The cycle repeats until an entire financial life story has unfolded.

The gameplay loop should always remain simple on the surface while being powered by a sophisticated decision engine underneath.

End of Chapter 2

This chapter establishes the complete gameplay flow for SPIKE LIFE and serves as the implementation reference for the Game Engine, UI/UX, multiplayer synchronization, and Financial Decision Engine integration. Subsequent chapters will expand on each subsystem without altering this core gameplay loop.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume I – Gameplay Foundation

Chapter 3 – Planning Cycle Engine

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1. Purpose

This chapter defines the operational engine that drives the gameplay rhythm of SPIKE LIFE.

The Planning Cycle Engine governs:

- Time progression
- Simultaneous gameplay
- Event generation
- Decision timing
- Financial processing
- Annual transitions

It is the heartbeat of the game.

---

2. Design Philosophy

The Planning Cycle represents one six-month period of a player's life.

Players cannot control what happens during each cycle.

Players only control how they respond.

The game intentionally mirrors real life:

Life presents situations.

Players make financial decisions.

Consequences accumulate.

Life continues.

---

3. Planning Cycle Structure

Every year consists of:

Planning Cycle 1

January – June

Planning Cycle 2

July – December

↓

13th Month Pay

↓

Annual Financial Planning

↓

Age +1

A ten-year campaign therefore contains:

20 Planning Cycles

---

4. Planning Cycle State Machine

Every Planning Cycle always follows the same sequence.

Initialize Cycle

↓

Generate Life Domain

↓

Generate Situation

↓

Reveal Situation

↓

Decision Timer Starts

↓

Player Decision

↓

Financial Engine Processing

↓

Consequence Reveal

↓

Cycle Complete

This sequence must never change.

---

5. Simultaneous Multiplayer

SPIKE LIFE uses simultaneous gameplay.

All players participate together.

There are **no sequential turns**.

Every Planning Cycle:

- All players receive a Life Domain.
- All players receive a Situation.
- All players receive a timer.
- Everyone decides simultaneously.
- Results are revealed together.

Benefits:

- Eliminates waiting
- Creates social discussion
- Keeps everyone engaged
- Reduces total game duration

---

6. Cycle Initialization

At the beginning of every Planning Cycle:

The engine performs:

- Advance simulation clock
- Load player state
- Check pending delayed consequences
- Update market conditions
- Update economic variables
- Determine event probabilities

No player interaction occurs during initialization.

---

7. Life Domain Animation

The twelve Life Domains remain visible throughout the game.

When a Planning Cycle begins:

All domains pulse softly.

A rapid highlight animation begins.

The highlight randomly cycles through domains.

Speed gradually slows.

One domain stops glowing brightly.

Example

Career

↓

Selected

The animation lasts approximately:

2–3 seconds

The animation should build anticipation.

---

8. Weighted Domain Selection

Selection is random but not uniform.

The engine adjusts probabilities according to:

Current Age

Career

Marital Status

Children

Income

Previous Decisions

Current Goals

Previous Events

Example

Age 22

Career

Higher probability

Housing

Lower probability

Retirement

Very low probability

Age 45

Retirement

Higher probability

Health

Higher probability

Education

Lower probability

This creates realistic life progression.

---

9. Situation Generation

After selecting a Life Domain,

the Situation Engine loads an event.

Every domain contains multiple event pools.

Example

Career

- First Job
- Promotion
- Demotion
- Resignation
- Company Closure
- Salary Increase
- Salary Reduction

One event is randomly selected.

The player cannot predict which event will occur.

---

10. Event Classification

Every situation belongs to one of five classes.

Positive

Example

Promotion

Negative

Example

Hospitalization

Opportunity

Example

Business Partner

Crisis

Example

Typhoon

Milestone

Example

Marriage

Each class influences the Financial Engine differently.

---

11. Situation Reveal

The selected situation expands into a large encounter card.

The encounter card displays:

Title

Illustration

Short Description

Immediate Context

Three Decisions

No financial calculations are displayed.

---

12. Decision Timer

Immediately after the situation appears:

The countdown begins.

Default:

15 Seconds

The timer is displayed as:

Circular countdown

Remaining seconds

Color progression

Green

↓

Yellow

↓

Red

At:

3 seconds

The timer pulses.

At:

0

The Auto Advisor activates.

---

13. Auto Advisor

Failure to decide does not create a random outcome.

Instead,

the Advisor chooses the safest balanced recommendation.

This reflects an important principle:

Doing nothing is itself a financial decision.

The Advisor always selects:

Balanced Growth

unless a crisis requires immediate protection.

---

14. Advisor Assistance

Players may request help before the timer expires.

Advisor interaction pauses the timer.

Maximum advisor consultation:

20 seconds

The Advisor explains:

Benefits

Costs

Trade-offs

Potential long-term effects

The Advisor never reveals future events.

The Advisor never guarantees success.

---

15. Player Decision

Every Planning Cycle requires:

Exactly one major decision.

Examples

Save

Invest

Protect

Borrow

Delay

Decline

The game intentionally limits decision complexity.

Players should never:

Allocate percentages

Choose interest rates

Perform calculations

Build financial plans manually

The Financial Decision Engine handles these tasks.

---

16. Decision Lock

Once submitted,

the decision becomes final.

No undo.

No revisions.

This encourages commitment.

---

17. Financial Engine Processing

After all players have locked decisions,

the Financial Engine executes.

Calculations include:

Income

Expenses

Living Costs

Savings

Emergency Fund

Investments

Debt

Protection

Business

Goal Funding

Inflation

Risk Exposure

Random Events

Life Score

All processing occurs simultaneously.

---

18. Hidden Delayed Effects

Not every consequence appears immediately.

Some effects remain dormant.

Examples

Bought Health Protection

↓

No visible effect

↓

Three cycles later

↓

Hospitalization

↓

Insurance pays benefits

The delayed consequence reinforces long-term planning.

---

19. Consequence Reveal

When processing completes,

all players receive results simultaneously.

Displayed information:

Cash

Net Worth

Emergency Fund

Goal Progress

Protection

Life Score

Financial Health

Changes animate clearly.

Positive values:

Green

Negative values:

Red

---

20. Financial Health Indicator

Each player maintains a Financial Health meter.

The indicator summarizes:

Cash Flow

Debt

Emergency Fund

Protection

Goal Progress

Stability

Displayed as:

Excellent

Good

Stable

Vulnerable

Critical

Players understand their financial condition without reading reports.

---

21. Mini Discussions

After results are revealed,

allow a brief discussion period.

Recommended:

5–10 seconds

Players naturally compare outcomes.

The game should encourage conversation.

---

22. End of Cycle

When every player is ready,

the Planning Cycle ends.

The board returns to its neutral state.

The next Planning Cycle begins.

---

23. End of Year Sequence

After Planning Cycle 2,

the game enters Year-End Mode.

Sequence

Receive 13th Month Pay

↓

Annual Financial Planning

↓

Annual Financial Review

↓

Age +1

↓

Begin New Year

---

24. 13th Month Planning

Players receive:

Mandatory Philippine

13th Month Pay

Players choose one primary allocation.

Possible allocations:

Emergency Fund

House

Travel

Business

Retirement

Education

Protection

Debt Reduction

Lifestyle

Only one major allocation is allowed.

This creates meaningful prioritization.

---

25. Annual Financial Review

At year-end,

display:

Current Age

Cash Flow

Net Worth

Emergency Fund Progress

Goal Completion

Protection Readiness

Financial Health

Advisor Insight

The review should complete within:

10 seconds

The pace of the game should never slow dramatically.

---

26. Planning Cycle Duration

Target duration per Planning Cycle:

25–35 seconds

Target annual duration:

60–70 seconds

Target complete game:

45–60 minutes

with

2–6 players.

---

27. Cursor Implementation Rules

Mandatory implementation requirements:

- Entire Planning Cycle controlled by a finite state machine.

- All players synchronized before processing.
- No player may advance independently.
- Financial calculations occur only after all decisions are locked.
- UI must remain responsive during processing.
- Animations should never delay gameplay.
- Maximum animation duration: 3 seconds.
- Planning Cycle must remain deterministic once decisions are locked.

---

28. Future Expansion

The Planning Cycle Engine is designed for future enhancements.

Examples:

Market cycles

Election years

Pandemics

Economic recessions

Natural disasters

AI-generated situations

Financial World localization

Seasonal events

These extensions must integrate into the existing Planning Cycle without altering its structure.

---

29. Chapter Summary

The Planning Cycle Engine establishes the rhythm of SPIKE LIFE.

Every six months:

Life presents one meaningful situation.

The player makes one important financial decision.

The Financial Decision Engine evaluates the consequences.

The player experiences the results.

Life moves forward.

This simple, repeatable cycle keeps gameplay engaging while allowing increasingly sophisticated financial simulations beneath the surface.

The Planning Cycle Engine should remain consistent throughout all future versions of SPIKE LIFE.

End of Chapter 3

This chapter defines the temporal and operational framework of SPIKE LIFE. Together with Chapters 1 and 2, it forms the immutable gameplay foundation upon which all financial systems, UI/UX interactions, and technical implementations will be built.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume I – Gameplay Foundation

Chapter 4 – Life Domain Engine

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Priority: Highest

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1. Purpose

The Life Domain Engine is responsible for determining **what area of life** the player will experience during each Planning Cycle.

Unlike traditional board games where players physically move around a board, SPIKE LIFE treats life as unpredictable.

Players do not choose what happens.

Life chooses.

Players choose only how they respond.

This distinction defines the identity of SPIKE LIFE.

---

2. Design Philosophy

The Life Domain Engine exists to simulate reality.

People rarely wake up choosing whether this year will be about:

- Career
- Health
- Housing
- Family

Life decides.

The game should recreate this uncertainty.

Every Planning Cycle begins with the question:

"What part of your life will require your attention this time?"

---

3. The Life Domain Board

The game board is **not** a movement track.

It is a permanent Life Dashboard.

It remains visible throughout the entire game.

The player never moves around it.

Instead, the board represents the twelve major areas of financial life.

---

4. Life Domains

The MVP contains twelve Life Domains.

1. Career

Employment

Promotion

Salary

Professional Growth

Resignation

Termination

---

2. Opportunity

Unexpected opportunities.

Scholarships

Side Hustles

Freelancing

Mentorship

Special Offers

Career Transfers

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3. Family

Marriage

Children

Parents

Family Support

Celebrations

Family Emergencies

Relationships

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4. Health

Preventive Care

Hospitalization

Critical Illness

Mental Health

Accidents

Medical Expenses

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5. Business

Startup

Expansion

Partnership

Business Failure

Business Loans

Market Growth

---

6. Lifestyle

Travel

Entertainment

Luxury Purchases

Hobbies

Dining

Impulse Spending

"Doodads"

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7. Housing

Rent

House Purchase

Condominium

Repairs

Typhoon Damage

Eviction

Renovation

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8. Education

Continuing Education

Board Exams

Professional Certifications

Graduate School

Child Education

Skills Development

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9. Government

Taxes

Government Benefits

SSS

PhilHealth

Pag-IBIG

Permits

Regulatory Changes

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10. Community

Charity

Church

Volunteer Work

Community Leadership

Neighborhood Issues

Social Responsibility

---

11. Investment & Finance

Savings

Emergency Fund

Bonds

Stocks

Mutual Funds

Credit Cards

Debt

Scams

---

12. Milestone

Birthdays

Marriage

Retirement

Inheritance

Major Achievements

Life Celebrations

Legacy Events

---

5. Domain Categories

For balancing purposes,

domains belong to four groups.

Income & Growth

Career

Opportunity

Business

Investment & Finance

Personal Life

Family

Health

Lifestyle

Housing

Development

Education

Government

Community

Major Life Events

Milestone

The engine should maintain a healthy distribution across these groups.

---

6. Domain Selection Algorithm

The player does **not** choose the Life Domain.

The engine selects it automatically.

Selection uses:

Weighted Random Probability.

Every Planning Cycle recalculates the probability of every domain.

---

7. Probability Factors

Domain selection is influenced by:

Current Age

Career

Income Level

Marital Status

Children

Current Goals

Previous Situations

Protection Status

Business Ownership

Financial Health

No two players should receive identical probabilities.

---

8. Age-Based Weighting

The engine gradually changes priorities as players age.

Early Career (22–29)

Higher Probability

Career

Education

Opportunity

Lifestyle

Lower Probability

Retirement

Legacy

Building Years (30–39)

Higher Probability

Housing

Marriage

Children

Business

Protection

Lower Probability

Education

Mid-Career (40–49)

Higher Probability

Health

Business

Education Funding

Investments

Promotion

Pre-Retirement (50–60)

Higher Probability

Retirement

Health

Legacy

Community

Investment Preservation

The transition should feel natural.

---

9. Goal-Based Weighting

Player goals influence future events.

Example

Player selected:

House Goal

Target Age

30

As age approaches:

Housing probability gradually increases.

Similarly,

Business Goal

↓

Business situations become more frequent.

This creates the feeling that life is responding to the player's ambitions.

---

10. Situation History

The engine maintains memory.

Recently selected domains become slightly less likely.

Example

Career

↓

Career

↓

Career

↓

Career

should almost never happen.

Instead,

the engine promotes diversity.

This improves replayability.

---

11. Domain Cooldown

Every selected domain enters a temporary cooldown.

Example

Housing selected.

Next Planning Cycle

Housing probability decreases.

After several Planning Cycles,

normal probability returns.

This prevents repetitive gameplay.

---

12. Event Intensity

Each domain supports multiple intensity levels.

Level 1

Minor

Level 2

Moderate

Level 3

Major

Level 4

Critical

Example

Health

Level 1

Annual Check-up

Level 2

Minor Injury

Level 3

Hospitalization

Level 4

Critical Illness

The Financial Engine determines appropriate intensity.

---

13. Positive vs Negative Balance

The game should never feel unfair.

The engine balances:

Positive Opportunities

Negative Challenges

Neutral Events

Growth Events

Recovery Events

Recommended distribution

Positive

35%

Negative

30%

Neutral

20%

Opportunity

15%

The balance should be dynamic.

Players should never experience long streaks of only negative events.

---

14. Philippine Flavor

The Philippine Financial World includes culturally relevant situations.

Examples

Typhoon Season

Family Reunion

Fiesta

Christmas

OFW Opportunities

Community Activities

Election Years

Barangay Events

Family Financial Support

These should appear naturally throughout gameplay.

---

15. Visual Presentation

The twelve domains remain visible throughout gameplay.

Default

Soft colors.

Inactive.

Planning Cycle

Domains begin pulsing.

Animation accelerates.

The selected domain glows brightly.

All other domains dim.

The chosen domain expands.

The Situation Card appears.

Total animation time:

2–3 seconds.

---

16. Audio Design

Each domain has its own subtle sound.

Examples

Career

Office notification.

Health

Heartbeat.

Business

Cash register.

Housing

Door opening.

Family

Children laughing.

These audio cues strengthen immersion without becoming distracting.

---

17. Future Expansion

Every domain supports unlimited situations.

Example

Career

Version 1

25 Situations

Version 2

75 Situations

Version 5

250 Situations

The engine should never require modification when additional situations are added.

Content grows independently.

---

18. Financial World Compatibility

The twelve domains remain constant across all countries.

Only the situations change.

Example

Housing

Philippines

Typhoon Damage

Australia

Bushfire Damage

Japan

Earthquake Damage

USA

Property Tax Increase

The Core Engine remains identical.

---

19. AI Compatibility

Future AI-generated situations must always belong to one Life Domain.

The AI should never invent new domains.

This maintains game consistency.

---

20. Replayability Strategy

Replayability comes from:

Weighted Random Selection

Situation Diversity

Player Goals

Age Progression

Previous Decisions

Economic Conditions

Delayed Consequences

Random Opportunities

Business Outcomes

No two games should feel identical.

---

21. Cursor Implementation Rules

Mandatory implementation requirements:

- The Life Domain Board remains fixed throughout gameplay.
- Players never move around the board.

- One Life Domain is selected per Planning Cycle.
- Selection uses weighted randomness.
- Cooldown prevents repetition.
- Goal weighting influences future events.
- Animations complete within three seconds.
- The board remains responsive on desktop and tablet.
- The Life Domain Engine must be entirely data-driven.
- New domains or situations should be configurable through JSON without changing engine logic.

---

22. Chapter Summary

The Life Domain Engine replaces the traditional movement mechanics found in board games.

Instead of moving around a board, life itself moves around the player.

Every Planning Cycle introduces uncertainty by selecting one meaningful area of life requiring attention.

The player cannot control what life presents.

The player controls only the quality of the decisions they make.

This philosophy forms one of the defining innovations of SPIKE LIFE and differentiates it from traditional financial education games.

End of Chapter 4

This chapter defines the Life Domain Engine, one of the core systems that gives SPIKE LIFE its unique identity. Together with the Planning Cycle Engine and the Financial Decision Engine, it forms the foundation of the game's simulation architecture. Future chapters will define the Situation Engine, which populates each Life Domain with realistic Philippine life events and decision scenarios.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume I – Gameplay Foundation

Chapter 5 – Situation Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Highest

---

1. Purpose

The Situation Engine is responsible for generating the life events that players experience during every Planning Cycle.

It answers one question:

"What exactly happened during this period of your life?"

The Situation Engine transforms a selected Life Domain into a meaningful life event that requires a financial decision.

It is the primary storytelling engine of SPIKE LIFE.

---

2. Design Philosophy

The Situation Engine must simulate **real life**, not merely finance.

Situations should feel familiar, believable, and emotionally engaging.

Players should frequently react with statements such as:

- "That happened to me."
- "That's exactly what happened to my parents."
- "I know someone who experienced this."

Financial education becomes memorable when situations feel authentic.

---

3. Situation Lifecycle

Every Planning Cycle follows this sequence:

Life Domain Selected

↓

Situation Generated

↓

Situation Revealed

↓

Player Decision

↓

Financial Engine Evaluation

↓

Immediate Consequence

↓

Hidden Future Consequences Stored

Each situation exists only for one Planning Cycle but may produce effects that last for years.

---

4. Situation Structure

Every situation contains the following elements:

Situation ID

Unique identifier.

Example

CAREER_001

Domain

Example

Career

Title

Example

"Congratulations! You Got Promoted"

Narrative

A short story describing the situation.

Maximum:

100 words.

The tone should remain conversational and optimistic, avoiding excessive technical language.

Financial Context

Hidden from the player.

Used internally by the Financial Decision Engine.

Example:

Salary Increase

+15%

Promotion Probability Modifier

+5%

Lifestyle Inflation Risk

+10%

Decision Set

Exactly **three** player choices.

No more.

Immediate Consequences

Applied immediately.

Hidden Consequences

Stored for future Planning Cycles.

---

5. Situation Categories

Every situation belongs to one primary category.

Opportunity

Creates new possibilities.

Examples:

Promotion

Scholarship

Business Partner

Investment Opportunity

Challenge

Creates financial pressure.

Examples:

Medical Expense

Flood

Vehicle Repair

Unexpected Tax

Milestone

Major life event.

Examples:

Marriage

Birth of Child

Home Purchase

Retirement

Lifestyle

Personal spending choices.

Examples:

Travel

Luxury Watch

Gaming PC

Motorcycle

Concert

Financial

Money management.

Examples:

Savings Opportunity

Debt Offer

Business Loan

Investment Choice

Social

Community and family obligations.

Examples:

Wedding Invitation

Support Parents

Family Reunion

Community Donation

---

6. Situation Difficulty

Each situation has an intensity level.

Level	Description
1	Minor
2	Moderate
3	Major
4	Critical

Example

Health

Level 1

Annual Physical Exam

Level 2

Minor Surgery

Level 3

Hospitalization

Level 4

Critical Illness

Intensity affects financial impact, not decision complexity.

---

7. Situation Frequency

The engine balances variety.

Recommended distribution:

Common

60%

Uncommon

25%

Rare

10%

Legendary

5%

Legendary situations become memorable moments.

Example

Winning a National Business Award.

Receiving a Large Inheritance.

---

8. Philippine Authenticity

All MVP situations must feel familiar to Filipino players.

Examples:

Typhoon damaged your roof.

Your parents ask for financial help.

Your company announces a 13th Month Bonus.

Your sibling is getting married.

Your child starts college.

You receive an OFW job offer.

Barangay imposes new permit requirements.

Christmas spending increases.

Avoid situations that are uncommon in the Philippine setting unless clearly marked for future Financial Worlds.

---

9. Decision Design

Every situation presents **exactly three strategic choices**.

The player never adjusts percentages.

The player never edits budgets.

The player simply chooses a life strategy.

Example

Promotion

Option A

Increase Lifestyle

Option B

Balanced Growth

Option C

Invest in Your Future

Each option represents a different financial philosophy.

---

10. Decision Principles

No option is completely right.

No option is completely wrong.

Every decision should create:

Benefits

Costs

Trade-offs

Delayed effects

The player should feel ownership of the outcome.

---

11. Immediate Consequences

Immediately after a decision, the player sees changes to:

Cash Flow

Savings

Emergency Fund

Debt

Protection

Goal Progress

Life Score

The feedback must be concise and easy to understand.

---

12. Hidden Consequences

Many decisions create delayed outcomes.

Example

Player buys Health Protection.

Immediate effect:

Monthly disposable income decreases.

Three Planning Cycles later:

Hospitalization occurs.

Medical expenses are largely covered.

The hidden benefit becomes visible only when relevant.

This is one of the core educational mechanics of SPIKE LIFE.

---

13. Cascading Events

Some situations unlock future situations.

Example

Player starts a business.

Future possibilities include:

Business Expansion

Business Loan

Hiring Employees

Market Crash

Business Acquisition

Similarly,

Marriage unlocks:

Children

Education Planning

Housing Upgrade

Family Expenses

The player's life evolves based on previous choices.

---

14. Randomness

Situations are random but intelligent.

The engine considers:

Age

Career

Income

Goals

Previous Decisions

Protection

Business Ownership

Marital Status

Children

No player should receive unrealistic combinations.

Example

A player without children should not receive "University Tuition Payment."

---

15. Life Stage Progression

Situations evolve naturally.

Ages 22–29

First Job

Promotion

Travel

Dating

Graduate School

Renting

Side Hustle

Ages 30–39

Marriage

Children

Home Purchase

Business Startup

Health Protection

Career Growth

Ages 40–49

Education Funding

Business Expansion

Leadership Roles

Health Concerns

Investment Growth

Ages 50–60

Retirement Planning

Legacy

Business Exit

Community Leadership

Health Management

This progression creates a believable life story.

---

16. Doodads

Impulse purchases form a dedicated situation group.

Examples:

Latest Smartphone

Luxury Watch

Motorcycle

Designer Bag

Gaming Console

High-End Laptop

New Car

Premium Vacation

These situations create temptation.

There is no universally correct answer.

Buying the item may increase short-term happiness but delay long-term goals.

---

17. Family Obligations

Family situations are essential.

Examples:

Support Parents

Sibling Needs Tuition

Medical Assistance

Wedding Contribution

Birthday Celebration

Christmas Expenses

Filipino financial life often includes shared family responsibilities.

The game should reflect this reality respectfully.

---

18. Advisor Integration

Players may request Advisor guidance.

The Advisor explains:

Potential trade-offs

Possible risks

Possible long-term effects

The Advisor never predicts future random events.

The player remains responsible for the final decision.

---

19. Situation Database

The Situation Engine must be fully data-driven.

Every situation should be stored as structured content rather than hard-coded logic.

Recommended structure:

Situation

↓

Decision Options

↓

Immediate Effects

↓

Hidden Effects

↓

Probability Rules

↓

Unlock Rules

↓

Advisor Guidance

This allows hundreds of situations to be added without changing the engine.

---

20. Replayability

The engine minimizes repetition.

Mechanisms include:

Situation cooldowns

Weighted randomness

Age progression

Goal weighting

Previous decisions

Unlocked situations

Economic conditions

No two games should tell the same story.

---

21. Content Targets

MVP recommendation:

Minimum situations per Life Domain:

25

Total MVP situations:

300

Future releases:

1,000+

The engine should scale without architectural changes.

---

22. Cursor Implementation Rules

Mandatory implementation requirements:

- Exactly one situation per Planning Cycle.
- Exactly three decision choices.
- Situation text should remain under 100 words.
- No manual financial calculations presented to players.
- All financial values computed by the Financial Decision Engine.
- Hidden consequences stored separately from immediate effects.
- Situation database must be configurable via JSON.
- New situations should require no code changes.

---

23. Chapter Summary

The Situation Engine transforms the abstract concept of "life" into a sequence of meaningful experiences.

It creates the emotional connection between the player and the Financial Decision Engine.

Every Planning Cycle tells one small story.

Over twenty Planning Cycles, those stories combine into a unique financial life journey.

The Situation Engine is therefore more than a random event generator—it is the narrative heart of SPIKE LIFE.

It should consistently deliver situations that are authentic, emotionally engaging, financially meaningful, and uniquely relevant to the Philippine experience while remaining extensible to future Financial Worlds.

End of Chapter 5

This chapter defines the Situation Engine that powers the narrative experience of SPIKE LIFE. Together with the Life Domain Engine and the Planning Cycle Engine, it creates a flexible, scalable framework capable of supporting thousands of realistic life situations while preserving a simple and engaging player experience.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume I – Gameplay Foundation

Chapter 6 – Decision Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Highest

---

1. Purpose

The Decision Engine is the heart of SPIKE LIFE.

If the Life Domain Engine answers:

"What part of life requires your attention?"

and the Situation Engine answers:

"What happened?"

then the Decision Engine answers:

"What will you do about it?"

Every Planning Cycle ultimately exists to present **one meaningful financial decision**.

Everything else supports that single moment.

---

2. Design Philosophy

SPIKE LIFE is **not** a budgeting simulator.

It is **not** an accounting simulator.

It is a **financial decision simulator**.

Players should spend their time making life choices—not adjusting percentages, calculating interest, or creating spreadsheets.

The Decision Engine translates simple human choices into sophisticated financial outcomes.

---

3. Core Principle

One Planning Cycle

↓

One Situation

↓

One Major Decision

↓

One Immediate Outcome

↓

One Future Story

The player should never be overwhelmed.

---

4. The One Decision Rule

Every Planning Cycle presents exactly **one major decision**.

No exceptions.

The player should never encounter:

- Multiple simultaneous financial decisions
- Budget worksheets
- Complex forms
- Manual allocation percentages

Decision fatigue is intentionally avoided.

---

5. Decision Architecture

Every decision consists of three layers.

Layer 1

Situation

Example

Promotion

Layer 2

Decision Options

Exactly three choices.

Layer 3

Financial Engine

Hidden calculations.

The player only sees Layers 1 and 2.

---

6. Decision Categories

Every decision belongs to one of the following strategic categories.

Security

Protect the future.

Examples

Emergency Fund

Insurance

Debt Reduction

Savings

Growth

Increase future wealth.

Examples

Invest

Business

Education

Career Development

Lifestyle

Enjoy today.

Examples

Travel

Luxury Purchases

Entertainment

Celebrations

Responsibility

Support others.

Examples

Parents

Children

Community

Family Obligations

Opportunity

Accept or reject new opportunities.

Examples

Business

Promotion

Property

Partnership

---

7. Three-Choice Framework

Every situation offers **exactly three choices**.

This standardization keeps gameplay fast and intuitive.

The recommended pattern is:

Option A

Secure Future

Low Risk

High Stability

Slower Goal Achievement

Option B

Balanced Growth

Moderate Risk

Balanced Progress

Recommended by Advisor

Option C

Pursue Opportunity

Higher Risk

Higher Reward

Greater Uncertainty

Not every situation uses these exact labels, but the underlying philosophy remains consistent.

---

8. Decision Examples

Promotion

Situation

You have been promoted with a ₱15,000 monthly salary increase.

Choose one strategy.

Option A

Strengthen Your Foundation

- Increase Emergency Fund
- Increase Retirement
- Delay Lifestyle Upgrades

Option B

Balanced Progress

- Improve Lifestyle Moderately
- Continue Saving
- Maintain Protection

Option C

Upgrade Your Lifestyle

- Buy a Car
- Increase Spending
- Delay Long-Term Goals

Hospitalization

Situation

A family member requires hospitalization.

Option A

Use Emergency Fund

Option B

Borrow Money

Option C

Liquidate Investments

Each choice produces different long-term consequences.

---

9. Trade-Off Philosophy

Every decision must involve sacrifice.

There are **no perfect choices**.

Examples:

Buying Health Protection

↓

Travel delayed

Buying a House

↓

Emergency Fund grows more slowly

Starting a Business

↓

Higher risk

↓

Potentially higher wealth

Supporting Parents

↓

Retirement delayed

Players learn that financial planning is the art of balancing competing priorities.

---

10. Immediate Consequences

After every decision,

the player immediately sees:

Cash Flow

Savings

Emergency Fund

Debt

Goal Progress

Protection

Life Score

The feedback should be concise and visual.

---

11. Delayed Consequences

Many decisions create hidden future effects.

Example

Decision

Purchase Health Protection

Immediate

Disposable Income decreases.

Future

Hospitalization

↓

Medical expenses largely covered.

The game rewards long-term thinking without revealing future events.

---

12. Opportunity Cost

Every decision should implicitly answer:

"What am I giving up?"

Examples:

Travel

↓

Retirement delayed.

Luxury Watch

↓

Emergency Fund delayed.

Business Investment

↓

Higher future uncertainty.

The engine should make opportunity costs visible after the decision.

---

13. Financial Decision Engine Integration

The player never allocates money manually.

The Financial Decision Engine interprets every choice.

Example

Player chooses

Strengthen Your Foundation

The engine automatically determines:

Emergency Fund Contribution

Retirement Contribution

Protection Increase

Savings Allocation

Remaining Disposable Income

The complexity remains hidden.

---

14. Annual Financial Allocation Rules

The Financial Engine follows these baseline principles.

Living Expenses

Dynamic

30–50%

of Monthly Income.

Long-Term Financial Security

Recommended maximum

25%

of Gross Monthly Income

allocated across:

Emergency Fund

Family Protection Plan

Health Protection Plan

Education Plan

Retirement Security Plan

Savings

The remaining disposable income is managed according to the selected decision strategy.

Players never manipulate these percentages directly.

---

15. Emergency Fund Logic

Emergency Fund Target

Six months of current monthly income.

The fund grows gradually through automatic monthly contributions.

It cannot be completed instantly.

When emergencies occur,

the Emergency Fund becomes the first financial buffer before debt or investments are used.

---

16. Goal Trade-Off Engine

Every player maintains multiple life goals.

Examples

House

Travel

Business

Education

Retirement

Every financial decision influences one or more goals.

Examples

Buying Health Protection

↓

House delayed

Starting Business

↓

Travel delayed

Funding Retirement

↓

Business capital reduced

The game teaches prioritization rather than optimization.

---

17. Decision Timer

Every decision uses a countdown timer.

Default

15 seconds.

Available settings

No Timer

20 Seconds

15 Seconds

10 Seconds

5 Seconds

The timer prevents analysis paralysis and maintains gameplay pace.

---

18. Auto Advisor

If the timer expires,

the player does **not** receive a random decision.

Instead,

the Advisor automatically chooses the Balanced Growth option.

This represents a reasonable, conservative financial recommendation.

The player is informed that:

Failure to decide is itself a decision.

---

19. Advisor Consultation

Players may consult the Advisor before making a decision.

The timer pauses during consultation.

Maximum consultation time

20 seconds.

The Advisor explains:

Advantages

Disadvantages

Potential trade-offs

Goal impact

Financial Health impact

The Advisor never predicts future random events.

---

20. Financial Health Impact

Every decision influences the player's Financial Health.

Examples

Increasing Protection

Financial Health improves.

Large Luxury Purchase

Financial Health decreases slightly.

Debt Reduction

Financial Health improves.

Ignoring Emergencies

Financial Health declines.

Financial Health summarizes the overall resilience of the player's financial position.

---

21. Life Score Integration

Every decision contributes to Life Score.

Life Score considers:

Financial Security

Protection

Goal Achievement

Wealth Growth

Life Stability

The player sees only the resulting score change.

The underlying calculations remain hidden.

---

22. Decision Memory

The engine remembers previous decisions.

Repeated behaviors influence future situations.

Examples

Repeated luxury spending

↓

Higher financial stress

Consistent saving

↓

Greater resilience

Business success

↓

Larger investment opportunities

The player's history shapes future gameplay.

---

23. Decision Diversity

The engine should discourage repetitive strategies.

If the player always chooses aggressive investment,

future risks increase.

If the player always avoids risk,

wealth growth slows.

Balanced play is rewarded over time.

---

24. User Experience Rules

Decision screens should remain clean.

Display only:

Situation

Illustration

Three Choices

Timer

Advisor Button

Expected trade-off summary

Never display:

Financial formulas

Interest calculations

Investment mathematics

Budget tables

The interface should encourage intuitive decisions.

---

25. Cursor Implementation Rules

Mandatory implementation requirements:

- Exactly three decision choices per situation.
- Exactly one decision per Planning Cycle.
- Financial calculations handled exclusively by the Financial Decision Engine.
- No manual allocation interfaces.
- Timer configurable by difficulty.
- Advisor consultation pauses timer.
- Auto Advisor selects the Balanced option on timeout.
- Decision consequences stored as both immediate and delayed effects.
- Decision templates configurable through JSON.
- Decision Engine fully data-driven.

---

26. Future Expansion

The Decision Engine is designed for future enhancements, including:

- AI-generated decision options

- Personality-based recommendations
- Cooperative decision events
- Ethical dilemmas
- Family voting mechanics
- Business board decisions
- Multi-step strategic scenarios

These additions must preserve the One Decision Rule and the simplicity of the player experience.

---

27. Chapter Summary

The Decision Engine is the central interaction point of SPIKE LIFE.

Players never compete by solving financial equations.

They compete by making wiser life decisions.

Every Planning Cycle presents a realistic situation.

Every situation demands one meaningful choice.

The Financial Decision Engine quietly transforms that choice into a chain of financial consequences that may unfold immediately or many years later.

The player experiences the results—not the calculations.

This separation between **simple decisions** and **sophisticated financial simulation** is the defining characteristic of SPIKE LIFE and ensures that the game remains engaging, educational, and accessible to players regardless of their financial background.

End of Chapter 6

This chapter establishes the Decision Engine as the primary gameplay interaction system. Together with the Life Domain Engine and the Situation Engine, it completes the Core Gameplay Loop. The next volume will build upon this foundation by defining the Financial Planning Engine, where every decision is translated into realistic cash flow, protection, investments, goals, and long-term financial outcomes.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume II – Financial Engine

Chapter 7 – Financial Planning Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

The Financial Planning Engine is the mathematical and simulation core of SPIKE LIFE.

It translates every player decision into realistic financial outcomes while keeping all calculations invisible to the player.

The Financial Planning Engine should feel like a professional financial planner quietly working behind the scenes.

The player experiences life.

The engine manages the finances.

---

2. Design Philosophy

Players should never feel like accountants.

They should never:

- Create spreadsheets
- Compute interest
- Compute inflation
- Calculate future values
- Build budgets manually
- Decide allocation percentages

Instead, players make life decisions.

The Financial Planning Engine automatically converts those decisions into sound financial planning mechanics.

---

3. Financial Simulation Cycle

Every Planning Cycle executes the following sequence.

Load Current Financial State

↓

Update Age

↓

Update Income

↓

Update Living Expenses

↓

Compute Financial Capacity

↓

Apply Player Decision

↓

Allocate Funds

↓

Apply Investment Growth

↓

Apply Inflation

↓

Apply Random Events

↓

Update Goals

↓

Update Protection

↓

Update Net Worth

↓

Update Financial Health

↓

Update Life Score

↓

Save New Financial State

Every Planning Cycle ends with a complete financial recalculation.

---

4. Core Financial Variables

Every player maintains the following financial profile.

Personal Information

Current Age

Career

Civil Status

Dependents

Employment Status

Income

Monthly Income

Annual Income

13th Month Pay

Bonuses

Business Income

Passive Income

Expenses

Living Expenses

Debt Payments

Insurance Premiums

Education Funding

Retirement Contributions

Emergency Fund Contributions

Lifestyle Spending

Assets

Cash

Savings

Emergency Fund

Investments

Business Value

Property Value

Retirement Fund

Education Fund

Liabilities

Credit Card Debt

Personal Loans

Housing Loans

Business Loans

Emergency Debt

Goals

House

Travel

Business

Education

Retirement

Emergency Fund

---

5. Income Engine

Income is calculated monthly but processed automatically inside each Planning Cycle.

Annual Income

Monthly Income × 12

Planning Cycle Income

Annual Income ÷ 2

Income may increase due to:

Promotion

Salary Adjustment

Business Growth

Side Hustles

Bonuses

Income may decrease due to:

Demotion

Job Loss

Reduced Business Revenue

Medical Leave

---

6. Living Expense Engine

Living expenses represent essential household costs.

Default Range

30–50%

of Monthly Income.

Living expenses adjust dynamically based on:

Age

Marriage

Children

Housing

Lifestyle

Inflation

Employment Status

Example

Single

≈30%

Married

≈38%

Two Children

≈45%

Three Children

≈50%

This percentage is managed automatically.

---

7. Financial Capacity

Financial Capacity represents money available after basic living expenses.

Formula

Income

− Living Expenses

=

Financial Capacity

Every financial decision draws from this capacity.

---

8. Automatic Allocation Engine

Players never distribute money manually.

The engine automatically allocates funds according to the selected strategy.

Allocation priorities include:

Emergency Fund

Protection

Retirement

Education

Savings

Investments

Debt Reduction

Lifestyle

Each decision modifies the priority order rather than the percentages.

---

9. Financial Security Rule

As a general guideline,

long-term financial security should not normally exceed:

25%

of Gross Monthly Income.

This allocation includes:

Emergency Fund

Family Protection Plan

Health Protection Plan

Retirement Security Plan

Education Plan

Long-Term Savings

Players may occasionally exceed this level through special situations, but the Advisor will identify potential cash flow strain.

---

10. Emergency Fund Engine

Emergency Fund Target

6 × Current Monthly Income

Example

Monthly Income

₱50,000

Target

₱300,000

Emergency Fund growth occurs gradually through automatic monthly contributions.

Emergency Funds are used first during:

Medical Emergencies

Job Loss

Unexpected Repairs

Natural Disasters

Business Interruptions

Only after depletion does the engine consider borrowing or liquidation.

---

11. Savings Engine

Savings represent low-risk liquidity.

Default Annual Growth

4%

Savings provide:

Liquidity

Security

Emergency Support

Goal Funding

Savings remain separate from investments.

---

12. Investment Engine

Default investment assumptions.

Investment	Annual Growth
Savings	4%
Government / Bonds	5%
Balanced Portfolio	6%
Equity Investments	8%

Growth compounds automatically.

Players choose strategies rather than financial instruments.

---

13. Business Engine

Business behaves differently.

Growth is variable.

Possible outcomes

Loss

−100%

Stable

0%

Moderate Growth

+20%

Strong Growth

+80%

Exceptional Growth

+300%

Business outcomes are influenced by:

Market Conditions

Experience

Economic Cycles

Random Events

Previous Decisions

Business introduces the highest volatility in the game.

---

14. Inflation Engine

Inflation is fixed for MVP.

Annual Inflation

4%

Inflation automatically increases:

Housing Costs

Education Costs

Travel Costs

Lifestyle Costs

Business Costs

Retirement Requirements

Goal Funding Targets

Players experience inflation without calculating it.

---

15. Goal Inflation

Every Dream Board goal maintains two values.

Present Value

Future Value

Example

House Today

₱5,000,000

Target Age

35

Current Age

22

Future Cost

Automatically calculated

The engine recalculates every Planning Cycle.

---

16. Debt Engine

Debt is introduced only through situations.

Debt may include:

Credit Cards

Housing Loans

Business Loans

Emergency Loans

Debt automatically affects:

Cash Flow

Financial Health

Life Score

Future Borrowing Capacity

Debt should feel useful but dangerous.

---

17. Protection Engine

Protection consists of five planning categories.

Family Protection Plan

Income Protection Plan

Health Protection Plan

Education Protection Plan

Retirement Security Plan

These replace generic insurance terminology throughout the game.

Protection reduces future losses rather than creating immediate rewards.

---

18. Goal Funding Engine

Every Planning Cycle,

the engine updates progress toward:

House

Travel

Business

Education

Retirement

Emergency Fund

Progress is displayed visually.

Example

House

████████░░

80%

Goal completion should feel motivating.

---

19. Random Financial Events

Unexpected financial events may occur.

Examples

Medical Bills

Flood

Typhoon

Investment Boom

Market Crash

Wedding

Family Support

Vehicle Repairs

Christmas Expenses

These events interact directly with the Financial Planning Engine.

---

20. Financial Health Engine

Financial Health summarizes overall financial resilience.

Calculated from:

Cash Flow

Emergency Fund

Protection

Debt

Goal Progress

Investment Balance

Displayed as:

Excellent

Good

Stable

Vulnerable

Critical

The player should understand their financial condition at a glance.

---

21. Net Worth Engine

Net Worth is calculated automatically.

Formula

Assets

−

Liabilities

=

Net Worth

Net Worth contributes to Life Score but does not determine victory.

---

22. Life Score Integration

The Financial Planning Engine contributes approximately:

25%

Financial Security

20%

Protection

20%

Goal Achievement

20%

Wealth Creation

15%

Life Stability

These components combine to produce the player's Life Score.

---

23. Annual Financial Planning

After Planning Cycle 2,

players receive:

Mandatory Philippine

13th Month Pay

Players select one primary allocation.

Examples

Emergency Fund

House

Business

Travel

Retirement

Education

Protection

Debt Reduction

Lifestyle

The Financial Engine performs all required reallocations automatically.

---

24. Advisor Integration

The Advisor reads the Financial Engine.

Advice may include:

Emergency Fund below target.

Protection gap increasing.

Retirement behind schedule.

Business concentration risk.

House goal delayed.

Excellent savings discipline.

Advice is always personalized.

---

25. Data Persistence

Every Planning Cycle stores:

Income

Expenses

Assets

Liabilities

Goals

Protection

Emergency Fund

Financial Health

Life Score

The Financial Planning Engine is fully stateful.

Every decision permanently influences future gameplay.

---

26. Cursor Implementation Rules

Mandatory implementation requirements:

- The Financial Planning Engine must be completely separated from the UI.
- All calculations occur server-side or within the simulation engine.
- No financial formulas are exposed to players.
- Inflation applies automatically every year.

- Investment returns compound automatically.
- Emergency Funds deplete before borrowing.
- Goal funding updates every Planning Cycle.
- Every financial variable must be serializable to JSON.
- The engine must remain deterministic when given the same inputs and random seed.
- Financial assumptions must be configurable through a centralized configuration file for future localization.

---

27. Future Expansion

The Financial Planning Engine is designed to support:

Country-specific inflation

Variable interest rates

Tax systems

Currency conversion

Housing loans

Mortgage amortization

Investment products

Cryptocurrency

Business sectors

Economic recessions

Central bank policy

Real-world economic scenarios

These extensions should require configuration changes rather than architectural redesign.

---

28. Chapter Summary

The Financial Planning Engine is the invisible intelligence behind SPIKE LIFE.

Players experience simple life decisions.

The engine performs sophisticated financial planning.

Every promotion, purchase, emergency, investment, or family obligation updates a living financial model that evolves over the player's simulated lifetime.

The engine should always remain:

- Accurate
- Transparent in outcomes
- Invisible in calculations
- Consistent
- Extensible

It is this separation between intuitive gameplay and rigorous financial simulation that enables SPIKE LIFE to teach sound financial planning without ever feeling like a spreadsheet.

End of Chapter 7

This chapter establishes the Financial Planning Engine as the computational core of SPIKE LIFE. It converts human decisions into realistic financial outcomes while preserving a clean and engaging player experience. Subsequent chapters will build upon this engine by defining the Goal Engine, Wealth Engine, Risk & Protection Engine, and Life Score algorithm in greater detail.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume II – Financial Engine

Chapter 8 – Goal Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

The Goal Engine transforms financial planning into a game.

Rather than asking players to simply accumulate wealth, SPIKE LIFE asks them to pursue meaningful life goals.

The Goal Engine continuously answers one question:

"Am I getting closer to the life I want?"

Goals become the emotional driver of the game.

Money becomes the tool.

---

2. Design Philosophy

The Goal Engine is based on one fundamental belief:

People do not save money because they love saving money.

People save money because they have dreams.

The Financial Engine exists to fund those dreams.

The Goal Engine exists to measure progress toward them.

---

3. Goal Lifecycle

Every goal follows the same lifecycle.

Dream Created

↓

Future Cost Calculated

↓

Funding Begins

↓

Progress Updated

↓

Unexpected Events

↓

Goal Delayed or Accelerated

↓

Goal Achieved

↓

Life Score Updated

---

4. Goal Categories

The MVP includes six primary life goals.

🏠 Home Ownership

Purchase of:

House

Townhouse

Condominium

Lot

Home Construction

✈️ Travel

Domestic

International

Family Vacation

Pilgrimage

Retirement Travel

🚀 Business

Startup

Expansion

Second Branch

Equipment Purchase

Franchise

🎓 Education

Child Education

Graduate School

Professional Certifications

Skills Development

🏖️ Retirement

Desired Retirement Income

Desired Retirement Age

Desired Retirement Lifestyle

🛡️ Emergency Fund

Automatically created.

Target:

Six months of current monthly income.

---

5. Dream Board Initialization

Before gameplay begins,

every player creates a Dream Board.

For each goal:

Current Cost

↓

Target Age

↓

Priority

↓

Financial Engine computes Future Value

The player never calculates future values manually.

---

6. Philippine Default Values

These defaults are configurable.

Home

Present Cost

₱5,000,000

Travel

Present Cost

₱300,000

Business Startup

₱500,000

Child Education

₱500,000

(at today's value)

Retirement

Player selects desired monthly retirement income.

Default:

₱80,000/month

Emergency Fund

Automatically computed.

6 × Monthly Income.

---

7. Inflation Engine

All goals inflate automatically.

Default inflation

4% annually.

Formula

Future Cost

=

Present Value

×

(1 + Inflation)^Years

Players never see the formula.

They simply experience:

"The longer I wait, the more expensive my dream becomes."

---

8. Goal Funding

Every Planning Cycle,

the Financial Planning Engine contributes automatically toward goals based on the player's decisions.

Players never manually allocate money.

Instead,

their strategic choices influence:

Funding Priority

Contribution Amount

Funding Speed

9. Goal Priority

Each goal has one of three priorities.

High

Primary Life Goal

Medium

Important

Low

Optional

Priority influences automatic funding.

Players may change priorities during Annual Planning.

---

10. Goal Progress

Every goal maintains:

Current Cost

Future Cost

Current Funding

Funding %

Estimated Completion Age

Displayed visually.

Example

🏠 HOME

72%

Target Age

35

Projected Completion

36

Simple.

Easy to understand.

---

11. Goal Delays

One of the defining mechanics of SPIKE LIFE.

Good financial decisions may delay goals.

Examples

Purchased Health Protection

↓

House delayed

12 months

Started Business

↓

Travel delayed

18 months

Supported Parents

↓

Retirement delayed

6 months

The player learns that responsible decisions often involve sacrifice.

---

12. Goal Acceleration

Positive events may accelerate progress.

Examples

Promotion

↓

House Goal earlier

Business Success

↓

Retirement earlier

Investment Growth

↓

Travel Goal funded

Unexpected Bonus

↓

Education Fund completed

The game rewards discipline.

---

13. Goal Failure

Some goals may not be achieved.

Example

House Goal

Target Age

35

Actual

Age 40

The goal remains active.

The player may still complete it later.

The game never permanently removes a dream.

---

14. Goal Dependencies

Certain goals influence others.

Examples

Marriage

↓

Housing becomes more important.

Child Born

↓

Education Goal activated.

Business Success

↓

Retirement accelerates.

House Purchased

↓

Living Expenses change.

Goals interact naturally.

---

15. Dynamic Goals

Life changes.

Players may modify goals during Year-End Planning.

Examples

Increase Retirement Target

Delay Travel

Cancel Business Plan

Upgrade House

Reduce Lifestyle

Dreams evolve.

The game should reflect this.

---

16. Annual Planning

After receiving the 13th Month Pay,

players review their Dream Board.

Possible actions:

Change Priority

Adjust Target Age

Increase Goal Amount

Cancel Goal

Create New Goal

No financial calculations are required.

---

17. Goal Completion

When funding reaches the required future value,

the goal is completed.

Example

🏠

House Purchased!

The game celebrates with:

Animation

Sound

Achievement Badge

Life Score Bonus

Financial Health Update

Goal completion should feel rewarding.

---

18. Missed Goals

Missing a goal should not feel like failure.

Instead,

the Advisor explains.

Example

"Your retirement target has moved two years later due to higher housing expenses."

The player understands why.

---

19. Goal Trade-Off Engine

Every major financial decision updates one or more goals.

Example

Situation

Promotion

Choice

Upgrade Lifestyle

Effects

House

−6 months

Emergency Fund

−12 months

Travel

+3 months

Retirement

−9 months

The engine computes these automatically.

---

20. Hidden Opportunity Cost

Players should see:

Goal Progress

but not the detailed mathematics.

The Financial Engine quietly computes:

Inflation

Compound Growth

Funding Rates

Opportunity Costs

Cash Flow

The interface remains simple.

---

21. Goal Notifications

The player receives milestone updates.

Examples

House Fund

50%

Emergency Fund Complete

Retirement Halfway Funded

Business Capital Ready

Education Fund Behind Schedule

Notifications should be encouraging rather than punitive.

---

22. Goal Completion Rewards

Completing a goal increases:

Life Score

Financial Health

Player Satisfaction

Examples

House

+25 Life Score

Emergency Fund

+15

Retirement

+30

Business

Variable

Education

+20

Exact values remain configurable.

---

23. Advisor Integration

The Advisor continuously evaluates goal progress.

Examples

"You are ahead of schedule."

"Inflation is increasing your housing cost."

"Reducing discretionary spending could achieve your travel goal one year earlier."

Advice explains trade-offs.

The Advisor never dictates decisions.

---

24. Replayability

Different Dream Boards create different games.

Player A

House First

Player B

Business First

Player C

Travel First

Player D

Retirement First

The same situations produce completely different outcomes.

---

25. Cursor Implementation Rules

Mandatory implementation requirements:

- Every goal stored independently.
- Future Value recalculated annually.
- Inflation configurable globally.
- Goal funding automatic.

- Progress bars animated.
- Goal completion generates celebration events.
- Goal priorities editable during Annual Planning.
- Goals serialized in JSON.
- Goal Engine completely independent from UI.
- No manual financial calculations exposed.

---

26. Future Expansion

Future Goal categories may include:

Wedding

Vehicle

Second Home

Farm

Overseas Migration

Legacy Giving

Charitable Foundation

Passive Income Target

Early Retirement

The Goal Engine should support unlimited goal templates.

---

27. Chapter Summary

The Goal Engine gives purpose to financial planning.

Players are not simply collecting money.

They are building the life they envisioned at the beginning of the game.

Every promotion, emergency, investment, sacrifice, and celebration moves them closer to—or farther from—their dreams.

This transforms financial planning from a numerical exercise into a deeply personal journey.

The Goal Engine is therefore the emotional core of SPIKE LIFE.

It ensures that every financial decision is connected to something meaningful, reinforcing the central philosophy of the game:

Money is not the destination. It is the vehicle that helps people achieve the life they aspire to live.

End of Chapter 8

This chapter defines the Goal Engine, which translates aspirations into measurable progress and integrates seamlessly with the Financial Planning Engine. Together they create a simulation where players experience the real-world tension between dreams, limited resources, time, inflation, and life's unexpected events.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume II – Financial Engine

Chapter 9 – Wealth Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

The Wealth Engine is responsible for simulating how players build, preserve, lose, and transfer wealth throughout their financial lives.

Unlike many financial games, wealth in SPIKE LIFE is **not** the objective.

Wealth is a resource that enables players to achieve life goals, protect their families, and create long-term financial security.

The Wealth Engine answers one question:

"How financially strong are you becoming?"

---

2. Design Philosophy

SPIKE LIFE deliberately separates **wealth** from **success**.

A player with the highest Net Worth may not win.

Conversely, a player with modest wealth but excellent protection, completed goals, and strong financial discipline may achieve a higher Life Score.

The Wealth Engine exists to support life—not define it.

---

3. Definition of Wealth

Wealth is calculated as:

Assets

−

Liabilities

=

Net Worth

This calculation is performed automatically every Planning Cycle.

Players never compute Net Worth manually.

---

4. Wealth Components

The Wealth Engine tracks the following asset classes.

Liquid Assets

Cash

Savings

Emergency Fund

Money Market Holdings

Investment Assets

Government Bonds

Balanced Investments

Equity Investments

Future Investment Products

Property Assets

House

Condominium

Land

Investment Property

Business Assets

Business Equity

Business Cash

Business Equipment

Business Goodwill

Business Expansion Value

Retirement Assets

Retirement Fund

Employer Retirement Benefits

Private Retirement Accounts

Education Assets

Education Fund

Scholarship Benefits

Education Savings

---

5. Liabilities

The engine tracks all major obligations.

Housing Loan

Business Loan

Personal Loan

Credit Card Debt

Emergency Loan

Family Debt

Outstanding liabilities reduce:

Net Worth

Financial Health

Borrowing Capacity

Life Score

---

6. Net Worth Calculation

Every Planning Cycle,

the Wealth Engine automatically computes:

Cash

+

Savings

+

Investments

+

Property

+

Business Value

+

Retirement Fund

+

Education Fund

–

Outstanding Debt

=

Net Worth

Displayed as:

Current Net Worth

Net Worth Change

Historical Trend

---

7. Wealth Growth Sources

Players build wealth through:

Salary Growth

Business Growth

Investment Returns

Property Appreciation

Debt Reduction

Disciplined Saving

Inheritance (Rare)

Unexpected Opportunities

Each source contributes differently over time.

---

8. Wealth Erosion

Wealth may decrease due to:

Medical Expenses

Natural Disasters

Business Losses

Market Declines

Fraud

Scams

Poor Financial Decisions

Lifestyle Inflation

Excessive Debt

The engine reflects the reality that wealth is difficult to build and easy to lose.

---

9. Asset Appreciation

Assets appreciate differently.

Savings

4% annually

Government / Bond Investments

5%

Balanced Portfolio

6%

Equity Investments

8%

Residential Property

Variable

Typically follows inflation with modest long-term appreciation.

Business

Highly variable

Possible annual outcomes:

−100%

to

+300%

The Wealth Engine automatically applies appreciation every year.

---

10. Property Ownership

Property ownership is a major wealth milestone.

Property may be acquired through:

House Goal Completion

Special Opportunity

Business Success

Inheritance

Property ownership changes:

Living Expenses

Net Worth

Financial Health

Future Housing Events

Property Maintenance Costs

Property should feel valuable but not risk-free.

---

11. Business Valuation

Businesses are dynamic assets.

Business Value depends on:

Capital Invested

Years in Operation

Market Conditions

Owner Decisions

Random Events

Business Expansion

Business Reputation

Business value is recalculated every Planning Cycle.

---

12. Investment Portfolio

Players never purchase individual financial products.

Instead,

the Decision Engine chooses an investment strategy.

Strategies include:

Conservative

Balanced

Growth

Aggressive

The Wealth Engine converts these strategies into realistic investment allocations.

---

13. Liquidity

Not all wealth is immediately available.

Liquidity ranking:

Cash

↓

Savings

↓

Emergency Fund

↓

Investments

↓

Property

↓

Business Equity

The engine prioritizes liquid assets before forcing asset liquidation.

---

14. Asset Liquidation

When financial emergencies occur,

assets are liquidated in this order:

Cash

↓

Emergency Fund

↓

Savings

↓

Investments

↓

Business Assets

↓

Property

The player is informed of the financial consequences.

---

15. Lifestyle Inflation

One of the key educational mechanics.

As income increases,

players face pressure to increase spending.

Examples

Luxury Car

Bigger House

Premium Travel

Designer Goods

Fine Dining

Lifestyle inflation reduces long-term wealth if unmanaged.

The Wealth Engine models this automatically.

---

16. Wealth Milestones

The engine celebrates important achievements.

Examples

First ₱100,000 Saved

Emergency Fund Completed

First Million Net Worth

Debt-Free

Business Owner

Property Owner

Retirement Fully Funded

Milestones provide:

Achievements

Animations

Life Score Bonuses

Positive reinforcement

---

17. Wealth Distribution

The engine evaluates diversification.

Example

Player owns:

90% Business

10% Cash

Advisor warns:

"Your wealth is highly concentrated."

Balanced portfolios improve Financial Health.

Diversification reduces future risk.

---

18. Wealth Concentration Risk

High concentration creates vulnerability.

Examples

Only Business

↓

Business Failure

↓

Major Wealth Loss

Only Property

↓

Typhoon Damage

↓

Large Financial Impact

Diversification becomes an important educational lesson.

---

19. Inflation Impact

Inflation quietly erodes purchasing power.

Although Net Worth may increase,

real purchasing power may not.

The Advisor occasionally explains this concept using plain language.

Example

"Your investments grew, but housing costs increased faster than expected."

---

20. Wealth Timeline

The engine records wealth progression throughout the game.

Displayed graphically.

Example

Age 22

↓

₱50,000

↓

Age 30

↓

₱850,000

↓

Age 40

↓

₱4,200,000

↓

Age 50

↓

₱11,800,000

Players see the story of their financial growth.

---

21. Wealth and Life Score

Net Worth contributes to Life Score,

but only as one component.

Approximate contribution:

20%

Players who sacrifice everything for wealth should not automatically win.

Balanced planning remains the objective.

---

22. Advisor Integration

The Advisor evaluates wealth quality.

Example guidance:

"You have accumulated significant wealth, but your emergency fund remains below target."

or

"Your investment portfolio is growing well, but your retirement goal is underfunded."

Advice focuses on balance rather than maximizing assets.

---

23. Replayability

Different decisions produce different wealth paths.

Examples

High Saving

↓

Slow but stable wealth

Business Focus

↓

High volatility

Aggressive Investing

↓

Large gains

Large losses

Lifestyle Spending

↓

Lower wealth

Higher immediate satisfaction

Every player develops a unique financial journey.

---

24. Future Expansion

The Wealth Engine supports future asset classes.

Examples

Real Estate Investment Trusts (REITs)

Cooperative Shares

Mutual Funds

Exchange-Traded Funds

Digital Assets

Agricultural Investments

Rental Properties

International Investments

These additions require only configuration updates, not architectural changes.

---

25. Cursor Implementation Rules

Mandatory implementation requirements:

- Net Worth recalculated every Planning Cycle.
- Asset values stored independently.
- Liabilities tracked independently.
- Appreciation rates configurable.
- Business valuation event-driven.
- Property values inflation-adjusted.
- Wealth history persisted.
- Wealth graphs generated dynamically.
- No manual calculations displayed.
- Wealth Engine completely separated from the UI.

---

26. Chapter Summary

The Wealth Engine transforms financial decisions into long-term financial outcomes.

Players experience the gradual accumulation—or erosion—of wealth as a natural consequence of their choices.

The engine teaches that wealth is created through:

- Consistent saving
- Disciplined investing
- Responsible borrowing

- Business growth
- Long-term planning

It also demonstrates that wealth can be lost through:

- Poor decisions
- Excessive debt
- Lifestyle inflation
- Lack of protection
- Unmanaged risk

Most importantly, SPIKE LIFE teaches that wealth is **not the destination**.

Wealth is one of several pillars supporting a financially balanced life.

Players ultimately win not by becoming the richest person at the table, but by using wealth wisely to achieve their dreams, protect what matters most, and build lasting financial security.

End of Chapter 9

This chapter establishes the Wealth Engine as the asset management layer of SPIKE LIFE. Working alongside the Financial Planning Engine and Goal Engine, it models the creation, preservation, and responsible use of wealth while reinforcing the game's central philosophy that financial success is measured by balance, resilience, and life fulfillment—not by Net Worth alone.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume II – Financial Engine

Chapter 10 – Risk & Protection Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

The Risk & Protection Engine simulates one of the most important realities of financial life:

Life is uncertain.

No amount of planning can prevent unexpected events.

However,

good financial planning can reduce their financial impact.

The purpose of this engine is to demonstrate that protection is not an expense—

it is a strategy for preserving wealth, protecting goals, and maintaining financial stability.

---

2. Design Philosophy

SPIKE LIFE does **not** teach insurance.

SPIKE LIFE teaches **risk management**.

Players should naturally discover why protection matters through experience.

The game never forces protection.

Instead,

players experience the consequences of being prepared—or unprepared.

---

3. Risk Categories

The engine models five major categories of financial risk.

Family Risk

Death of income earner

Loss of financial support

Family obligations

Dependent care

Income Risk

Job loss

Disability

Reduced working capacity

Career disruption

Business interruption

Health Risk

Hospitalization

Critical illness

Accident

Long-term treatment

Mental health recovery

Property Risk

Typhoon

Fire

Flood

Earthquake

Theft

Major repairs

Wealth Risk

Investment losses

Business failure

Scams

Fraud

Economic recession

Inflation

---

4. Protection Philosophy

Instead of generic insurance products,

SPIKE LIFE uses Financial Planning categories.

Players never buy:

❌ Life Insurance

❌ Critical Illness Insurance

❌ Accident Insurance

Instead,

they invest in:

✅ Family Protection Plan

✅ Income Protection Plan

✅ Health Protection Plan

✅ Education Protection Plan

✅ Retirement Security Plan

This aligns protection with financial goals rather than products.

---

5. Protection Plans

Family Protection Plan

Purpose

Protect family financial stability.

Reduces losses from:

Death

Loss of income

Family dependency

Income Protection Plan

Purpose

Protect earning ability.

Reduces impact from:

Disability

Job interruption

Income loss

Business interruption

Health Protection Plan

Purpose

Reduce medical expenses.

Protects against:

Hospitalization

Critical illness

Major surgery

Medical emergencies

Education Protection Plan

Purpose

Preserve education funding.

Protects education goals when:

Parents lose income

Unexpected financial crises occur

Business losses happen

Retirement Security Plan

Purpose

Protect retirement objectives.

Helps preserve retirement savings despite:

Inflation

Unexpected expenses

Delayed retirement

Financial setbacks

---

6. Protection Levels

Each category has four readiness levels.

None

Basic

Adequate

Comprehensive

The Financial Engine continuously evaluates readiness.

---

7. Protection Readiness Score

Every Planning Cycle,

the engine computes:

Protection Readiness

Based on:

Income

Dependents

Assets

Goals

Emergency Fund

Current Protection

Age

Displayed simply as:

Excellent

Good

Needs Improvement

High Risk

Critical

Players never see complex formulas.

---

8. Protection Gap

The engine identifies financial gaps.

Example

Player

Married

Two children

Minimal Family Protection

Advisor

"You currently have a significant Family Protection gap."

The gap is explained,

not calculated.

---

9. Risk Events

The engine continuously generates possible risks.

Examples

Hospitalization

Critical Illness

Job Loss

Business Failure

Flood

Typhoon

House Fire

Vehicle Accident

Family Emergency

Unexpected Funeral Expenses

Economic Recession

Investment Crash

These events are independent from player choices.

Life remains uncertain.

---

10. Probability Engine

Risk events are weighted.

Factors include:

Age

Career

Business Ownership

Lifestyle

Health Decisions

Previous Events

Economic Conditions

Protection Status

The engine avoids unrealistic repetition.

---

11. Risk Severity

Every event has four levels.

Minor

Moderate

Major

Catastrophic

Example

Health

Minor

Outpatient Consultation

Moderate

Hospital Admission

Major

Critical Illness

Catastrophic

Long-Term Disability

Severity determines financial impact.

---

12. Protection Benefits

Protection never prevents events.

Protection reduces consequences.

Example

Hospitalization

Without Health Protection

Medical Cost

₱850,000

Player Pays

₱850,000

With Health Protection

Medical Cost

₱850,000

Player Pays

₱35,000

The event still happens.

The financial outcome changes.

---

13. Emergency Fund Interaction

Before protection activates,

the Financial Engine evaluates:

Emergency Fund

↓

Protection

↓

Savings

↓

Investments

↓

Debt

↓

Asset Liquidation

Protection is part of a complete financial strategy,

not a replacement for emergency savings.

---

14. Risk Chain

Some risks trigger additional events.

Example

Job Loss

↓

Emergency Fund Used

↓

Retirement Contributions Stop

↓

House Goal Delayed

↓

Financial Health Declines

The engine models cascading consequences.

---

15. Delayed Value

Protection often appears "expensive."

The engine intentionally delays its visible benefit.

Example

Age 25

Player buys Health Protection.

Nothing happens.

Age 32

Major Surgery

↓

Medical costs largely covered.

Players learn that protection creates value before the crisis occurs.

---

16. Advisor Recommendations

The Advisor evaluates protection continuously.

Examples

"Completing your Emergency Fund should be your first priority."

"Your family's financial protection appears insufficient."

"Your retirement is well funded, but your health protection is below recommended levels."

Advice is always contextual.

---

17. Family Responsibility

Protection recommendations increase when players:

Marry

Have children

Support parents

Start businesses

Take housing loans

Responsibility increases the importance of protection.

---

18. Economic Risks

The engine also models macroeconomic events.

Examples

High Inflation

Economic Slowdown

Interest Rate Increase

Business Recession

Pandemic

Government Policy Change

These affect multiple players simultaneously.

---

19. Natural Disasters

Philippine-specific events include:

Typhoon

Flood

Landslide

Earthquake

Volcanic Activity

These primarily affect:

Housing

Business

Income

Community

The engine reflects Philippine realities.

---

20. Financial Resilience

The Risk Engine contributes to Financial Health.

High resilience includes:

Emergency Fund Complete

Adequate Protection

Diversified Assets

Low Debt

Stable Cash Flow

Strong resilience improves Life Score.

---

21. Life Score Integration

Protection contributes approximately

20%

of the total Life Score.

Players who ignore protection may accumulate wealth,

but remain financially vulnerable.

Balanced planning is consistently rewarded.

---

22. Replayability

Different players experience different risks.

Examples

Entrepreneur

Business Risks

Professional

Career Risks

Family Builder

Education Risks

Retiree

Health Risks

No two protection journeys are identical.

---

23. Future Expansion

The Risk & Protection Engine supports:

Climate Risk

Cyber Fraud

Identity Theft

Long-Term Care

Pandemic Models

Country-Specific Risk Libraries

Advanced Estate Planning

Business Succession

Future Financial Worlds will customize risks without changing engine architecture.

---

24. Cursor Implementation Rules

Mandatory implementation requirements:

- Risk events generated independently from player decisions.
- Protection reduces financial impact, never eliminates the event.
- Protection categories replace insurance product terminology.
- Risk severity configurable through JSON.
- Probability engine data-driven.
- Protection calculations hidden from players.
- Emergency Fund evaluated before debt.
- Cascading consequences supported.
- Advisor reads Protection Readiness dynamically.

- Entire Risk Engine independent from UI.

---

25. Educational Objectives

By the end of a game,

players should naturally understand:

Why Emergency Funds matter.

Why protection should be established before wealth accumulation.

Why risks cannot be predicted.

Why preparation is more effective than reaction.

Why protecting a family differs from protecting assets.

These lessons emerge through gameplay,

not lectures.

---

26. Chapter Summary

The Risk & Protection Engine introduces uncertainty into SPIKE LIFE while reinforcing one of the game's central philosophies:

You cannot control what happens in life. You can control how prepared you are.

Unexpected events continue to occur regardless of planning.

Prepared players recover faster.

Unprepared players sacrifice wealth, delay dreams, accumulate debt, and experience lower financial resilience.

By replacing product-oriented language with financial planning categories, SPIKE LIFE teaches protection as part of an integrated financial strategy rather than as a sales exercise.

Together with the Financial Planning Engine, Goal Engine, and Wealth Engine, the Risk & Protection Engine completes the four foundational pillars of long-term financial well-being.

End of Chapter 10

This chapter establishes the Risk & Protection Engine as the resilience layer of SPIKE LIFE. It ensures that uncertainty remains a constant force throughout the simulation while demonstrating that thoughtful preparation—not luck—is what allows players to achieve lasting financial security and a balanced life.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume II – Financial Engine

Chapter 11 – Advisor Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

The Advisor Engine is the intelligence layer that helps players understand the financial implications of their decisions without taking away their freedom to choose.

Unlike traditional educational software, the Advisor is **not a teacher**.

It is **not a salesperson**.

It is **not an instruction manual**.

It behaves like a trusted professional financial advisor who explains consequences, highlights trade-offs, and encourages better thinking.

The player always remains the decision maker.

---

2. Design Philosophy

The Advisor Engine follows one fundamental principle:

Advise. Never Decide.

Players must never feel that there is only one "correct" answer.

Real financial planning is about making informed trade-offs based on priorities, responsibilities, and risk tolerance.

The Advisor provides clarity—not certainty.

---

3. Educational Philosophy

The Advisor does **not** teach financial concepts directly.

Instead, it asks and answers questions such as:

- What happens if I choose this?
- What am I giving up?
- How will this affect my future?
- What risk am I taking?
- How will this affect my family?

Learning happens naturally through conversation.

---

4. Advisor Responsibilities

The Advisor continuously evaluates:

- Cash Flow
- Living Expenses
- Emergency Fund
- Protection Readiness
- Investment Allocation
- Goal Progress
- Debt
- Financial Health
- Life Score Trends

The Advisor transforms these calculations into plain-language guidance.

---

5. What the Advisor Should Never Do

The Advisor must never:

❌ Force decisions

❌ Reveal future random events

❌ Reveal hidden probabilities

❌ Guarantee success

❌ Perform financial calculations on screen

❌ Sell financial products

❌ Replace player judgment

The Advisor protects player agency.

---

6. Advisor Interaction Modes

The Advisor operates in four modes.

Mode 1

Passive Observation

The Advisor silently monitors the player's financial condition.

No interruptions occur.

Mode 2

Optional Consultation

The player clicks:

💬 **Ask Advisor**

The timer pauses.

The Advisor provides guidance.

The player resumes play.

Mode 3

Automatic Warning

The Advisor automatically intervenes only when major financial risks exist.

Examples:

Emergency Fund critically low

Debt becoming excessive

Retirement severely behind

Protection gaps becoming dangerous

The intervention should remain brief.

Mode 4

Auto Advisor

If the player fails to decide before the timer expires,

the Advisor selects the **Balanced Growth** option.

The player is informed that:

"Time expired. Your Advisor selected a balanced strategy."

---

7. Advisor Consultation Flow

Player clicks:

Ask Advisor

↓

Decision Timer Pauses

↓

Advisor analyzes situation

↓

Advice displayed

↓

Player closes Advisor

↓

Decision Timer resumes

Maximum consultation:

20 seconds

---

8. Advisor Communication Style

The Advisor should communicate like an experienced financial planner.

Characteristics:

Professional

Friendly

Encouraging

Objective

Respectful

Never judgmental

Never uses fear

Never pressures the player

---

9. Advisor Language

Advice should explain consequences rather than recommendations.

Example

Instead of:

"Buy Health Protection."

Use:

"Choosing Health Protection may delay your home purchase by approximately one year, but it significantly reduces the financial impact of future medical emergencies."

The player decides.

---

10. Advisor Categories

The Advisor provides guidance in six areas.

Cash Flow

Income

Expenses

Living Costs

Goals

Home

Travel

Education

Business

Retirement

Protection

Family Protection Plan

Income Protection Plan

Health Protection Plan

Education Protection Plan

Retirement Security Plan

Investments

Diversification

Growth

Risk

Liquidity

Debt

Borrowing

Repayment

Debt Burden

Financial Health

Overall financial resilience

Life Score trends

---

11. Advisor Timing

The Advisor should not appear every Planning Cycle.

Recommended frequency:

Optional

Always available

Automatic Advice

Only when necessary

Major Warnings

Rare

This prevents players from ignoring repeated messages.

---

12. Context Awareness

Advice changes according to player circumstances.

Example

Age 24

Advice focuses on:

Career

Emergency Fund

Skill Development

Savings

Age 35

Advice focuses on:

Housing

Children

Education

Protection

Age 55

Advice focuses on:

Retirement

Healthcare

Legacy

Income Stability

The Advisor evolves naturally throughout life.

---

13. Goal Awareness

The Advisor continuously evaluates Dream Board progress.

Examples

"You are ahead of schedule for your retirement goal."

or

"Inflation is increasing the future cost of your housing goal."

Advice always relates back to player aspirations.

---

14. Financial Health Awareness

The Advisor continuously evaluates:

Emergency Fund

Debt

Protection

Goal Funding

Cash Flow

Investment Balance

Business Exposure

Advice should summarize overall financial resilience.

---

15. Opportunity Cost Guidance

One of the Advisor's most important responsibilities.

Example

Player wants:

Luxury Motorcycle

Advisor

"This purchase is affordable today, but it may postpone your Emergency Fund by approximately one year."

The player clearly understands the trade-off.

---

16. Positive Reinforcement

The Advisor should celebrate good decisions.

Examples

"Excellent discipline."

"Your Emergency Fund is now fully funded."

"Your retirement planning is ahead of schedule."

Positive reinforcement encourages continued good behavior.

---

17. Recovery Guidance

Players inevitably make mistakes.

The Advisor helps them recover.

Example

High Debt

Advisor

"You've accumulated significant debt. Prioritizing repayment over discretionary spending for the next few Planning Cycles could improve your Financial Health."

The tone remains constructive.

---

18. Advisor Memory

The Advisor remembers previous decisions.

Examples

Repeated luxury purchases

↓

Advice shifts toward spending discipline.

Repeated conservative choices

↓

Advisor may encourage measured investment growth.

Business success

↓

Advisor discusses diversification.

The Advisor should feel like a long-term coach.

---

19. Advisor Intelligence

The Advisor combines information from:

Financial Planning Engine

Goal Engine

Wealth Engine

Risk & Protection Engine

Life Score Engine

Player History

Current Situation

This creates highly personalized advice.

---

20. Auto Advisor Decision Rules

When the timer expires,

the Advisor chooses according to this priority:

1.

Prevent Financial Collapse

↓

2.

Maintain Positive Cash Flow

↓

3.

Protect Existing Goals

↓

4.

Strengthen Financial Health

↓

5.

Support Long-Term Growth

The Auto Advisor never selects high-risk strategies by default.

---

21. Advisor Insights at Year-End

After every year,

the Advisor summarizes:

Strengths

Weaknesses

Emerging Risks

Progress Toward Dreams

One Recommended Focus

Example

"This year you strengthened your Emergency Fund, but your retirement savings are beginning to fall behind inflation."

---

22. Multiplayer Advisor

The Advisor evaluates every player independently.

Advice is private.

Other players cannot view another player's financial advice.

Players remain free to discuss decisions voluntarily.

---

23. Advisor Reputation (Future Expansion)

Future versions may introduce:

Advisor Reputation

Players earn points by giving good financial advice to others.

Poor advice reduces reputation.

This supports financial advisor training without changing gameplay.

---

24. AI Compatibility

Future AI models may replace rule-based advice.

Requirements:

Never reveal hidden probabilities.

Never reveal future events.

Never override player decisions.

Never produce inconsistent financial advice.

All AI guidance must remain grounded in the Financial Planning Engine.

---

25. Accessibility

Advice should be:

Under 120 words

Plain English

Free of technical jargon

Understandable by first-time players

Professional but conversational

The Advisor should educate through clarity.

---

26. Cursor Implementation Rules

Mandatory implementation requirements:

- Advisor Engine independent from UI.
- Advice generated from current financial state.
- Timer pauses during consultation.
- Maximum consultation duration configurable.
- Auto Advisor selects Balanced Growth strategy.
- Advice templates stored separately from financial logic.
- Rule-based engine replaceable with AI in future.
- No financial formulas displayed.
- Advice personalized using player history.
- Advisor output deterministic when using the same player state.

---

27. Future Expansion

The Advisor Engine is designed to support:

AI Voice Advisor

Regional Languages

Voice Conversations

Career Coaching

Business Mentoring

Retirement Coaching

Estate Planning

University Mode

Professional Financial Advisor Certification Mode

These additions extend the delivery of advice without changing the underlying philosophy.

---

28. Chapter Summary

The Advisor Engine is the human face of SPIKE LIFE.

While the Financial Planning Engine performs sophisticated calculations behind the scenes, the Advisor transforms those calculations into understandable guidance that helps players become better decision makers.

The Advisor never tells players what to do.

Instead, it helps them understand the consequences of what they are about to do.

This preserves player agency while reinforcing the educational mission of SPIKE LIFE.

Ultimately, the Advisor is not measured by how often players follow its suggestions.

It is measured by whether players gradually begin making wiser decisions on their own.

That is the true objective of financial education.

End of Chapter 11

This chapter establishes the Advisor Engine as the interpretive layer between the Financial Planning Engine and the player. Together with the Goal Engine, Wealth Engine, and Risk & Protection Engine, it completes the core Financial Engine of SPIKE LIFE, ensuring that sophisticated financial simulations remain approachable, understandable, and engaging without sacrificing player freedom or educational value.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume III – Game Systems

Chapter 12 – Multiplayer Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

The Multiplayer Engine transforms SPIKE LIFE from a single-player financial simulator into a highly engaging social experience.

Unlike traditional board games where players wait for individual turns, SPIKE LIFE is designed around **simultaneous participation**.

Every player remains engaged throughout the entire session.

The Multiplayer Engine answers one question:

"How do multiple people experience life together while each living a unique financial journey?"

---

2. Design Philosophy

SPIKE LIFE is designed around **shared experiences, individual decisions**.

Everyone experiences time together.

Everyone receives different life situations.

Everyone decides simultaneously.

Everyone watches the consequences unfold together.

The objective is to eliminate downtime while maximizing discussion, laughter, coaching, and learning.

---

3. Supported Players

MVP

Minimum

2 Players

Maximum

6 Players

Recommended

4–6 Players

The system architecture must support expansion to larger facilitated classroom modes in future versions.

---

4. Multiplayer Principles

The Multiplayer Engine follows five principles.

Principle 1

No Waiting

Players should never spend long periods watching someone else play.

Principle 2

Shared Time

Everyone experiences the same Planning Cycle.

Principle 3

Independent Lives

Each player receives different situations.

No player shares identical gameplay.

Principle 4

Shared Learning

Players naturally discuss decisions and outcomes.

Learning occurs socially.

Principle 5

Fast Pace

The entire game should finish within

45–60 minutes.

---

5. Multiplayer Game Flow

Create Game

↓

Host Configures Settings

↓

Players Join Lobby

↓

Ready Check

↓

Character Creation

↓

Dream Board

↓

Simulation Starts

↓

20 Shared Planning Cycles

↓

Game Ends

↓

Life Summary

↓

Winner

---

6. Game Lobby

The lobby allows players to prepare before gameplay.

Each player sees:

Avatar

Player Name

Ready Status

Career

Connection Status

Dream Board Completion

Host Controls

Start Game

Kick Player

Difficulty

Decision Timer

Campaign Length

---

7. Host Responsibilities

The Host controls:

Start Game

Pause Game

Resume Game

Restart Game

Difficulty

Decision Timer

Campaign Length

Future options:

Spectator Mode

AI Players

Tournament Mode

---

8. Planning Cycle Synchronization

Every Planning Cycle begins simultaneously.

The sequence is:

Planning Cycle Begins

↓

All Boards Animate

↓

Every Player Receives

Life Domain

↓

Every Player Receives

Situation

↓

Decision Timer Starts

↓

Players Decide

↓

System Waits

↓

Financial Engine Computes

↓

Results Revealed

↓

Next Planning Cycle

No player advances independently.

---

9. Individual Situations

Every player receives:

A different Life Domain.

A different Situation.

Example

Player 1

Career

Promotion

Player 2

Housing

Typhoon Damage

Player 3

Business

Major Investor

Player 4

Health

Hospitalization

Player 5

Lifestyle

Luxury Motorcycle

Player 6

Family

Parents Need Financial Help

Each player's story remains unique.

---

10. Decision Timer

All players receive the timer simultaneously.

Default

15 seconds

Options

No Timer

20 Seconds

15 Seconds

10 Seconds

5 Seconds

The countdown is synchronized across all players.

---

11. Waiting State

When a player finishes early,

they enter:

Decision Locked

The interface changes to:

✓ Decision Submitted

Waiting for other players...

During this period,

players may observe the board,

but cannot change decisions.

---

12. Auto Advisor Timeout

If a player fails to decide before time expires,

the Auto Advisor immediately selects the Balanced option.

Gameplay never pauses because one player failed to respond.

---

13. Simultaneous Resolution

When all players are ready,

or timers expire,

the Financial Engine processes every player simultaneously.

Only after all calculations finish

are results displayed.

This ensures fairness.

---

14. Consequence Reveal

Results appear simultaneously.

Example

Player 1

Promotion

Salary +₱15,000

____

Player 2

Typhoon Damage

Repair Cost

₱120,000

____

Player 3

Business Expanded

Revenue +20%

Player 4

Hospitalization

Health Protection Activated

Player 5

Bought Motorcycle

Emergency Fund Delayed

Player 6

Supported Parents

Travel Goal Delayed

Players naturally react together.

---

15. Social Learning

Discussion is encouraged.

Examples

"I would have chosen differently."

"You should build your Emergency Fund first."

"Good thing you bought Health Protection."

The game intentionally encourages conversation.

---

16. Privacy Rules

Each player owns private financial information.

Visible to others:

Avatar

Name

Age

Life Score

Achievements

Major Life Events

Hidden:

Cash

Net Worth

Debt

Protection Details

Emergency Fund

Goal Funding

Detailed Advisor Messages

Players may voluntarily share information.

---

17. Advisor Privacy

Advisor consultations remain private.

Other players cannot see:

Advice

Recommendations

Protection Gaps

Financial Health

Advisor messages.

---

18. Annual Review

At the end of every year,

all players simultaneously receive:

Age

Life Score

Goal Progress

Financial Health

Achievements

Major Milestones

The interface then proceeds together into the next year.

---

19. Pause System

Only the Host may pause gameplay.

Reasons:

Discussion

Break

Technical Issue

Facilitator Instruction

Paused games freeze:

Timers

Animations

Financial Processing

---

20. Reconnection

Future online versions must support:

Temporary disconnect

Reconnect

State recovery

No gameplay progress should be lost.

For MVP,

single-device local multiplayer is sufficient.

---

21. Local Multiplayer

The recommended MVP supports:

One shared display.

Each player uses:

Mouse

Touch

or

Dedicated player panel.

Future versions may support:

Individual mobile devices.

---

22. Classroom Mode (Future)

Supports:

Instructor Dashboard

20–40 Participants

Multiple Tables

Leaderboard

Facilitator Controls

Discussion Prompts

This mode extends the Multiplayer Engine without changing gameplay.

---

23. Cooperative Mode (Future)

Optional game mode.

Players cooperate toward:

Community Wealth

Disaster Recovery

Family Legacy

Business Partnership

Not part of MVP.

---

24. Competitive Mode

The default mode.

Each player pursues:

Individual Dreams

Individual Goals

Individual Life Score

Players compete while learning from one another.

---

25. Spectator Mode (Future)

Allows observers to:

Watch gameplay

Review decisions

Replay simulations

Educational institutions may use this for demonstrations.

---

26. Multiplayer Performance Requirements

Planning Cycle synchronization:

Maximum waiting time

<1 second

Financial calculations

<500 ms

Animation

2–3 seconds

Target frame rate

60 FPS

Gameplay must remain smooth with six players.

---

27. Multiplayer State Machine

Lobby

↓

Ready

↓

Character Creation

↓

Dream Board

↓

Planning Cycle

↓

Decision

↓

Locked

↓

Processing

↓

Results

↓

Annual Review

↓

Next Cycle

↓

End Game

No player may skip states.

---

28. Fairness Rules

The Multiplayer Engine guarantees:

Identical timer lengths.

Independent random situations.

Equal Financial Engine processing.

Synchronized consequence reveal.

No player advantage from device speed.

Deterministic processing using the same random seed.

---

29. Cursor Implementation Rules

Mandatory implementation requirements:

- Support 2–6 players.
- Planning Cycles synchronized.
- One shared simulation clock.
- Independent Life Domains.
- Independent Situations.
- Independent Financial States.
- Simultaneous Financial Engine processing.
- Results revealed simultaneously.
- Auto Advisor handles timeouts.
- Multiplayer state machine implemented separately from UI.
- Future online multiplayer must not require redesign of the engine.

---

30. Future Expansion

The Multiplayer Engine is designed to support:

Online Multiplayer

Cross-platform Play

Mobile Companion App

Cloud Save

Voice Chat

Team Play

Tournament Mode

University Classroom Mode

Corporate Training Mode

Financial Advisor Certification Mode

Global Leaderboards

These features should extend the existing architecture rather than replace it.

---

31. Chapter Summary

The Multiplayer Engine is what transforms SPIKE LIFE into a shared life experience rather than a solitary simulation.

Players live through the same years together, but each follows a unique financial journey shaped by different opportunities, setbacks, and decisions.

The simultaneous Planning Cycle eliminates downtime, encourages conversation, and creates memorable moments when everyone's outcomes are revealed together.

This system supports the central educational philosophy of SPIKE LIFE:

People learn financial planning best not only by making their own decisions, but also by observing, discussing, and reflecting on the decisions of others.

The Multiplayer Engine therefore serves not only as a technical synchronization layer, but also as the foundation for collaborative learning, replayability, and long-term engagement.

End of Chapter 12

This chapter establishes the Multiplayer Engine as the synchronization and social interaction layer of SPIKE LIFE. Together with the Gameplay Foundation and Financial Engine, it completes the core architecture required for the MVP and provides a scalable foundation for future online multiplayer, classroom deployments, and cooperative game modes.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume III – Game Systems

Chapter 13 – Life Score Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

The Life Score Engine is the master evaluation system of SPIKE LIFE.

It determines how well a player has managed their financial life over the course of the simulation.

Unlike traditional financial games where the richest player automatically wins, SPIKE LIFE rewards **balanced financial decision-making**.

The Life Score answers one question:

"How successfully did you build a financially resilient and meaningful life?"

---

2. Design Philosophy

Money is **not** the objective.

Money is one of many resources used to achieve a fulfilling and financially secure life.

A player with:

- Lower Net Worth
- Better Protection
- Completed Goals
- Strong Financial Discipline

may outperform

a player with

- Higher Net Worth

- High Debt
- No Emergency Fund
- Poor Protection
- Missed Life Goals.

The Life Score rewards balance over excess.

---

3. Educational Philosophy

The scoring system should reinforce proper financial behavior.

Players should naturally discover that:

- Saving consistently is better than saving occasionally.
- Protection matters before emergencies happen.
- Diversification reduces long-term risk.
- Debt should be managed carefully.
- Wealth should support life goals.
- Financial resilience matters more than financial appearance.

The score teaches through reinforcement.

---

4. Winning the Game

The winner is the player with the highest **Life Score**.

Not necessarily the player with:

- Highest salary
- Biggest business
- Largest investment portfolio
- Most expensive house

SPIKE LIFE rewards sustainable financial success.

---

5. Score Architecture

Life Score is composed of multiple pillars.

Each pillar represents one dimension of financial well-being.

Life Score

=

Financial Health

+

Goal Achievement

+

Protection

+

Wealth

+

Life Stability

+

Decision Quality

+

Growth

-

Penalties

Each component is calculated independently.

---

6. Score Range

Recommended MVP Scale

Minimum

0

Maximum

1000

Typical Results

Score	Interpretation
900–1000	Exceptional Financial Life
800–899	Excellent
700–799	Strong
600–699	Stable
500–599	Vulnerable

Score	Interpretation
Below 500	Financially At Risk

The exact numerical value is less important than the accompanying explanation.

---

7. Financial Health Component

Approximate Weight

25%

Factors include

Positive Cash Flow

Emergency Fund

Debt Management

Protection

Income Stability

Expense Control

Financial Capacity

This is the single largest scoring component.

---

8. Goal Achievement Component

Approximate Weight

20%

The engine evaluates

Home Goal

Travel Goal

Business Goal

Education Goal

Retirement Goal

Emergency Fund Goal

Scoring rewards

Early completion

On-time completion

Partial completion

Delayed completion

Missed goals receive reduced points but are never punitive.

---

9. Protection Component

Approximate Weight

20%

Measures

Family Protection

Income Protection

Health Protection

Education Protection

Retirement Security

Protection Readiness

Risk Exposure

Protection before crises is rewarded more than reacting afterward.

---

10. Wealth Component

Approximate Weight

15%

Factors

Net Worth

Investment Growth

Property Ownership

Business Value

Savings Growth

Diversification

Wealth contributes,

but never dominates.

---

11. Life Stability Component

Approximate Weight

10%

Measures

Income Stability

Low Financial Stress

Emergency Preparedness

Controlled Debt

Stable Housing

Financial Resilience

Stable lives receive higher scores.

---

12. Decision Quality Component

Approximate Weight

5%

The engine evaluates long-term decision patterns.

Examples

Repeated impulsive purchases

↓

Lower score

Consistent long-term planning

↓

Higher score

Balanced decision-making is rewarded.

---

13. Personal Growth Component

Approximate Weight

5%

Examples

Career Progression

Education

Business Development

Professional Certifications

Improved Financial Habits

The game recognizes continuous improvement.

---

14. Penalty Engine

The score may decrease due to:

Persistent Negative Cash Flow

High Debt

No Emergency Fund

Repeated High-Risk Decisions

Protection Gaps

Missed Major Goals

Business Collapse

Asset Concentration

Penalties should educate,

not punish.

---

15. Dynamic Scoring

Scores update every Planning Cycle.

Players receive immediate feedback.

Example

Life Score

742

↑ +12

The increase is shown,

but the mathematical formula remains hidden.

---

16. Milestone Bonuses

Major achievements grant bonus points.

Examples

Emergency Fund Completed

+15

Debt-Free

+20

House Purchased

+25

Business Started

+20

Business Successfully Expanded

+25

Retirement Fully Funded

+35

Bonuses celebrate positive financial behavior.

---

17. Opportunity Cost Recognition

The engine rewards good trade-offs.

Example

Player delays travel

↓

Funds Emergency Fund

↓

Higher Life Score

Choosing long-term security over short-term gratification is rewarded.

---

18. Risk Awareness

The score recognizes preparation.

Example

Hospitalization occurs.

Prepared player

↓

Minimal score reduction.

Unprepared player

↓

Large reduction.

The lesson:

Preparation matters.

---

19. Recovery Recognition

The game rewards recovery,

not perfection.

Example

Player accumulates debt.

Later

Repays debt.

Rebuilds Emergency Fund.

Improves Financial Health.

Life Score gradually recovers.

Mistakes are part of learning.

---

20. End-of-Year Evaluation

Every year,

players receive a summary.

Example

Excellent

Emergency Fund

↑

Needs Attention

Retirement

↓

Improving

Debt

↓

Outstanding

Goal Progress

↑

The summary explains why the score changed.

---

21. End-of-Game Evaluation

At the conclusion of ten years,

the player receives a comprehensive Life Report.

Includes

Final Life Score

Financial Health

Net Worth

Goals Achieved

Protection Readiness

Debt Status

Investment Growth

Major Milestones

Advisor Summary

Life Story Timeline

The report should feel like a graduation ceremony rather than a report card.

---

22. Leaderboard

Multiplayer leaderboard displays

Player

Life Score

Achievements

Major Milestones

Not displayed

Cash

Debt

Net Worth

Detailed finances

Privacy remains protected.

---

23. Advisor Integration

The Advisor continuously interprets Life Score.

Examples

"Your financial resilience is improving."

"You've accumulated wealth, but your protection remains below recommended levels."

"Your long-term planning has become more balanced."

Advice explains trends,

not numbers.

---

24. Anti-Exploitation Rules

Players should not maximize one metric to win.

Examples

Saving everything

↓

Travel never completed

↓

Lower Goal Score

Investing everything

↓

Emergency Fund ignored

↓

Protection weak

↓

Lower Financial Health

The engine encourages balance.

---

25. Replayability

Different strategies produce different scores.

Examples

Conservative Saver

Balanced Planner

Entrepreneur

Investor

Family Builder

Every strategy can succeed,

provided it remains financially balanced.

---

26. Score Transparency

Players should understand

why

their score changes,

but never the exact mathematical formula.

This prevents optimization through gaming the algorithm.

Players should optimize their financial behavior,

not the scoring system.

---

27. Life Story Recognition

The engine generates narrative achievements.

Examples

Financial Comeback

Resilient Planner

Family First

Disciplined Investor

Debt-Free Achiever

Community Builder

These achievements personalize every playthrough.

---

28. Future Expansion

Future versions may include

Community Impact

Environmental Responsibility

Legacy Planning

Estate Planning

Generational Wealth

Business Employment Creation

Philanthropy

Social Entrepreneurship

These dimensions can extend Life Score without changing the architecture.

---

29. Cursor Implementation Rules

Mandatory implementation requirements

- Life Score calculated every Planning Cycle.
- Weightings configurable.
- Scoring independent from UI.
- Formula hidden from players.
- Life Report generated automatically.
- Milestone achievements event-driven.
- Score history persisted.
- Leaderboard privacy enforced.
- Life Story generated from player history.
- Entire scoring engine deterministic using the same simulation state.

---

30. Example Life Score Calculation

Illustrative only.

Financial Health 218

Goal Achievement 172

Protection 181

Wealth 134

Life Stability 81

Decision Quality 47

Growth 43

Penalties -18

____

TOTAL LIFE SCORE

858

The player sees the categories,

not the hidden internal calculations.

---

31. Chapter Summary

The Life Score Engine is the philosophical heart of SPIKE LIFE.

It ensures that players are rewarded not for accumulating the greatest amount of money, but for making thoughtful, disciplined, and balanced financial decisions over the course of their simulated lives.

The engine reinforces every major lesson taught by the game:

- Build resilience before pursuing wealth.
- Protect what matters most.
- Balance today's enjoyment with tomorrow's security.
- Pursue dreams responsibly.
- Recover from setbacks.

- Learn continuously.
- Make decisions that create long-term stability.

By evaluating the entire financial journey rather than isolated financial outcomes, the Life Score Engine transforms SPIKE LIFE from a wealth-building game into a life-building game.

It is the final measure of how successfully a player has navigated the opportunities, risks, sacrifices, and rewards of life.

End of Chapter 13

This chapter establishes the Life Score Engine as the master evaluation framework of SPIKE LIFE. It integrates outputs from the Financial Planning Engine, Goal Engine, Wealth Engine, Risk & Protection Engine, and Advisor Engine into a single holistic measure of financial well-being. It ensures that the game's ultimate objective remains faithful to its central philosophy: **the best financial life is not the wealthiest one, but the one built through wise decisions, resilience, balance, and purpose.**

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume III – Game Systems

Chapter 14 – Difficulty Engine

Document Version: 1.0

Status: Approved Foundation

Priority: High

---

1. Purpose

The Difficulty Engine controls how challenging the simulation feels without changing the educational objectives of SPIKE LIFE.

Difficulty should never make the game unfair.

Instead, it adjusts:

- Decision pressure
- Financial uncertainty
- Frequency of unexpected events
- Amount of advisor assistance
- Recovery opportunities

Every difficulty level teaches the same financial principles while providing different player experiences.

---

2. Design Philosophy

Difficulty should change **how players experience the game**, not **what they learn**.

Whether a player chooses Beginner or Expert mode, the correct financial principles remain unchanged.

Difficulty modifies:

- Pace
- Complexity
- Forgiveness
- Uncertainty

It does **not** change the core Financial Engine.

---

3. Design Goals

The Difficulty Engine must satisfy five objectives.

Accessibility

New players should never feel overwhelmed.

Replayability

Experienced players should continue finding the game challenging.

Fairness

Randomness should never feel unfair.

Educational Value

Financial lessons must remain consistent.

Scalability

Future Financial Worlds should inherit the same difficulty framework.

---

4. Difficulty Levels

The MVP includes four difficulty levels.

Beginner

Recommended for:

Students

Families

First-time players

Characteristics

- No decision timer
- Frequent Advisor guidance
- Fewer severe crises
- Higher recovery opportunities
- Lower investment volatility
- More forgiving economy

Educational focus:

Understanding financial fundamentals.

Standard

(Default)

Recommended for:

General players

Universities

SPIKE workshops

Characteristics

- 15-second decision timer
- Balanced event distribution
- Normal inflation
- Standard business volatility
- Moderate Advisor intervention

Educational focus:

Balanced financial planning.

Advanced

Recommended for:

Financial professionals

Advisors

Experienced players

Characteristics

- 10-second timer
- Increased uncertainty
- Larger market fluctuations

- More realistic business risk
- Reduced Advisor intervention

Educational focus:

Managing complex financial trade-offs.

Expert

Recommended for:

Financial planning enthusiasts

Competitive players

Certification programs

Characteristics

- 5-second timer
- Minimal Advisor support
- High economic volatility
- More cascading risks
- Smaller recovery windows

Educational focus:

Long-term resilience under pressure.

---

5. Difficulty Matrix

System	Beginner	Standard	Advanced	Expert
Decision Timer	None	15 sec	10 sec	5 sec
Advisor Help	Very High	Normal	Limited	Minimal
Economic Volatility	Low	Medium	High	Very High
Business Risk	Low	Medium	High	Extreme

System	Beginner	Standard	Advanced	Expert
Recovery Events	Frequent	Normal	Limited	Rare
Event Complexity	Simple	Moderate	High	Highest

---

6. Decision Timer

The timer is the most visible difficulty setting.

Beginner

No Timer

Standard

15 seconds

Advanced

10 seconds

Expert

5 seconds

Players should feel increasing urgency,

not frustration.

---

7. Advisor Assistance

Difficulty controls Advisor involvement.

Beginner

Advisor proactively offers suggestions.

Explains basic concepts.

Highlights trade-offs.

Standard

Advisor available on request.

Provides balanced guidance.

Advanced

Advisor responds only when asked.

Less detailed explanations.

Expert

Advisor provides concise strategic observations.

Player expected to understand implications.

---

8. Economic Volatility

Difficulty influences macroeconomic variation.

Examples

Inflation

Business Growth

Investment Returns

Market Downturns

Employment Stability

Expert mode introduces wider fluctuations.

---

9. Random Event Frequency

Difficulty affects event intensity,

not randomness.

Examples

Beginner

Minor Medical Expense

Expert

Major Hospitalization

Beginner

Small Business Opportunity

Expert

High-Risk Venture Capital Opportunity

The same event categories exist,

but outcomes become more consequential.

---

10. Recovery Opportunities

Financial recovery is easier on lower difficulties.

Examples

Unexpected Bonus

Family Assistance

Government Aid

Business Recovery

Beginner players receive more opportunities to recover.

Expert players must rely on stronger planning.

---

11. Business Difficulty

Business risk scales with difficulty.

Beginner

Stable Growth

Standard

Balanced Market

Advanced

Competitive Market

Expert

Highly Volatile Market

Business should remain rewarding,

but never guaranteed.

---

12. Inflation

Default inflation remains:

4%

However,

Advanced and Expert modes introduce temporary inflation shocks.

Examples

6%

8%

10%

These are event-driven rather than permanent.

---

13. Protection Importance

Protection becomes increasingly valuable on higher difficulties.

Examples

Beginner

Medical Event

↓

Small Financial Impact

Expert

Medical Event

↓

Large Financial Impact

↓

Health Protection dramatically reduces losses.

The lesson becomes more apparent.

---

14. Decision Complexity

Difficulty does **not** increase the number of choices.

Every Planning Cycle still offers:

Exactly three decisions.

Complexity increases through consequences,

not interface design.

---

15. Financial Transparency

Beginner mode displays additional explanations.

Examples

"Emergency Fund helps pay unexpected expenses."

Advanced mode assumes prior knowledge.

The interface remains identical.

Only explanations differ.

---

16. Advisor Education

Beginner

Advisor teaches.

Example

"Saving regularly helps prepare for emergencies."

Expert

Advisor coaches.

Example

"Your liquidity remains below recommended levels."

---

17. Multiplayer Difficulty

The Host selects one difficulty for the entire table.

All players use identical settings.

This maintains fairness.

Future versions may support:

Individual adaptive difficulty.

---

18. Adaptive Difficulty (Future)

Future versions may automatically adjust difficulty based on:

Financial Literacy

Player Performance

Previous Games

Learning Progress

Certification Level

The adjustment should remain invisible.

---

19. Accessibility Mode

Separate from difficulty.

Accessibility options include:

Colorblind Mode

Large Text

Reduced Motion

Audio Assistance

Screen Reader Support

Longer Decision Timer

Accessibility is not considered an easier difficulty.

---

20. Practice Mode

A special learning mode.

Features

Unlimited Advisor

No Timer

Pause Anytime

Replay Decisions

Undo (optional)

Educational explanations

Not scored.

Ideal for onboarding.

---

21. Competitive Mode

Designed for tournaments.

Fixed settings.

Standard timer.

No pauses.

No practice tools.

No undo.

Identical random seed generation rules.

Maximum fairness.

---

22. Difficulty Independence

Difficulty never changes:

Financial formulas

Goal Engine

Protection calculations

Life Score architecture

Only the player's experience changes.

The underlying financial model remains identical.

---

23. Performance Requirements

Changing difficulty should never affect:

Simulation speed

Frame rate

Network synchronization

Financial accuracy

Difficulty parameters are loaded during game initialization.

---

24. Cursor Implementation Rules

Mandatory implementation requirements:

- Difficulty configuration loaded at game start.
- Difficulty stored as configuration rather than hard-coded logic.
- Financial Engine independent from difficulty.
- Decision timer configurable.

- Advisor behavior configurable.
- Event probabilities configurable.
- Recovery probabilities configurable.
- Accessibility settings independent from difficulty.
- Multiplayer uses a shared difficulty profile.
- Future Financial Worlds inherit the same difficulty architecture.

---

25. Future Expansion

The Difficulty Engine supports future modes such as:

Financial Advisor Certification

University Examination Mode

Corporate Wellness Challenge

Family Cooperative Mode

Economic Crisis Mode

Retirement Challenge

Entrepreneur Mode

OFW Challenge

Single Parent Challenge

Each mode extends the framework without changing the core architecture.

---

26. Educational Outcomes

By exposing players to increasing levels of uncertainty,

the Difficulty Engine reinforces:

Financial resilience

Preparedness

Adaptability

Long-term planning

Risk management

Disciplined decision-making

Players should gradually become comfortable making thoughtful financial decisions under pressure.

---

27. Chapter Summary

The Difficulty Engine ensures that SPIKE LIFE remains approachable for beginners while providing meaningful challenges for experienced players.

Rather than changing financial truths, it changes the environment in which those truths are experienced.

This preserves the educational integrity of the simulation while greatly improving replayability.

The engine demonstrates an important principle that mirrors real life:

Financial planning does not become easier because life becomes easier. Good planning becomes more valuable as life becomes more uncertain.

This philosophy allows players of all experience levels to engage with the same core financial concepts while continually discovering new challenges and deeper strategic decisions.

End of Chapter 14

This chapter establishes the Difficulty Engine as the pacing and challenge layer of SPIKE LIFE. It controls the player's experience without compromising the consistency of the Financial Planning Engine, ensuring that every game—whether played by a student, a financial advisor, or a family—remains engaging, educational, fair, and highly replayable.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume III – Game Systems

Chapter 15 – Randomization Engine

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

The Randomization Engine is responsible for creating uncertainty, surprise, replayability, and realism throughout SPIKE LIFE.

Its objective is **not** to create chaos.

Its objective is to simulate life.

Life is unpredictable.

The player cannot predict:

- when opportunities appear,
- when crises occur,
- when markets rise,
- when illness strikes,
- when luck changes.

The Randomization Engine reproduces this uncertainty while remaining statistically fair.

---

2. Design Philosophy

Randomness must never feel unfair.

Instead,

players should consistently feel:

"That could really happen."

rather than

"The game cheated."

Randomness must always be:

- believable,
- explainable,
- balanced,
- educational.

---

3. Guiding Principles

The engine follows six principles.

Principle 1

Random, but Logical.

Events should always make sense.

A 24-year-old fresh graduate should not suddenly receive an inheritance worth ₱100 million.

Principle 2

Random, not Repetitive.

Players should rarely experience the same event repeatedly.

Principle 3

Random, but Recoverable.

Bad luck should never permanently eliminate a player from competition.

Principle 4

Random, but Weighted.

Events must consider:

Age

Career

Goals

Income

Life Stage

Business Ownership

Protection

Family Status

Principle 5

Random, but Educational.

Every event should teach a financial lesson.

Principle 6

Stories over Statistics.

The engine exists to create memorable life stories,

not mathematical randomness.

---

4. Sources of Randomness

The Randomization Engine controls:

Life Domains

Situations

Business Outcomes

Investment Returns

Economic Events

Natural Disasters

Health Events

Career Events

Lifestyle Temptations

Family Events

Opportunity Events

Advisor Story Variations

Achievement Unlocks

---

5. Weighted Random Selection

SPIKE LIFE never uses pure random selection.

Instead,

every event has a dynamic probability.

Example

Career

Age 23

40%

Age 55

5%

Retirement

Age 23

2%

Age 58

40%

This creates believable life progression.

---

6. Event Pools

Every Life Domain contains four event pools.

Common

Everyday life.

Examples

Salary increase

Family celebration

Small repair

Weekend trip

Uncommon

Important life events.

Examples

Promotion

House purchase

Business loan

Major vacation

Rare

Life-changing events.

Examples

Major inheritance

Business acquisition

Large investment opportunity

Natural disaster

Legendary

Extremely rare.

Examples

National business award

Unexpected major fortune

Successful company exit

These become memorable moments.

---

7. Probability Layers

Every Planning Cycle evaluates multiple probability layers.

Layer 1

Life Stage

↓

Layer 2

Current Financial Status

↓

Layer 3

Previous Decisions

↓

Layer 4

Economic Environment

↓

Layer 5

Random Variation

↓

Final Situation

This creates highly believable randomness.

---

8. Situation Cooldown

Recently experienced situations become temporarily less likely.

Example

Promotion

↓

Cooldown

↓

Cannot immediately repeat.

Likewise,

Major Hospitalization

↓

Cooldown

↓

Reduced probability.

The cooldown system prevents repetitive gameplay.

---

9. Domain Cooldown

Entire Life Domains also use cooldowns.

Example

Career

↓

Career

↓

Career

↓

Unlikely

The engine encourages variety.

---

10. Opportunity Balancing

The engine tracks recent player experiences.

If several negative events occur,

future probabilities gradually favor:

Recovery

Opportunities

Positive Milestones

Conversely,

long streaks of positive outcomes increase the likelihood of future challenges.

This creates emotional balance.

---

11. Economic Cycles

The simulation contains hidden economic conditions.

Examples

Economic Growth

Stable Economy

Slowdown

Recession

Recovery

Economic conditions influence:

Employment

Business

Investments

Housing

Government

Players experience changing markets naturally.

12. Business Randomization

Businesses evolve through weighted randomness.

Possible outcomes

Business Expansion

Steady Growth

Flat Revenue

Market Competition

Supply Problems

Customer Boom

Technology Shift

Business Failure

The probability depends on:

Industry

Investment

Experience

Economic Cycle

Previous Decisions

13. Investment Randomization

Investment returns fluctuate naturally.

Examples

Savings

Very Stable

Government Bonds

Stable

Balanced Investments

Moderate

Equities

High Volatility

Players learn diversification through experience.

---

14. Natural Disaster Engine

Philippine-specific events include:

Typhoon

Flood

Earthquake

Volcanic Activity

Landslide

These affect:

Housing

Business

Community

Income

Severity varies each game.

---

15. Family Event Engine

Examples

Marriage

Birth of Child

Parents Need Support

Wedding Invitation

Sibling Graduation

Family Medical Emergency

Christmas Expenses

These events become increasingly likely as players age.

---

16. Lifestyle Temptation Engine

Lifestyle opportunities appear randomly.

Examples

Luxury Phone

Motorcycle

Designer Watch

Gaming Console

Car Upgrade

International Travel

The player must continually balance:

Enjoyment

versus

Long-term goals.

---

17. Hidden Seeds

Every new game generates a unique Simulation Seed.

The seed determines:

Economic Cycle

Rare Events

Market Conditions

Opportunity Timing

Disaster Timing

Business Volatility

Using the same seed always produces identical results,

provided player decisions remain the same.

This supports deterministic testing.

---

18. Dynamic Probability

Probabilities evolve continuously.

Example

Player repeatedly ignores Health Protection.

↓

Health event probability gradually increases.

Player builds strong Emergency Fund.

↓

Minor emergencies become less financially disruptive.

The world responds to player behavior.

---

19. Player Independence

Every player receives an independent random stream.

Example

Player 1

Promotion

Player 2

Hospitalization

Player 3

Business Opportunity

Player 4

Flood Damage

Player 5

Wedding

Player 6

Investment Boom

The simulation creates unique life stories.

---

20. Shared World Events

Occasionally,

all players experience the same macro event.

Examples

Economic Recession

Pandemic

Election Year

Interest Rate Increase

Fuel Price Spike

Inflation Shock

Typhoon Season

Each player's financial outcome differs according to their preparedness.

---

21. Replayability Strategy

Replayability is created through:

Simulation Seeds

Weighted Probabilities

Business Outcomes

Different Careers

Different Dream Boards

Economic Cycles

Delayed Consequences

Player Decisions

No two games should feel identical.

---

22. Fairness Rules

Randomness must never:

Target one player repeatedly.

Destroy a player's game permanently.

Prevent recovery.

Create impossible situations.

Every player should always have a realistic path toward success.

---

23. Random Event Memory

The engine remembers:

Recent Events

Major Crises

Business Success

Protection Decisions

Goal Progress

Financial Health

Future probabilities adapt accordingly.

---

24. Randomness Visibility

Players should experience randomness.

They should never see:

Percentages

Probability Tables

Random Seeds

Event Weights

Internal calculations remain hidden.

---

25. Future AI Integration

Future AI may generate dynamic situations.

Requirements:

Must belong to an existing Life Domain.

Must follow financial rules.

Must respect life stage.

Must maintain educational objectives.

Must remain explainable.

AI may generate stories,

not financial logic.

---

26. Cursor Implementation Rules

Mandatory implementation requirements:

- Use weighted random selection throughout.
- All probability tables configurable through JSON.
- Event cooldowns mandatory.
- Domain cooldowns mandatory.
- Simulation Seed generated at game start.

- Seed persisted throughout the game.
- Economic Cycle generated once per game.
- Random Engine isolated from Financial Engine.
- Deterministic replay supported.
- Future AI events plug into existing event pools.

---

27. Performance Requirements

Randomization processing should complete within:

100 milliseconds

per Planning Cycle.

The engine should support:

10,000+

possible situations

without measurable performance degradation.

All probability calculations should occur before the Situation Reveal animation begins.

---

28. Educational Objectives

Players should naturally discover that:

Life is unpredictable.

Preparation matters more than prediction.

Diversification reduces risk.

Protection reduces financial damage.

Emergency Funds improve resilience.

Good decisions increase recovery capacity.

Bad luck can happen to anyone.

Good planning improves outcomes.

---

29. Chapter Summary

The Randomization Engine gives SPIKE LIFE its sense of realism and replayability.

Rather than producing arbitrary surprises, it creates believable life stories shaped by age, goals, financial condition, economic cycles, and previous decisions.

Every Planning Cycle feels fresh because the player cannot predict what life will bring next—but they can influence how well prepared they are to respond.

This reflects one of the central philosophies of SPIKE LIFE:

You cannot control uncertainty. You can only improve your readiness.

The Randomization Engine ensures that no two lives unfold in exactly the same way while preserving fairness, educational value, and long-term engagement.

End of Chapter 15

This chapter establishes the Randomization Engine as the uncertainty layer of SPIKE LIFE. Together with the Planning Cycle Engine, Life Domain Engine, Situation Engine, and Financial Planning Engine, it creates a dynamic simulation where every game tells a different story while remaining realistic, balanced, and faithful to the educational goals of the platform.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume IV – User Experience

Chapter 16 – UX Philosophy & Design System

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

This chapter defines the User Experience philosophy of SPIKE LIFE.

Unlike conventional financial software, SPIKE LIFE should never feel like:

- Banking software
- Accounting software
- Excel
- Insurance quotation software
- Investment dashboard

SPIKE LIFE must feel like a **beautiful, modern strategy game** where players naturally learn financial planning through interaction.

Every UI decision must support this philosophy.

---

2. UX Vision

The player should feel like they are playing

Civilization meets Monopoly meets The Sims meets a modern digital board game

—not using financial software.

The interface should create:

- Curiosity
- Excitement
- Simplicity
- Confidence
- Delight

The UI must never intimidate new users.

---

3. Design Philosophy

The UX follows five principles.

Principle 1

Life First

The interface should present

Life

before

Money.

Players should first notice

their dreams,

their journey,

their decisions,

their achievements.

Financial numbers are secondary.

Principle 2

One Decision at a Time

Never overwhelm players.

Only one meaningful decision should occupy the player's attention.

Everything else becomes supporting information.

Principle 3

Immediate Understanding

Every screen should answer

"What should I do now?"

within

3 seconds.

If a new player needs explanation,

the interface has failed.

Principle 4

Emotion Before Data

Players should remember

stories,

not spreadsheets.

Every Planning Cycle should feel like

another chapter of their life.

Principle 5

Beautiful Simplicity

Minimal text.

Large typography.

Generous spacing.

Strong visual hierarchy.

No clutter.

---

4. Overall UX Architecture

SPIKE LIFE uses

One Workspace

instead of multiple application screens.

The player never feels like they are navigating software.

Instead,

they remain inside one living game board.

---

5. Five Lenses Architecture

The interface is built around one workspace viewed through five perspectives.

LIFE

Overall life journey

Age

Career

Family

Timeline

Achievements

Planning Cycle

PLAN

Dream Board

Goals

Funding Progress

Advisor

Life Score

PROTECT

Protection Readiness

Emergency Fund

Risk Exposure

Financial Health

GROW

Income

Savings

Investments

Business

Net Worth

Cash Flow

JOURNEY

Life Story

Major Decisions

Achievements

Timeline

Replay

Reflection

Players never "change screens."

They simply change perspective.

---

6. Workspace Philosophy

The center of the screen always belongs to gameplay.

Supporting information surrounds the experience.

The interface should never resemble dashboards with dozens of widgets.

---

7. Screen Hierarchy

Priority

1

Current Situation

Priority

2

Decision Options

Priority

3

Planning Cycle Status

Priority

4

Life Dashboard

Priority

5

Supporting Information

Everything else remains secondary.

---

8. The Living Board

The board never disappears.

Players remain connected to the simulation.

Visible throughout gameplay:

Life Domains

Current Age

Planning Cycle

Player Avatar

Life Score

Financial Health

Goal Progress

The board itself becomes the game's identity.

---

9. Life Domain Grid

The twelve Life Domains occupy the center.

Career

Opportunity

Family

Health

Business

Lifestyle

Housing

Education

Government

Community

Investment

Milestone

Planning Cycles animate this board.

The selected domain expands.

The Situation Card appears.

---

10. Visual Hierarchy

Every screen follows this order.

Largest

Situation

↓

Decision Buttons

↓

Illustration

↓

Timer

↓

Advisor Button

↓

Player Dashboard

↓

Secondary Statistics

Players always know where to look.

---

11. Typography

Typography should prioritize readability.

Recommended

Desktop

Title

40–56 px

Heading

28–36 px

Body

18–22 px

Buttons

18–20 px

Never use text below

16 px.

The interface should remain comfortable on laptops.

---

12. Color Philosophy

Colors communicate meaning.

Green

Growth

Savings

Positive

Blue

Planning

Stability

Information

Gold

Dreams

Goals

Achievements

Orange

Business

Opportunities

Purple

Education

Growth

Red

Risk

Loss

Urgency

Gray

Inactive

Muted

The palette should remain elegant.

Avoid overly saturated colors.

---

13. Motion Philosophy

Animations communicate state.

Never decorate.

Every animation should answer

"What changed?"

Examples

Life Domain Selection

Situation Reveal

Decision Lock

Financial Gain

Goal Completion

Achievement Unlock

Animations should remain

200–500 ms

except Planning Cycle reveals

which may last

2–3 seconds.

---

14. Information Density

The interface should never display:

Large spreadsheets

Financial tables

Long reports

Tiny text

Players should see only

the information required

to make their current decision.

---

15. Dream Board UX

The Dream Board should resemble

a vision board,

not a budgeting worksheet.

Examples

🏠 Dream Home

✈️ Japan Trip

🚀 Coffee Shop

🎓 Child's Education

⛱️ Retirement

Each goal displays:

Illustration

Progress Ring

Target Age

Status

No spreadsheets.

---

16. Decision Screen

The Decision Screen becomes the focus of every Planning Cycle.

Structure

Situation

↓

Illustration

↓

Three Choices

↓

Advisor

↓

Countdown Timer

↓

Player Confirms

Nothing else competes for attention.

---

17. Countdown Timer

The timer should feel exciting.

Circular animation.

Large numbers.

Color transition

Green

↓

Amber

↓

Red

At

3 seconds,

the timer pulses.

At

0,

Auto Advisor activates.

---

18. Advisor Experience

The Advisor opens as

a floating conversation panel.

It should resemble

a trusted financial coach,

not a chatbot.

Tone

Professional

Friendly

Calm

Encouraging

The panel never covers

the entire screen.

---

19. Financial Dashboard

Financial information remains compact.

Visible

Income

Cash

Emergency Fund

Net Worth

Life Score

Financial Health

Hidden

Detailed formulas

Interest

Inflation calculations

Allocation percentages

The player focuses on decisions,

not accounting.

---

20. End-of-Year Review

Should resemble

an awards ceremony.

Displays

Achievements

Dream Progress

Life Score

Advisor Summary

Major Events

New Year's Planning

Duration

10–15 seconds.

Players should feel excited for the next year.

---

21. Multiplayer Experience

Every player should remain engaged.

When waiting,

players see:

Other players

Decision status

Planning Cycle

Board animation

They never stare at a blank screen.

---

22. Emotional Journey

The UX should create this emotional rhythm.

Anticipation

↓

Surprise

↓

Decision

↓

Suspense

↓

Outcome

↓

Reflection

↓

Celebration

↓

Repeat

This rhythm defines the SPIKE LIFE experience.

---

23. Accessibility

Mandatory support includes:

High Contrast Mode

Large Text Mode

Colorblind Safe Palette

Reduced Motion

Keyboard Navigation

Screen Reader Labels

Accessibility should never reduce gameplay quality.

---

24. Responsive Design

Primary Target

Desktop

Laptop

Minimum Width

1366 px

Secondary

Tablet

Portrait

Landscape

Future

Mobile Companion

The MVP is optimized for classroom and workshop use.

---

25. Performance Goals

Screen transitions

<300 ms

Planning Cycle animation

<3 sec

Financial updates

<500 ms

UI

60 FPS

The interface should always feel responsive.

---

26. Cursor Implementation Rules

Mandatory implementation requirements:

- One Workspace architecture.
- Five Lens navigation.
- Life Domain Board always visible.
- Situation always centered.

- Three decision buttons only.
- No spreadsheet-style layouts.
- Typography never below 16 px.
- Responsive for laptops without vertical scrolling.
- Financial dashboard collapsible.
- Animations GPU accelerated.
- Accessibility implemented from MVP.

---

27. UX Anti-Patterns

The following are prohibited.

❌ Tiny text.

❌ Multiple sidebars.

❌ Scroll-heavy interfaces.

❌ Financial tables during gameplay.

❌ Modal overload.

❌ More than one major decision.

❌ More than six visible statistics at once.

❌ Hidden navigation.

❌ Deep menu structures.

---

28. Design Inspiration

The visual language should combine ideas from:

Apple Human Interface Guidelines

Monument Valley

Civilization VI

The Sims

Modern digital board games

Premium fintech applications

The result should feel premium, playful, and timeless.

---

29. Success Metrics

The UX succeeds if:

A first-time player understands the interface within three minutes.

Players rarely ask where to click next.

Most interactions require only one click.

Gameplay remains immersive.

The interface disappears into the experience.

Players remember the story—not the software.

---

30. Chapter Summary

The UX of SPIKE LIFE is intentionally designed to disappear.

Players should never feel that they are operating financial software.

Instead, they should feel immersed in a living world where dreams evolve, life unfolds, opportunities appear, crises emerge, and every Planning Cycle presents one meaningful financial decision.

The interface exists to make those decisions effortless.

Everything else—financial calculations, risk modeling, investment growth, inflation, and scoring—operates invisibly beneath the surface.

The greatest compliment the UX can receive is not:

"The interface is beautiful."

It is:

"I forgot I was learning finance. I was simply living my life."

End of Chapter 16

This chapter establishes the UX philosophy and design system that governs every screen, animation, interaction, and visual element of SPIKE LIFE. Together with the Gameplay Foundation and Financial Engine, it ensures that the product remains approachable, emotionally engaging, visually elegant, and educational without ever feeling like traditional financial software.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume IV – User Experience

Chapter 17 – Screen Specifications & User Interface Architecture

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

This chapter defines every major screen, interface, navigation flow, layout, and interaction within SPIKE LIFE.

Unlike conventional applications,

SPIKE LIFE is designed as **one continuous interactive game world**, not a collection of disconnected screens.

Players should feel they are progressing through life—not navigating software.

---

2. UX Architecture

SPIKE LIFE follows a **One Workspace Architecture**.

There are **no traditional pages** during gameplay.

Instead:

Workspace

↓

Planning Cycle

↓

Situation

↓

Decision

↓

Consequence

↓

Repeat

Everything happens inside the same visual environment.

---

3. Complete Screen Map

Launch Screen

↓

Sign In

↓

Main Menu

↓

Create / Join Game

↓

Lobby

↓

Character Creation

↓

Dream Board

↓

Game Tutorial (Optional)

↓

Gameplay Workspace

↓

Annual Review

↓

Game Summary

↓

Life Report

↓

Leaderboard

No screen should feel disconnected from the previous one.

---

4. Launch Screen

Purpose

Introduce the game.

Contents

- Animated SPIKE LIFE Logo
- Background Animation
- Version Number
- Loading Status

Buttons

- Start
- Settings
- Credits

Design Goal

Simple.

Elegant.

Premium.

Loading time should feel intentional.

---

5. Sign-In Screen

Purpose

Identify the player.

Elements

Player Name

Avatar

Local Profile

Cloud Login (Future)

Guest Login

Remember Me

No unnecessary registration.

Players should reach gameplay quickly.

---

6. Main Menu

Sections

Continue Game

New Game

Load Game

Tutorial

Settings

Achievements

Exit

Background

Animated Financial World.

Subtle motion.

---

7. Create Game Screen

Host configures:

Players

Difficulty

Decision Timer

Campaign Length

Financial World

Private / Public (Future)

Buttons

Create

Cancel

---

8. Join Game Screen

Displays

Available Games

Host Name

Players Connected

Ping (Future)

Join Button

Refresh

Future

QR Code Join

---

9. Multiplayer Lobby

Displays

Player Avatars

Player Names

Ready Status

Career Selection

Dream Board Status

Host Controls

Start Game

Kick Player

Change Settings

The lobby should feel energetic.

---

10. Character Creation

Simple.

Fast.

Less than two minutes.

Player selects:

Avatar

Name

Starting Career

Optional Appearance

Preview

Monthly Income

Career Description

Expected Growth

---

11. Dream Board Screen

One of the most important screens.

Players define:

🏠 Home

✈️ Travel

🚀 Business

🎓 Education

🏖️ Retirement

Emergency Fund

Each dream displayed as:

Large Card

Illustration

Target Age

Priority

Estimated Future Cost

Dreams should feel inspirational,

not financial.

---

12. Tutorial (Optional)

Interactive.

Maximum:

5 minutes.

Explains

Planning Cycle

Life Domains

Decision System

Advisor

Life Score

Players may skip anytime.

---

13. Gameplay Workspace

This is the permanent game screen.

The player remains here throughout gameplay.

Contains:

Life Domain Board

Player Dashboard

Planning Cycle Header

Decision Area

Advisor

Timer

Status Panel

No navigation away from this workspace.

---

14. Workspace Layout

Recommended layout

Planning Cycle Header

Life Domain Board

(Currently Animated)

Situation Card

Decision Buttons

Player Dashboard

____

Simple.

Focused.

---

15. Planning Cycle Header

Displays

Current Year

Current Age

Planning Cycle

Jan–Jun

or

Jul–Dec

Financial World

Difficulty

Player Turn Status

---

16. Life Domain Board

Always visible.

Twelve animated domains.

Inactive

Muted

Selected

Expanded

Active

Highlighted

Animation

Pulse

Glow

Expansion

Duration

2–3 seconds.

---

17. Situation Card

Appears after Life Domain selection.

Contains

Illustration

Situation Title

Narrative

Three Decisions

Advisor Button

Timer

This becomes the visual focus.

---

18. Decision Panel

Exactly three buttons.

Large.

Easy to click.

Example

Secure Future

Balanced Growth

Enjoy Today

Buttons should occupy most of the available width.

No scrolling.

---

19. Advisor Panel

Floating side panel.

Appears on request.

Displays

Advice

Trade-offs

Potential Impacts

Current Financial Health

Advisor closes with one click.

---

20. Countdown Timer

Circular.

Large.

Centered.

Transitions

Green

↓

Amber

↓

Red

At

3 seconds

Timer pulses.

At

0

Auto Advisor activates.

---

21. Player Dashboard

Compact.

Visible throughout gameplay.

Displays

Age

Life Score

Financial Health

Emergency Fund

Goal Progress

Current Income

No detailed financial reports.

---

22. Goal Panel

Displays Dream Board progress.

Example

🏠 House

72%

✈️ Travel

41%

🚀 Business

18%

Progress Rings preferred over bars.

---

23. Consequence Screen

Appears immediately after processing.

Displays

Income Change

Cash Flow

Net Worth

Goal Changes

Life Score

Achievements

Duration

4–6 seconds.

Then automatically returns to gameplay.

---

24. Annual Review Screen

Appears after:

13th Month Pay.

Displays

Year Summary

Dream Progress

Financial Health

Life Score

Major Events

Advisor Summary

Annual Awards

Buttons

Continue

Review Goals

---

25. 13th Month Planning Screen

Special annual event.

Player chooses

one primary allocation.

Large cards.

Examples

Emergency Fund

Business

Travel

House

Retirement

Education

Lifestyle

This screen should feel rewarding.

---

26. Achievement Screen

Triggered instantly.

Examples

Emergency Fund Complete

Debt-Free

Millionaire

Business Owner

Homeowner

Retirement Ready

Animation

Confetti

Sound

Badge

---

27. End Game Screen

Displays

Winner

Final Life Score

Life Story

Major Achievements

Dream Completion

Financial Health

Buttons

Replay

Save

View Report

Exit

---

28. Life Report Screen

A personalized summary.

Includes

Timeline

Dreams Achieved

Career Progression

Business Growth

Net Worth History

Protection Journey

Advisor Summary

Major Decisions

Life Score Breakdown

Export PDF (Future)

---

29. Leaderboard

Displays

Rank

Player

Life Score

Achievements

Major Milestones

Never display

Cash

Debt

Net Worth

Detailed finances

---

30. Settings Screen

Categories

Gameplay

Audio

Graphics

Accessibility

Language

Notifications

Future

Financial World Packs

---

31. Pause Screen

Displays

Resume

Restart

Settings

Quit

Host Controls (Multiplayer)

Pause freezes:

Animations

Timers

Financial Processing

---

32. Responsive Behavior

Desktop

1920×

1080

Primary target.

Laptop

1366×

768

No vertical scrolling.

Tablet

Responsive layout.

Future

Mobile Companion.

---

33. Screen Transition Philosophy

Transitions should feel cinematic.

Examples

Fade

Slide

Expand

Glow

Zoom

Avoid

Hard cuts.

---

34. Cursor Implementation Rules

Mandatory implementation requirements:

- One Workspace architecture.
- No page reloads during gameplay.
- React state controls screen transitions.
- No vertical scrolling during gameplay.
- Situation always centered.

- Three decision buttons only.
- Dashboard always visible.
- Life Domain Board persistent.
- Responsive layout for laptops.
- GPU-accelerated animations.
- Maximum transition time: 300 ms (except Planning Cycle reveal).

---

35. Future Screens

Reserved for future versions.

Financial World Selection

Career Campaign

Advisor Certification

Family Cooperative Mode

University Mode

Corporate Wellness

Marketplace

Replay Theatre

Statistics Dashboard

Replay Analyzer

---

36. Screen Navigation Diagram

Launch

↓

Sign In

↓

Main Menu

↓

Lobby

↓

Character Creation

↓

Dream Board

↓

Tutorial

↓

Gameplay Workspace

↓

Annual Review

↓

Gameplay Workspace

↓

Annual Review

↓

...

↓

End Game

↓

Life Report

↓

Leaderboard

↓

Main Menu

Gameplay should always return to the **Gameplay Workspace**.

Everything revolves around that central screen.

---

37. UX Success Criteria

The interface succeeds when:

A first-time player learns it within five minutes.

Players rarely ask where to click.

Gameplay never requires scrolling.

No screen feels cluttered.

Decisions require one click.

Animations feel meaningful.

Players remain immersed for the full session.

---

38. Chapter Summary

The screen architecture of SPIKE LIFE is intentionally designed around **one persistent gameplay workspace** rather than traditional application pages.

Players remain inside the simulation from beginning to end.

Supporting screens—such as the Dream Board, Annual Review, and Life Report—exist only to enrich the player's journey before seamlessly returning them to the core gameplay experience.

This architecture reinforces one of SPIKE LIFE's defining design principles:

The player should never feel like they are navigating software. They should feel like they are living a financial life.

Every screen, animation, transition, and layout decision should therefore minimize friction, maximize clarity, and keep the player's attention focused on the next meaningful decision.

End of Chapter 17

This chapter establishes the complete screen architecture and interaction flow for SPIKE LIFE. Together with the UX Philosophy, Gameplay Foundation, and Financial Engine, it provides a comprehensive blueprint for implementing the user interface in Cursor without ambiguity, ensuring a consistent, responsive, and immersive player experience across all future versions of the platform.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume IV – User Experience

Chapter 18 – Animation, Feedback & Interaction System

Document Version: 1.0

Status: Approved Foundation

Priority: Critical

---

1. Purpose

This chapter defines every animation, transition, visual effect, micro-interaction, and user feedback mechanism in SPIKE LIFE.

Animation is **not decoration**.

Animation is communication.

Every movement on screen must answer one question:

"What just happened?"

Animations should reinforce gameplay, increase excitement, reduce cognitive load, and create memorable emotional moments.

---

2. Design Philosophy

SPIKE LIFE should feel alive.

Nothing should appear instantly.

Nothing should disappear abruptly.

Every transition should feel intentional.

Players should subconsciously understand:

- where they are,
- what changed,
- what to do next,

without reading instructions.

---

3. Animation Principles

Every animation follows six principles.

Principle 1

Purpose Before Beauty

Animations must communicate information.

Never animate for decoration alone.

Principle 2

Fast

Animations should never slow gameplay.

Most animations

200–500 ms.

Major events

2–3 seconds maximum.

Principle 3

Smooth

Use easing curves.

Avoid linear motion.

Everything should feel natural.

Principle 4

Reward

Achievements should feel satisfying.

Progress should feel earned.

Principle 5

Consistency

The same action always produces the same animation.

Players learn through repetition.

Principle 6

Performance

Maintain

60 FPS

on recommended hardware.

---

4. Animation Categories

Animations fall into eight groups.

Planning Cycle

Board

Situation

Decision

Financial

Goals

Achievements

Transitions

---

5. Planning Cycle Opening

Every Planning Cycle begins identically.

Sequence

Planning Cycle Begins

↓

Header Updates

↓

Background subtly shifts

↓

Life Domains begin glowing

↓

Selection animation starts

↓

Selected Domain expands

↓

Situation appears

Duration

2.5–3 seconds.

Players should immediately recognize:

"A new chapter of life has begun."

---

6. Life Domain Animation

The Life Domain Board is the visual centerpiece.

Default State

Soft glow

Low motion

Planning Cycle

Domains pulse

↓

Highlight rotates rapidly

↓

Speed slows

↓

Selected domain flashes brightly

↓

Selected card expands

↓

Situation Card emerges

Inactive domains fade slightly.

---

7. Situation Reveal Animation

The selected domain opens into a Situation Card.

Sequence

Glow

↓

Expand

↓

Illustration fades in

↓

Situation title appears

↓

Narrative fades upward

↓

Decision buttons slide in

↓

Timer begins

Total duration

700–900 ms.

---

8. Decision Button Interactions

Buttons must feel responsive.

Hover

Slight elevation

Brightness increase

Click

Compress

↓

Glow

↓

Lock animation

Selected option becomes visually distinct.

Other choices fade.

No accidental double-clicks.

---

9. Decision Lock Animation

After choosing:

Button locks.

Checkmark appears.

Subtle confirmation sound.

Status changes to

✓ Decision Submitted

The player immediately understands:

"My decision has been recorded."

---

10. Timer Animation

The timer is central to gameplay.

Appearance

Large circular ring.

Animated countdown.

Color progression

Green

↓

Amber

↓

Red

At

3 seconds

Ring pulses.

At

1 second

Heartbeat animation.

At

0

Timer disappears.

Auto Advisor activates.

---

11. Auto Advisor Animation

Timeout sequence.

Timer expires.

↓

Soft advisor glow appears.

↓

Advisor icon activates.

↓

Message

"Balanced Decision Selected"

↓

Financial Engine continues.

No punishment animation.

The experience should feel supportive.

---

12. Advisor Panel Animation

Advisor slides in from the right.

Soft blur behind.

Avatar fades in.

Conversation appears line by line.

Closing

Panel slides away.

Timer resumes smoothly.

Never use abrupt modal windows.

---

13. Financial Update Animations

Every financial change animates.

Positive

Green

↑

Negative

Red

↓

Examples

Cash

Savings

Net Worth

Emergency Fund

Debt

Goal Funding

Life Score

Values count smoothly.

Never jump instantly.

---

14. Goal Progress Animation

Goal cards animate whenever progress changes.

Example

House

65%

↓

71%

Progress ring fills smoothly.

Celebratory sparkle when milestones reached.

---

15. Goal Completion Animation

When a goal completes:

Card expands.

Gold border appears.

Achievement badge unlocks.

Confetti.

Short celebration sound.

Duration

2 seconds.

Never interrupt gameplay for longer.

---

16. Achievement Animation

Achievements deserve emotional reward.

Sequence

Badge appears.

↓

Glow.

↓

Title revealed.

↓

Life Score bonus shown.

↓

Badge minimized into Journey timeline.

Examples

First ₱100,000 Saved

Emergency Fund Complete

Homeowner

Debt-Free

Retirement Ready

---

17. Consequence Reveal

After Financial Engine processing:

All player results appear simultaneously.

Each player panel animates independently.

Example

Cash

↑

Emergency Fund

↓

Business Value

↑

House Goal

Delayed

Results should appear in less than

2 seconds.

---

18. Multiplayer Reveal

One of the signature moments.

All players wait.

Screen displays:

Calculating Your Lives...

Animated processing.

↓

Results reveal simultaneously.

Players naturally react together.

This is one of the game's emotional highlights.

---

19. Annual Review Animation

Year ends.

Background transitions.

Snowfall (December theme)

or

Celebration lights.

13th Month Pay arrives.

Annual summary fades in.

Players feel closure before the next year.

---

20. Dream Board Animation

Dream cards feel alive.

Hover

Card gently lifts.

Progress

Ring fills.

Completed

Golden glow.

Future cost updates animate subtly.

Dreams should feel aspirational.

---

21. Financial Health Animation

Financial Health indicator changes smoothly.

Example

Stable

↓

Good

↓

Excellent

Transitions use color and icon changes.

Never sudden jumps.

---

22. Life Score Animation

Life Score updates after every Planning Cycle.

Example

742

↓

754

Score counts upward.

Positive gain highlighted.

Large score jumps produce celebratory effects.

---

23. Board Idle Animation

When players are thinking,

the board remains alive.

Subtle glow.

Slow movement.

Soft lighting.

Nothing distracting.

Avoid static interfaces.

---

24. Screen Transitions

Use only:

Fade

Slide

Scale

Expand

Blur

Never use:

Hard cuts

Flashing

Rapid zooms

Spinning transitions

---

25. Error Feedback

Errors should feel calm.

Example

Connection Lost

↓

Soft banner.

↓

Reconnect animation.

↓

Resume automatically.

Avoid alarming popups.

26. Sound Synchronization

Every major animation has synchronized audio.

Examples

Domain Selection

Soft click.

Goal Complete

Success chime.

Timer

Tick.

Achievement

Celebration.

Advisor

Warm notification.

Sounds remain subtle.

---

27. Haptic Feedback (Future)

Mobile Companion

Button press

Light vibration.

Achievement

Medium vibration.

Major milestone

Celebration vibration.

Desktop ignores haptics.

---

28. Accessibility

Animations must support:

Reduced Motion Mode.

Instant transitions.

No flashing lights.

Color-independent feedback.

Screen reader compatibility.

Accessibility always overrides visual effects.

---

29. Performance Targets

Target Frame Rate

60 FPS

Animation Start Delay

<50 ms

Planning Cycle Animation

<3 sec

Transition

<300 ms

Financial Updates

<500 ms

No dropped frames during multiplayer.

---

30. Cursor Implementation Rules

Mandatory implementation requirements:

- All animations implemented using Framer Motion.
- GPU-accelerated transforms only (transform, opacity, scale).

- Never animate layout properties that trigger expensive reflows.
- Respect the user's operating system prefers-reduced-motion setting.
- Animation durations configurable through a centralized theme file.
- Consequence animations synchronized across all players.
- No animation may block Financial Engine execution.
- Every animation interruptible if the game is paused.
- All animation states controlled by the finite state machine.
- Maintain 60 FPS on target hardware.

---

31. Emotional Design Timeline

Every Planning Cycle should produce the following emotional sequence:

Curiosity

↓

Suspense

↓

Discovery

↓

Decision

↓

Anticipation

↓

Consequence

↓

Reflection

↓

Progress

↓

Hope

↓

Next Planning Cycle

This rhythm should remain consistent throughout the game.

---

32. UX Success Criteria

The animation system succeeds when:

- Players instinctively understand what changed.
- Animations make the game feel alive without slowing it down.
- Major achievements feel rewarding.
- Financial updates are immediately understandable.
- Multiplayer reveals create excitement and conversation.
- Players remain emotionally engaged for the entire session.

Animations should enhance clarity first and delight second.

---

33. Chapter Summary

The Animation, Feedback & Interaction System is responsible for giving SPIKE LIFE its personality.

Without it, the game would simply be a sequence of calculations and decisions.

With it, every Planning Cycle becomes an engaging life episode filled with anticipation, consequence, celebration, and reflection.

Every glow, transition, sound, and movement should reinforce one central idea:

Life is always moving forward. Every decision leaves a mark.

The animation system should therefore make players feel that they are not merely interacting with software—they are watching the story of their financial life unfold in real time.

End of Chapter 18

This chapter defines the complete animation and interaction language of SPIKE LIFE. Together with the UX Philosophy and Screen Specifications, it forms the visual and emotional foundation of the game, ensuring that every interaction communicates meaning, supports learning, and creates a polished, premium player experience suitable for both education and entertainment.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume V – Technical Architecture

Chapter 19 – Software Architecture & Engine Design

Document Version: 1.0

Status: Architecture Approved

Priority: Critical

---

1. Purpose

This chapter defines the complete software architecture of SPIKE LIFE.

It serves as the technical constitution of the project and takes precedence over implementation decisions.

Whenever there is ambiguity between implementation details and gameplay specifications, **this architecture governs the final design**.

The architecture is built around one central objective:

Create a modular, deterministic, highly maintainable simulation engine capable of supporting decades of future expansion without requiring architectural redesign.

---

2. Architectural Philosophy

SPIKE LIFE is **not** a web application.

It is **not** a CRUD system.

It is **not** a dashboard.

It is a **real-time turn-based simulation engine**.

Therefore,

the architecture follows game-engine principles instead of traditional business software patterns.

---

3. Core Design Principles

The software follows ten architectural principles.

Principle 1

Simulation First

Everything exists to support the simulation.

The UI never owns business logic.

Principle 2

Engine Separation

Every engine is independent.

No engine should directly manipulate another engine's data.

Communication occurs only through the Game State.

Principle 3

Deterministic Simulation

Given:

- identical game state
- identical random seed
- identical player decisions

the engine must always produce identical outcomes.

Principle 4

Data Driven

Rules belong in configuration.

Never hardcode situations.

Never hardcode probabilities.

Never hardcode careers.

Never hardcode financial assumptions.

Principle 5

Stateless UI

The UI renders.

The Simulation decides.

Principle 6

Event Driven

Everything important is an event.

Promotion

Hospitalization

Goal Completed

Business Growth

Protection Activated

Everything becomes an event.

Principle 7

Single Source of Truth

The Game State is authoritative.

No duplicated state.

Principle 8

Composable Systems

Every subsystem can be replaced independently.

Principle 9

Future Proof

Every module should anticipate:

Financial Worlds

Online Multiplayer

AI Advisor

Localization

Expansion Packs

Principle 10

Educational Integrity

No technical optimization should compromise financial realism.

---

4. High-Level Architecture

Presentation Layer

|

▼

UI State Controller

|

▼

GAME ENGINE

Planning Cycle Engine

↓

Life Domain Engine

↓

Situation Engine

↓

Decision Engine

↓

Financial Engine

↓

Goal Engine

↓

Wealth Engine

↓

Risk Engine

↓

Advisor Engine

↓

Life Score Engine

↓

Randomization Engine

|

▼

Persistence Layer

↓

Cloud Sync

↓

Analytics

The Presentation Layer never bypasses the Game Engine.

---

5. Engine Responsibilities

Each engine has a single responsibility.

Engine	Responsibility
Planning Cycle	Controls game flow
Life Domain	Selects domain
Situation	Generates scenario
Decision	Validates player choices
Financial	Performs all calculations
Goal	Tracks dreams and progress
Wealth	Calculates assets and liabilities
Risk	Simulates uncertainty and protection
Advisor	Generates guidance
Life Score	Computes final score
Randomization	Controls probabilities

Each engine communicates only through the Game State.

---

6. Game State

The Game State is the heart of the architecture.

GameState

Players

Simulation

Economy

Planning Cycle

Random Seed

Events

Configuration

History

Metadata

Nothing exists outside the Game State.

---

7. State Immutability

Planning Cycles follow immutable updates.

Old State

↓

Simulation

↓

New State

Previous states remain recoverable.

Benefits:

Replay

Undo (Practice Mode)

Debugging

Testing

Analytics

---

8. Simulation Pipeline

Every Planning Cycle executes the exact same sequence.

Initialize

↓

Select Domain

↓

Generate Situation

↓

Accept Decision

↓

Financial Simulation

↓

Apply Risks

↓

Update Goals

↓

Update Wealth

↓

Update Protection

↓

Calculate Life Score

↓

Generate Events

↓

Persist State

↓

Render UI

The order is never changed.

---

9. Event Bus

All engines communicate through events.

Example

Promotion Event

↓

Financial Engine

↓

Income Updated

↓

Goal Engine

↓

House Goal Updated

↓

Life Score Engine

↓

Score Increased

↓

UI Updated

Loose coupling is mandatory.

---

10. Domain Model

Primary entities include:

Player

Career

Planning Cycle

Life Domain

Situation

Decision

Goal

Asset

Liability

Protection Plan

Investment

Business

Financial Health

Life Score

Achievement

Advisor Message

Simulation State

These entities are defined in the Domain Blueprint.

---

11. Technology Stack

Recommended stack.

Frontend

React

Next.js

TypeScript

Tailwind CSS

Framer Motion

PixiJS (optional for board animation)

Backend

Node.js

TypeScript

Hono

Simulation Engine

Pure TypeScript

Database

PostgreSQL

Prisma ORM

Caching

Redis (future online multiplayer)

Storage

Cloudflare R2 (future assets)

Authentication

Clerk or Auth.js

Deployment

Cloudflare Workers

Cloudflare Pages

---

12. Monorepo Structure

```
SPIKE_LIFE/

apps/
  web/
  admin/
  facilitator/

packages/
```

engine/

domain/

ui/

config/

advisor/

randomizer/

shared/

assets/

docs/

tests/

All shared logic belongs inside packages.

---

13. Engine Isolation

The following dependencies are prohibited.

Financial Engine

❌

UI Components

Advisor

❌

Database

Randomizer

❌

React

Goal Engine

❌

DOM

Every engine remains framework-independent.

---

14. Configuration System

Everything configurable.

Examples

Inflation

Investment Returns

Careers

Situations

Goals

Life Domains

Achievements

Difficulty

Advisor Rules

Localization

Configuration stored as JSON or YAML.

---

15. Save System

Every Planning Cycle generates a snapshot.

Snapshots include:

Complete Game State

Random Seed

Current Planning Cycle

Player States

Achievements

Event History

Replay becomes possible.

---

16. Replay Engine

Replay reconstructs gameplay.

Because simulation is deterministic,

only:

Random Seed

Player Decisions

Configuration Version

are required.

The engine rebuilds everything else.

---

17. Multiplayer Synchronization

Multiplayer uses:

Authoritative Simulation

Clients render only.

No client-side financial calculations.

Benefits:

Fairness

Security

Deterministic replay

Tournament support

---

18. Performance Targets

Simulation

<200 ms

Financial Engine

<100 ms

Planning Cycle

<500 ms

Rendering

60 FPS

Game initialization

<2 seconds

---

19. Logging

Structured logs only.

Examples

Planning Cycle Started

Decision Submitted

Goal Completed

Risk Triggered

Life Score Updated

Business Expanded

Logs support debugging and analytics.

---

20. Testing Strategy

Every engine must include:

Unit Tests

Integration Tests

Simulation Tests

Replay Tests

Property-Based Tests

Regression Tests

Deterministic replay tests are mandatory.

---

21. Versioning

Simulation configuration versioned independently.

Example

Simulation

v1.0.0

Career Library

v1.2.0

Situation Pack

v1.5.0

Financial Rules

v1.1.0

Old saved games remain playable.

22. Extensibility

Future expansion modules include:

Financial Worlds

Country Packs

New Careers

Investment Packs

Business Packs

Cooperative Mode

Campaign Mode

AI Story Engine

None should require rewriting the Game Engine.

---

23. Cursor Implementation Rules

Mandatory implementation requirements:

- Strict TypeScript.
- Functional programming preferred for simulation logic.
- Pure functions wherever possible.
- Engine packages contain zero UI code.
- All state updates immutable.
- Central Event Bus architecture.
- Simulation fully deterministic.
- JSON-driven configuration.
- Modular package boundaries enforced.
- Architecture Bible overrides implementation shortcuts.

---

24. Success Criteria

The architecture succeeds if:

- A new Financial World can be added without modifying the simulation core.
- A new Situation Pack requires no engine changes.
- Multiplayer and single-player use the same simulation logic.
- Replay reproduces identical results.
- Unit tests validate every financial calculation.
- UI can be completely redesigned without touching the Game Engine.

---

25. Chapter Summary

The Software Architecture defines SPIKE LIFE as a **simulation platform**, not merely a game.

By separating presentation, simulation, persistence, and configuration into independent layers, the architecture ensures long-term maintainability, extensibility, and correctness.

Every Planning Cycle follows a deterministic pipeline.

Every financial outcome is reproducible.

Every subsystem has a single responsibility.

This architecture allows SPIKE LIFE to evolve from a Philippine financial literacy game into a global Financial Worlds platform without sacrificing code quality or educational integrity.

The guiding architectural principle is:

Build the engine once. Expand the world forever.

End of Chapter 19

This chapter establishes the technical foundation for SPIKE LIFE. It defines the architectural rules that every developer, AI coding assistant, and future contributor must follow, ensuring that implementation remains faithful to the game's design philosophy while supporting scalability, security, testing, and future expansion.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume V – Technical Architecture

Chapter 20 – Data Architecture, Persistence & Security

Document Version: 1.0

Status: Architecture Approved

Priority: Critical

---

1. Purpose

This chapter defines how SPIKE LIFE stores, protects, synchronizes, secures, and preserves all game data.

The objective is not simply to save a game.

The objective is to preserve an entire financial life simulation with complete integrity.

Every Planning Cycle, every decision, every consequence, and every achievement must be recoverable, reproducible, and secure.

---

2. Design Philosophy

Data is the player's financial life.

Therefore, every saved game must be treated as a complete historical record rather than a collection of disconnected values.

The architecture follows four principles:

- Never lose player progress.
- Never corrupt financial calculations.
- Never trust the client.
- Never expose sensitive game logic.

---

3. Data Architecture

SPIKE LIFE uses a layered persistence model.

Player

↓

Game Session

↓

Simulation State

↓

Planning Cycle History

↓

Event Log

↓

Snapshots

↓

Analytics

Each layer has a distinct responsibility.

---

4. Primary Data Domains

The system stores the following major domains.

Player

Identity

Avatar

Preferences

Statistics

Achievements

Simulation

Current Age

Current Year

Planning Cycle

Difficulty

Financial World

Random Seed

Simulation Version

Financial State

Income

Expenses

Assets

Liabilities

Emergency Fund

Protection

Investments

Business

Cash Flow

Financial Health

Goal State

Dream Board

Goal Progress

Future Values

Funding Status

Target Ages

Gameplay State

Life Domain

Current Situation

Decision History

Achievements

Life Score

Timeline

Metadata

Save Version

Device

Language

Accessibility Settings

Configuration Version

---

5. Database Design Principles

The database is optimized for:

Consistency

Recoverability

Scalability

Analytics

Replay

not for reporting dashboards.

Normalization is preferred,

except where snapshot performance benefits from denormalization.

---

6. Storage Layers

Three persistence layers exist.

Layer 1

Live Game State

Fast read/write.

Current simulation only.

Layer 2

Planning Cycle History

Immutable history.

Stores every completed Planning Cycle.

Layer 3

Snapshots

Complete simulation checkpoints.

Supports:

Save

Resume

Replay

Recovery

---

7. Event Sourcing

Every major action generates an immutable event.

Examples

PLAYER_CREATED

GAME_STARTED

DOMAIN_SELECTED

SITUATION_REVEALED

DECISION_SUBMITTED

GOAL_COMPLETED

BUSINESS_STARTED

PROTECTION_ACTIVATED

LIFE_SCORE_UPDATED

GAME_FINISHED

The event history becomes the permanent record of the player's life.

---

8. Snapshot Strategy

A complete snapshot is stored:

- At game start
- After every Planning Cycle
- Before Annual Review
- At game completion

Snapshots allow instant recovery without replaying the entire simulation.

---

9. Save File Structure

Each save contains:

Game Metadata

↓

Simulation Configuration

↓

Player Profiles

↓

Financial State

↓

Goal State

↓

Random Seed

↓

Current Planning Cycle

↓

History

↓

Achievements

↓

Replay Metadata

Nothing required for replay exists outside the save.

---

10. Random Seed Persistence

The Simulation Seed is permanently stored.

This guarantees:

Deterministic Replay

Bug Reproduction

Tournament Verification

Simulation Validation

The seed never changes during a game.

---

11. Configuration Versioning

Every save stores:

Simulation Version

Financial Rules Version

Situation Library Version

Career Library Version

Localization Version

Future versions can migrate saves safely.

---

12. Replay Architecture

Replay requires only:

Simulation Seed

Player Decisions

Configuration Version

The engine reconstructs every Planning Cycle exactly.

Replay should never rely on recorded animations or screenshots.

---

13. Security Philosophy

SPIKE LIFE follows a Zero Trust Architecture.

The client is never trusted.

All authoritative calculations occur inside the Simulation Engine.

Clients receive results,

not authority.

---

14. Client Security

The client may never:

Calculate Life Score

Calculate Financial Health

Calculate Net Worth

Generate Situations

Determine Random Events

Modify Game State

The client only:

Displays

Animates

Collects player input

---

15. Server Authority

Only the Simulation Engine may:

Advance Planning Cycles

Generate Events

Calculate Wealth

Compute Protection

Generate Advisor Messages

Update Life Score

Persist Saves

---

16. Input Validation

Every client request is validated.

Examples

Invalid Planning Cycle

↓

Rejected

Invalid Decision

↓

Rejected

Duplicate Submission

↓

Ignored

Modified Payload

↓

Rejected

Never trust client data.

---

17. Anti-Cheat Strategy

Prevent:

Modified save files

Injected decisions

Timer manipulation

Life Score editing

Memory modification

Replay tampering

The authoritative server always validates game integrity.

---

18. SQL Injection Protection

All database access must use:

Parameterized queries

ORM validation

Prepared statements

Never construct SQL using string concatenation.

---

19. XSS Protection

All user-generated content must be:

Escaped

Sanitized

Validated

Examples

Player Name

Custom Dream Names

Chat (future)

Never render raw HTML.

---

20. CSRF Protection

Every authenticated request uses:

CSRF tokens

SameSite cookies

Origin validation

Session verification

---

21. Authentication

Future online mode supports:

Email

Google

Apple

Microsoft

Guest Mode

Authentication remains independent from gameplay.

---

22. Authorization

Roles include:

Player

Host

Facilitator

Administrator

Developer

Every API endpoint validates authorization.

---

23. Encryption

Sensitive information encrypted:

Passwords

Refresh Tokens

Session Keys

Saved Games (cloud)

TLS mandatory.

Passwords hashed using Argon2 or bcrypt.

Never store plain text credentials.

---

24. API Security

Every API request includes:

Authentication

Authorization

Rate Limiting

Validation

Audit Logging

Version Check

Invalid requests fail safely.

---

25. Malware Protection

Uploaded assets (future):

Virus scan

File type validation

Size validation

Storage isolation

Executable files prohibited.

---

26. Dependency Security

Mandatory:

Dependabot

Automated vulnerability scanning

Pinned versions

Security patch monitoring

No abandoned packages.

---

27. Logging & Auditing

Audit logs record:

Authentication

Game Creation

Host Actions

Simulation Errors

Configuration Changes

Administrative Actions

Financial calculations themselves are **not** logged verbatim to avoid excessive storage and protect player privacy.

---

28. Privacy

Player data should comply with modern privacy principles.

Support:

Account deletion

Data export

Cloud save removal

Minimal personal information collection

Future localization should accommodate regulations such as GDPR and the Philippine Data Privacy Act.

---

29. Backup Strategy

Cloud backups:

Daily incremental

Weekly full

Monthly archive

Game saves replicated across storage regions where infrastructure supports it.

No single point of failure.

---

30. Disaster Recovery

Recovery objectives:

Service restoration

<1 hour

Data loss

<5 minutes

Cloud saves restored automatically.

---

31. Performance Targets

Save Game

<100 ms

Load Game

<300 ms

Replay Initialization

<500 ms

Database Query

<50 ms

Planning Cycle Save

Background operation

Gameplay should never pause while saving.

---

32. Cursor Implementation Rules

Mandatory implementation requirements:

- PostgreSQL using Prisma ORM.
- Immutable Planning Cycle history.
- Snapshot after every Planning Cycle.
- Event sourcing for major gameplay events.
- Strict server-authoritative simulation.
- Parameterized database queries only.
- Input validation using shared schemas.
- Argon2 (preferred) or bcrypt for password hashing.
- TLS for all network communication.
- Security headers enabled (Content Security Policy, HSTS, X-Frame-Options, X-Content-Type-Options).
- Replay generated from Simulation Seed and player decisions.
- Save files versioned for forward compatibility.
- Complete separation between persistence layer and simulation engine.

---

33. Future Expansion

The Data Architecture supports:

Cloud Save

Cross-device Synchronization

Offline Mode

Replay Sharing

Tournament Validation

Classroom Analytics

AI Coaching History

Financial Worlds Marketplace

Simulation Benchmarking

Because the architecture is event-driven and deterministic, these features can be added without redesigning the core engine.

---

34. Chapter Summary

The Data Architecture, Persistence & Security layer is the foundation that protects the integrity of SPIKE LIFE.

It ensures that every financial life can be saved, replayed, audited, and recovered while remaining secure against tampering and common web application threats.

By combining immutable event history, deterministic simulation, server-authoritative processing, and modern security practices, SPIKE LIFE becomes more than a game—it becomes a reliable financial simulation platform.

The guiding principle of this chapter is:

Every financial life deserves to be preserved accurately, protected rigorously, and reproduced faithfully.

End of Chapter 20

This chapter completes the core technical architecture of SPIKE LIFE by defining how game data is stored, synchronized, secured, and recovered. Together with the Software Architecture chapter, it provides Cursor and future developers with a complete blueprint for implementing a scalable, secure, deterministic, and production-ready financial simulation platform capable of supporting future expansions, online multiplayer, Financial Worlds, and AI-driven features without architectural redesign.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume VI – Content Framework

Chapter 21 – Content Architecture & Financial Worlds System

Document Version: 1.0

Status: Architecture Approved

Priority: Critical

---

1. Purpose

This chapter defines how SPIKE LIFE grows beyond a single game into an expandable platform.

The MVP begins with one Financial World:

Philippines

However, the architecture must support unlimited future Financial Worlds without requiring changes to the Game Engine.

This chapter defines how countries, careers, economies, laws, cultures, currencies, and financial products are separated from the simulation itself.

---

2. Design Philosophy

One Engine.

Unlimited Worlds.

The simulation engine should never know whether it is running:

- Philippines
- Singapore

- Japan
- Australia
- Canada
- United States

The engine only understands:

- Income
- Inflation
- Protection
- Investments
- Goals
- Risks
- Decisions

Everything else belongs inside a Financial World Pack.

---

3. What is a Financial World?

A Financial World is a complete financial ecosystem.

It contains everything required to simulate life in a specific country or economic environment.

Examples include:

- Philippines
- Singapore
- Malaysia
- Indonesia
- Japan
- United Kingdom
- Australia
- Middle East

- United States

Each world provides its own localized rules while using the same simulation engine.

---

4. Financial World Components

Each Financial World contains:

Country Profile

↓

Currency

↓

Career Library

↓

Economic Rules

↓

Situation Library

↓

Investment Rules

↓

Protection Rules

↓

Government Rules

↓

Tax Rules

↓

Cultural Events

↓

Natural Disasters

↓

Localization

---

5. Financial World Structure

FinancialWorld/

config/

careers/

situations/

goals/

economy/

government/

investments/

protection/

events/

translations/

media/

Each world is packaged independently.

---

6. Philippine Financial World (MVP)

The first Financial World establishes the reference implementation.

Core characteristics:

Currency

Philippine Peso (PHP)

Inflation

4%

Mandatory 13th Month Pay

Enabled

Natural Disasters

Typhoon

Flood

Earthquake

Volcanic Activity

Common Family Structures

Multi-generational households

OFW families

Extended family support

Government Systems

SSS

PhilHealth

Pag-IBIG

Taxes (Future)

BIR

The MVP models these at a conceptual level rather than reproducing every legal detail.

---

7. Career Library

Every Financial World provides its own careers.

Philippines examples:

Teacher

Engineer

Call Center Professional

Nurse

OFW

Government Employee

Sales Professional

Financial Advisor

Entrepreneur

Freelancer

Small Business Owner

Each career defines:

Starting Salary

Growth Curve

Promotion Probability

Business Compatibility

Risk Factors

---

8. Situation Library

Each country owns its own situations.

Philippines:

Typhoon Damaged House

Barangay Election

Family Fiesta

13th Month Pay

Parents Need Support

OFW Opportunity

Christmas Spending

Community Fundraising

Japan:

Earthquake Preparedness

Bonus Season

Rail Relocation

Singapore:

CPF Changes

HDB Upgrade

National Service

The Game Engine remains unchanged.

---

9. Cultural Rules

Financial decisions are influenced by culture.

Examples:

Philippines

Strong family obligations

Frequent celebrations

Community participation

Extended family support

Future worlds replace these with local cultural behaviors.

---

10. Government Systems

Every Financial World defines:

Retirement programs

Healthcare systems

Housing programs

Tax systems

Government assistance

Business registration

Future versions may simulate these in greater detail.

---

11. Currency Engine

The Game Engine stores monetary values internally.

The Financial World controls:

Currency Symbol

Decimal Rules

Thousands Separator

Localization

Exchange Rates (Future)

Examples

PHP

JPY

USD

SGD

AUD

---

12. Inflation Profiles

Each Financial World defines:

Default Inflation

Inflation Variability

Economic Shocks

Government Response

The Financial Engine simply consumes these values.

---

13. Investment Library

Each world defines available investment types.

Philippines

Savings

Government Securities

Balanced Funds

Stocks

Cooperative Shares

Business

Future worlds introduce additional instruments without changing engine logic.

---

14. Protection Framework

Protection categories remain universal.

Family Protection Plan

Income Protection Plan

Health Protection Plan

Education Protection Plan

Retirement Security Plan

Only the implementation changes.

The educational philosophy remains consistent across all countries.

---

15. Goal Templates

Dreams vary by country.

Philippines

House

₱5M

Travel

₱300K

Business

₱500K

Japan

House values

Higher

Travel patterns

Different

The Goal Engine simply loads localized templates.

---

16. Economic Profiles

Each Financial World includes:

Interest Rates

Investment Returns

Salary Growth

Property Appreciation

Business Volatility

Employment Stability

Government Support

The Simulation Engine never hardcodes these values.

---

17. Localization

Every world includes:

Language

Date Format

Number Format

Currency

Icons

Images

Cultural References

Voice Pack (Future)

Localization extends beyond translation.

---

18. Educational Alignment

Although content changes,

the financial principles remain identical.

Every Financial World teaches:

Emergency Funds

Protection

Goal Planning

Risk Management

Investing

Long-Term Thinking

Balanced Decision-Making

Only examples and scenarios change.

---

19. Financial World Loader

When a game starts:

Select Financial World

↓

Load Configuration

↓

Load Careers

↓

Load Situations

↓

Load Economy

↓

Load Goals

↓

Load Government Rules

↓

Initialize Simulation

↓

Start Game

The engine should never require recompilation when a new world is added.

---

20. Expansion Packs

Future downloadable packs may include:

ASEAN Collection

North America

Europe

Middle East

Latin America

Africa

Fantasy Financial Worlds

Historical Worlds

Mars Colony

Educational Sandbox

Every expansion uses the same architecture.

---

21. Marketplace (Future)

Financial Worlds become downloadable modules.

Players may install:

Countries

Career Packs

Situation Packs

Achievement Packs

Business Packs

Educational Packs

This allows SPIKE LIFE to evolve into an ecosystem rather than a single title.

---

22. Content Authoring

New content should require **no programming**.

Content creators should build worlds using structured JSON/YAML definitions and media assets.

Required authoring tools (future):

World Editor

Situation Editor

Career Editor

Economy Editor

Translation Manager

This enables educators and financial experts to contribute without writing code.

---

23. Validation Rules

Every Financial World must pass automated validation.

Checks include:

- All careers have salary ranges.
- All situations belong to valid Life Domains.
- Inflation values are defined.
- Goal templates exist.
- Protection categories are complete.
- Localization files are valid.
- No missing media references.
- Probability distributions are balanced.

A world cannot be published until validation succeeds.

---

24. Cursor Implementation Rules

Mandatory implementation requirements:

- Financial Worlds loaded dynamically.
- World definitions stored outside application code.
- Engine consumes configuration only.
- Support hot-loading during development.
- World versioning independent of engine version.
- Content validation before loading.
- Localization separated from simulation.
- Asset references relative to the World package.
- New Financial Worlds require zero engine modifications.
- MVP ships with one fully implemented Philippine Financial World.

---

25. Future AI Integration

Future AI tools may generate:

New Situations

Career Descriptions

Advisor Dialogues

Localized Stories

Community Events

However:

AI must never modify:

Financial formulas

Probability rules

Simulation logic

Risk calculations

The engine remains authoritative.

---

26. Success Criteria

The Content Architecture succeeds if:

- A developer can add a new country without changing the Game Engine.
- An educator can create a Situation Pack without writing TypeScript.
- A localization team can translate the game without touching simulation code.
- Every Financial World feels culturally authentic.
- Financial lessons remain consistent worldwide.

---

27. Chapter Summary

The Financial Worlds System transforms SPIKE LIFE from a Philippine financial literacy game into a global financial education platform.

By separating content from simulation, the platform can grow indefinitely while preserving a single, stable architecture.

The engine teaches timeless financial principles.

Each Financial World teaches them through the lens of a specific country, economy, and culture.

This architecture fulfills one of the original long-term visions of SPIKE LIFE:

Build one world-class financial simulation engine. Let every nation tell its own financial story.

End of Chapter 21

This chapter establishes the Content Architecture and Financial Worlds framework, completing the separation between simulation logic and localized content. Together with the Software Architecture, Data Architecture, and UX framework, it provides the foundation for a scalable, globally extensible platform capable of supporting new countries, educational programs, enterprise deployments, and community-created content without requiring architectural redesign.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume VI – Content Framework

Chapter 22 – Career System & Life Progression Engine

Document Version: 1.0

Status: Architecture Approved

Priority: Critical

---

1. Purpose

The Career System is responsible for simulating one of the most important drivers of financial life:

How people earn income throughout their lives.

A player's career determines far more than salary.

It influences:

- Income growth
- Lifestyle
- Opportunities
- Risks
- Financial capacity
- Retirement
- Business opportunities
- Family decisions
- Life events

The Career Engine provides the economic foundation upon which every financial decision is built.

---

2. Design Philosophy

SPIKE LIFE does **not** simulate jobs.

It simulates **careers**.

A career is a journey.

Not merely a paycheck.

Players should feel that their professional life evolves naturally over the ten-year simulation.

---

3. Career Lifecycle

Every career follows the same progression.

Choose Career

↓

First Employment

↓

Performance

↓

Salary Growth

↓

Promotion Opportunities

↓

Career Decisions

↓

Career Risks

↓

Career Transition

↓

Late Career

↓

Retirement Preparation

Career progression is continuous.

---

4. Career Categories (Philippines MVP)

The MVP launches with approximately **20–30 careers**.

Examples:

Professional

Teacher

Engineer

Architect

Doctor

Nurse

Lawyer

CPA

Corporate

Call Center Professional

HR Manager

Marketing Professional

IT Professional

Operations Manager

Government

Government Employee

Uniformed Service

LGU Employee

Education

Teacher

Professor

Trainer

Entrepreneurial

Small Business Owner

Freelancer

Financial Advisor

Sales Professional

Online Seller

Creative

Designer

Content Creator

Photographer

Developer

Future Financial Worlds provide localized career libraries.

---

5. Career Attributes

Every career contains structured metadata.

Career ID

Career Name

Industry

Starting Salary

Salary Range

Growth Curve

Promotion Frequency

Promotion Amount

Business Compatibility

Stress Level

Stability Score

Risk Profile

Retirement Age

Localization

No career logic should be hardcoded.

---

6. Starting Salary

Every career begins with a configurable salary range.

Example

Teacher

₱28,000–₱35,000

Engineer

₱35,000–₱50,000

Financial Advisor

Variable

Freelancer

Variable

Business Owner

Variable

The exact salary is randomized within the configured range.

---

7. Salary Growth

Salary growth depends on:

Years of Experience

Performance

Economic Conditions

Career Type

Promotion

Career Decisions

The player never manually negotiates salary.

The simulation handles progression.

---

8. Promotion Engine

Promotion opportunities occur through the Situation Engine.

Examples

Junior Engineer

↓

Engineer

↓

Senior Engineer

↓

Manager

↓

Director

↓

Executive

Each promotion affects:

Income

Lifestyle Pressure

Tax Burden (future)

Leadership Opportunities

Business Opportunities

Life Score

---

9. Career Transitions

Players may change careers.

Examples

Employee

↓

Entrepreneur

Teacher

↓

School Administrator

Nurse

↓

OFW

Corporate Employee

↓

Financial Advisor

Every transition creates opportunities and trade-offs.

---

10. Career Stability

Each career includes a Stability Rating.

Very Stable

Government

Teacher

Healthcare

Moderately Stable

Engineering

IT

Corporate

Variable

Sales

Freelancer

Financial Advisor

High Risk

Entrepreneur

Startup Founder

Higher risk often creates higher reward.

---

11. Career Growth Profiles

Growth curves differ.

Examples

Teacher

Steady

Engineer

Moderate

Business Owner

Highly Variable

Financial Advisor

Performance Driven

Freelancer

Project Based

Growth curves shape long-term income.

---

12. Career Risks

Every profession has unique risks.

Examples

Corporate

Layoffs

Government

Policy Changes

Business

Market Downturn

Healthcare

Burnout

Construction

Economic Cycles

Freelancer

Income Volatility

These influence future situations.

---

13. Continuing Education

Players may invest in learning.

Examples

Graduate Degree

Professional Certification

Leadership Training

Technical Skills

Business Courses

Education increases:

Promotion probability

Income growth

Life Score

Future opportunities

---

14. Performance Engine

Performance evolves through player decisions.

Positive behaviors

Continuous Learning

Balanced Lifestyle

Professional Development

Negative behaviors

Burnout

Poor Financial Health

Repeated Risk Events

Performance affects promotion probability.

---

15. Career Satisfaction

Although hidden,

the engine tracks satisfaction.

Factors include:

Income

Work-Life Balance

Financial Stress

Business Success

Achievement of Dreams

Low satisfaction increases the probability of career changes.

---

16. Entrepreneurship Path

Players may establish businesses.

Possible paths:

Side Hustle

↓

Micro Business

↓

Growing Business

↓

Multi-Branch Business

↓

Successful Exit

Business ownership integrates with the Business Engine.

---

17. Financial Advisor Career

SPIKE LIFE includes Financial Advisor as a normal career,

not a privileged one.

Income depends on:

Client Growth

Performance

Economic Conditions

Persistence

Business Building

This career follows the same simulation rules as all others.

---

18. Career & Family Interaction

Career decisions influence:

Marriage

Children

Housing

Travel

Education

Retirement

Business

The engine models realistic work-life trade-offs.

---

19. Career & Health Interaction

High-stress careers increase probability of:

Burnout

Medical Events

Lifestyle Decisions

Career Breaks

Balanced planning improves resilience.

---

20. Retirement Transition

Late-career situations include:

Retirement Planning

Early Retirement

Business Succession

Mentoring

Consulting

Reduced Working Hours

Retirement is treated as a career transition,

not the end of gameplay.

---

21. Career Achievements

Examples

First Promotion

Manager

Business Owner

Industry Leader

Million Peso Earner

Mentor

Successful Entrepreneur

Career achievements contribute to Life Score.

---

22. Career Timeline

The Journey Lens records:

Career Changes

Promotions

Business Launches

Leadership Roles

Awards

Education

Retirement

Players can review their professional life story.

---

23. Career Balance

The engine intentionally balances:

Income

↓

Stress

↓

Lifestyle

↓

Health

↓

Family

↓

Financial Capacity

Higher salaries often create greater responsibilities.

---

24. Future Career Packs

Future expansions may include:

Healthcare Careers

Technology Careers

Creative Careers

Manufacturing

Agriculture

Military

Aviation

Maritime

International Careers

Digital Nomads

No engine changes required.

---

25. Career Configuration

Every career is defined entirely through data.

Example

```json
{
  "careerId": "teacher",
  "startingSalary": 32000,
  "growthCurve": "steady",
  "stability": 0.9,
  "promotionProbability": 0.18,
  "stress": 0.45,
  "businessCompatibility": 0.25
}
```

Simulation logic consumes this configuration.

---

26. Cursor Implementation Rules

Mandatory implementation requirements:

- Careers loaded from JSON configuration.
- Salary growth event-driven.
- Promotion probabilities configurable.
- Career transitions supported.
- Continuing education integrated with Situation Engine.
- Career Timeline persisted.
- No hardcoded salaries.
- Financial Worlds provide independent Career Libraries.
- Career Engine independent from UI.
- Career decisions integrated with Financial, Goal, and Risk Engines.

---

27. Educational Objectives

Players should understand that:

Career choices influence financial outcomes.

Income growth requires continuous development.

Higher income often brings greater responsibility.

Entrepreneurship offers opportunity and risk.

Professional learning creates long-term value.

Financial success depends on more than salary.

---

28. Future AI Integration

Future AI may generate:

Career advice

Interview opportunities

Promotion narratives

Performance reviews

Mentorship conversations

Career coaching

However,

salary calculations,

promotion rules,

and financial consequences remain controlled by the Simulation Engine.

---

29. Chapter Summary

The Career System transforms employment from a static income source into a living, evolving journey.

Players experience promotions, setbacks, career changes, entrepreneurship, education, and retirement as interconnected parts of their financial lives.

Rather than rewarding a single profession, SPIKE LIFE demonstrates that every career path can lead to financial success when paired with thoughtful planning, disciplined decision-making, and continuous personal growth.

This reinforces one of the platform's core philosophies:

Your career creates opportunities. Your financial decisions determine what those opportunities become.

End of Chapter 22

This chapter establishes the Career System and Life Progression Engine as the primary source of income, opportunity, and professional development within SPIKE LIFE. Together with the Financial Planning Engine, Goal Engine, Situation Engine, and Financial Worlds framework, it enables realistic career journeys that remain culturally localized, technically configurable, and fully extensible across future countries, industries, and educational programs.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume VI – Content Framework

Chapter 23 – Situation Library & Event Authoring System

Document Version: 1.0

Status: Architecture Approved

Priority: Critical

---

1. Purpose

The Situation Library is the storytelling engine of SPIKE LIFE.

It defines every meaningful life experience that a player may encounter throughout the simulation.

While the Financial Engine determines **how life changes**, the Situation Library determines **what life feels like**.

Every Planning Cycle is built around one carefully designed situation.

---

2. Design Philosophy

The Situation Library exists to answer one question:

"What happened during this period of your life?"

A good situation should:

- Feel authentic
- Create emotional engagement
- Require thoughtful financial decisions
- Teach one financial lesson
- Be replayable
- Be culturally relevant

The player should feel:

"That could happen to me."

---

3. Situation Authoring Principles

Every situation must satisfy six requirements.

Authentic

Based on realistic life experiences.

Relevant

Matches the player's current life stage.

Educational

Teaches one financial concept.

Balanced

No obviously correct answer.

Emotional

Creates curiosity, excitement, concern, or hope.

Concise

Readable within 15–20 seconds.

---

4. Situation Structure

Every situation follows the same architecture.

Situation ID

↓

Life Domain

↓

Title

↓

Narrative

↓

Illustration

↓

Decision A

↓

Decision B

↓

Decision C

↓

Immediate Effects

↓

Hidden Effects

↓

Advisor Notes

↓

Probability Rules

↓

Unlock Rules

Every situation follows this template.

---

5. Situation Metadata

Required metadata includes:

Situation ID

Life Domain

Financial World

Life Stage

Age Range

Difficulty

Rarity

Estimated Reading Time

Estimated Decision Time

Author Version

Localization

Every situation is independently versioned.

---

6. Life Domains

Each situation belongs to exactly one primary Life Domain.

Career

Opportunity

Family

Health

Business

Lifestyle

Housing

Education

Government

Community

Investment

Milestone

The Life Domain determines where it appears on the game board.

---

7. Situation Categories

Each situation also belongs to one secondary category.

Opportunity

Challenge

Milestone

Lifestyle

Financial

Relationship

Business

Community

Government

Emergency

This improves filtering and authoring.

---

8. Situation Lifecycle

Situation Created

↓

Validated

↓

Approved

↓

Published

↓

Loaded into Library

↓

Random Selection

↓

Player Decision

↓

Archived in Timeline

Situations are never modified during gameplay.

---

9. Narrative Guidelines

Narratives should:

- Use conversational language.
- Avoid financial jargon.
- Avoid legal terminology.
- Avoid product marketing.
- Stay under 100 words.
- End naturally before presenting choices.

Example:

"Your employer has offered you a promotion that comes with a significant salary increase—but also longer hours and greater responsibility. How would you like to approach this new chapter of your career?"

---

10. Decision Writing Guidelines

Every situation provides exactly three strategic options.

The options should represent different philosophies rather than different amounts of money.

Example

Promotion

Option A

Strengthen Financial Security

Option B

Maintain Balance

Option C

Upgrade Your Lifestyle

Avoid:

❌ Increase savings by 12%.

Instead:

✅ Prioritize your future.

---

11. Educational Objective

Every situation teaches exactly one primary lesson.

Examples

Emergency Fund

Insurance

Investment

Debt

Inflation

Opportunity Cost

Diversification

Lifestyle Inflation

Entrepreneurship

Career Growth

Goal Prioritization

Family Responsibility

Avoid teaching multiple unrelated lessons in one situation.

---

12. Situation Difficulty

Difficulty represents emotional complexity,

not mathematical complexity.

Levels:

1

Simple

2

Moderate

3

Complex

4

Life-Changing

All decisions remain easy to understand.

---

13. Rarity System

Each situation belongs to one rarity tier.

Common

60%

Uncommon

25%

Rare

10%

Legendary

5%

Legendary events create memorable stories,

not stronger rewards.

---

14. Unlock Conditions

Some situations become available only after certain conditions.

Examples

Marriage

↓

Child Born

↓

Education Planning

Business Started

↓

Business Expansion

↓

Business Acquisition

House Purchased

↓

Home Renovation

↓

Property Tax

Unlock rules create evolving life stories.

---

15. Hidden Consequences

Every situation supports delayed outcomes.

Example

Player ignores health check-up.

↓

Three years later

↓

Critical illness probability increases.

Players should rarely see immediate cause and effect.

The educational value lies in delayed consequences.

---

16. Situation Chains

Situations may form story arcs.

Example

Job Interview

↓

New Job

↓

Promotion

↓

Leadership

↓

Business Opportunity

↓

Entrepreneurship

Story arcs create narrative continuity.

---

17. Cultural Localization

Every Financial World maintains its own Situation Library.

Philippines

Typhoon

Fiesta

OFW Opportunity

Barangay Election

13th Month Pay

Family Reunion

Christmas Spending

Singapore

CPF

HDB

National Service

Japan

Bonus Season

Earthquake Preparedness

Public Transit Changes

The engine loads the appropriate library automatically.

---

18. Advisor Integration

Each situation includes Advisor content.

Examples

Key trade-offs

Financial concepts

Potential long-term impacts

Advisor guidance remains separate from financial logic.

---

19. Illustration Requirements

Every situation references artwork.

Preferred formats:

2D Illustration

Stylized Vector

Modern Flat Design

Soft Isometric

Avoid photorealistic imagery.

Illustrations should reinforce understanding at a glance.

---

20. Audio Hooks

Situations may define optional sounds.

Examples

Promotion

Celebration

Hospital

Soft alert

Business

Cash register

Typhoon

Wind ambience

Audio remains subtle.

---

21. Authoring Workflow

Content creators follow this pipeline.

Create Situation

↓

Assign Domain

↓

Write Narrative

↓

Define Decisions

↓

Configure Effects

↓

Configure Hidden Effects

↓

Assign Probability

↓

Assign Illustration

↓

Validate

↓

Publish

No programming required.

---

22. Validation Rules

Every situation must pass automated validation.

Checks include:

- Exactly one Life Domain.
- Exactly three decisions.
- Maximum narrative length.
- Valid illustration reference.
- Advisor content exists.
- Hidden effects valid.
- Immediate effects valid.
- Localization complete.
- Probability defined.
- Unlock rules valid.

Invalid situations cannot be published.

---

23. Situation Database

Recommended structure.

situations/

career/

family/

health/

business/

housing/

government/

community/

investment/

milestone/

opportunity/

lifestyle/

Each file contains one situation.

This minimizes merge conflicts.

---

24. JSON Schema Example

```
{
  "id": "career_promotion_001",
  "domain": "career",
  "category": "opportunity",
  "rarity": "common",
  "difficulty": 2,
  "title": "Promotion Opportunity",
  "decisions": [
    "Strengthen Your Future",
    "Maintain Balance",
    "Upgrade Your Lifestyle"
  ]
}
```

Simulation logic references IDs,

not text.

---

25. Future AI Authoring

Future AI tools may generate:

Situation drafts

Illustration prompts

Advisor dialogue

Localization

Difficulty suggestions

Human review remains mandatory before publication.

AI accelerates creation,

not validation.

---

26. Content Targets

Recommended MVP

12 Life Domains

×

25 Situations each

=

300 Situations

Future Release

1,000+

Enterprise Goal

10,000+

The architecture is designed for unlimited expansion.

---

27. Cursor Implementation Rules

Mandatory implementation requirements:

- Situations stored as individual JSON files.
- JSON schema validation on load.
- Separate localization files.
- Hidden effects isolated from visible effects.
- Story chains supported.
- Unlock conditions configurable.

- Situation cooldown supported.
- No hardcoded situation logic.
- Library loaded dynamically by Financial World.
- Future AI-generated situations use the same schema.

---

28. Educational Integrity

The Situation Library must never become:

A trivia game

A budgeting exercise

An insurance quiz

A compliance course

Every situation should teach naturally through experience,

never through lectures.

---

29. Success Criteria

The Situation Library succeeds if:

- Players frequently say, "That happened to me."
- Every Planning Cycle feels fresh.
- Situations remain culturally authentic.
- Different Financial Worlds feel genuinely different.
- Financial lessons emerge naturally from gameplay.
- New situations can be added without code changes.

---

30. Chapter Summary

The Situation Library is the narrative soul of SPIKE LIFE.

It transforms abstract financial concepts into relatable human experiences.

Every promotion, family obligation, business opportunity, medical emergency, celebration, and unexpected setback becomes a story that players remember long after the game ends.

Because every situation is fully data-driven, culturally localized, and architecturally independent, SPIKE LIFE can continuously expand with new countries, new careers, and new life experiences without modifying the simulation engine.

This fulfills one of the platform's defining principles:

People rarely remember financial formulas. They remember stories.

By teaching through stories instead of lectures, the Situation Library ensures that financial education becomes personal, memorable, and emotionally meaningful.

End of Chapter 23

This chapter establishes the Situation Library & Event Authoring System as the content creation framework for SPIKE LIFE. Together with the Financial Worlds architecture and the Game Engine, it enables an effectively unlimited catalog of realistic life experiences while maintaining educational consistency, cultural authenticity, technical scalability, and long-term maintainability.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume VI – Content Framework

Chapter 24 – Financial World Authoring Toolkit (FWAT)

Document Version: 1.0

Status: Architecture Approved

Priority: Critical

---

1. Purpose

This chapter defines the **Financial World Authoring Toolkit (FWAT)**.

The toolkit enables educators, economists, financial planners, game designers, universities, government agencies, and content creators to build entirely new Financial Worlds **without modifying the source code**.

Its purpose is simple:

Separate content creation from software development.

A new Financial World should be created by editing structured data—not by writing TypeScript.

---

2. Design Philosophy

The SPIKE LIFE engine should be written **once**.

Everything else should be content.

Examples:

- Countries
- Careers
- Situations
- Goals
- Economic Models
- Investments
- Government Programs
- Taxes
- Languages
- Images
- Sounds

These belong inside World Packs.

---

3. Vision

The long-term vision is to make SPIKE LIFE behave like:

Unreal Engine for Financial Education

where developers build the engine,

and educators build the worlds.

---

4. Toolkit Components

The Financial World Authoring Toolkit consists of:

World Builder

↓

Career Editor

↓

Situation Editor

↓

Goal Editor

↓

Economy Editor

↓

Government Editor

↓

Investment Editor

↓

Localization Manager

↓

Validation Engine

↓

Package Builder

↓

World Publisher

Every module is independent.

---

5. World Builder

The World Builder creates the foundation.

Configuration includes:

Country Name

Currency

Inflation

Language

Population

Flag

Map

Financial World ID

Version

Author

Description

Icon

Thumbnail

---

6. Career Editor

The Career Editor allows creation of careers without programming.

Each career defines:

Name

Description

Industry

Salary Range

Growth Curve

Promotion Frequency

Business Compatibility

Stress

Retirement Age

Localization

Graphical preview included.

---

7. Situation Editor

The Situation Editor is the most frequently used tool.

Users create:

Title

Narrative

Illustration

Advisor Notes

Three Decisions

Immediate Effects

Hidden Effects

Unlock Rules

Probability

Life Stage

Difficulty

Everything is form-based.

---

8. Goal Editor

Allows customization of Dream Board templates.

Examples

House

Travel

Business

Education

Retirement

Vehicle

Farm

Legacy

Wedding

Every goal defines:

Default Cost

Inflation Rule

Target Age

Priority

Completion Animation

---

9. Economy Editor

Defines macroeconomic assumptions.

Includes:

Inflation

Salary Growth

Investment Returns

Property Appreciation

Business Volatility

Government Support

Interest Rates

Economic Cycles

All values stored as configuration.

---

10. Government Editor

Defines country-specific systems.

Examples

Healthcare

Retirement

Housing

Education

Business Registration

Government Benefits

Taxes

These systems influence situations,

not engine architecture.

---

11. Investment Editor

Allows creation of investment products.

Examples

Savings

Government Bonds

Stocks

Cooperative Shares

Mutual Funds

Business

Future products

Configuration includes:

Growth

Volatility

Liquidity

Risk

Availability

---

12. Localization Manager

Every text element supports translation.

Examples

Career Names

Situations

Advisor Messages

Achievements

Buttons

Menus

Currency

Date Formats

Voice Packs (future)

No strings are hardcoded.

---

13. Media Manager

Supports:

Illustrations

Icons

Backgrounds

Animations

Sound Effects

Music

Voice Packs

Assets remain separate from gameplay logic.

---

14. Validation Engine

Before publishing,

the toolkit validates every asset.

Checks include:

Missing translations

Missing illustrations

Invalid probabilities

Broken references

Invalid JSON

Duplicate IDs

Missing advisor text

Missing decisions

Incomplete metadata

Publishing is blocked until validation passes.

---

15. Package Builder

Validated content is packaged into a World Pack.

Output:

Philippines_v1.0.spikeworld

The package contains:

Configuration

Careers

Situations

Goals

Media

Localization

Economy

Metadata

Checksum

Digital Signature

---

16. World Loader

The engine loads a package dynamically.

Sequence

Select World

↓

Validate Package

↓

Verify Signature

↓

Load Assets

↓

Load Configuration

↓

Initialize Simulation

↓

Start Game

No recompilation required.

---

17. World Versioning

Every package has independent version numbers.

Example

World

v1.0.0

Career Library

v1.3

Situation Pack

v2.0

Economy

v1.2

Localization

v5

Engine compatibility is automatically checked.

---

18. Dependency Rules

World Packs cannot modify:

Simulation Engine

Financial Engine

Life Score Engine

Randomization Engine

Security Layer

Networking

They may only provide content.

---

19. Author Profiles

Future versions support verified creators.

Roles include:

Official Publisher

University

Government

Financial Institution

Independent Creator

Community Contributor

Verified creators receive digital signatures.

---

20. Digital Signatures

Every published World Pack receives:

Checksum

Version

Publisher ID

Digital Signature

This prevents tampering and malicious modifications.

---

21. Marketplace Integration

Future marketplace categories:

Countries

Career Packs

Situation Packs

Educational Campaigns

Business Packs

Holiday Packs

Historical Packs

Fantasy Packs

Schools and universities may distribute private packs.

---

22. Educational Templates

The toolkit includes templates for:

High School

College

Corporate Wellness

Financial Advisor Training

Entrepreneurship

Retirement Planning

Family Finance

Each template preloads suitable content.

---

23. AI-Assisted Authoring

Future AI tools assist creators by generating:

Situation drafts

Career descriptions

Advisor conversations

Localization

Illustration prompts

Economic assumptions

AI never publishes directly.

Human approval remains mandatory.

---

24. World Testing Sandbox

Before publication,

authors can simulate:

10 Years

100 Years

1,000 Simulations

Outputs include:

Average Wealth

Goal Completion Rates

Financial Health Trends

Balance Issues

Probability Distribution

This ensures educational quality.

---

25. Quality Certification

World Packs receive quality ratings.

Bronze

Functional

Silver

Balanced

Gold

Educationally Validated

Platinum

Official SPIKE LIFE Certified

Certification builds trust.

---

26. Cursor Implementation Rules

Mandatory implementation requirements:

- Toolkit developed as a separate application.
- World Packs generated from structured configuration.
- Schema validation required before packaging.
- JSON Schema and TypeScript types generated from the same source.
- Digital signature verification before loading.
- World Packs sandboxed from engine code.
- Asset references validated automatically.
- Version compatibility enforced.

- Package builder deterministic.
- Toolkit extensible through plugins.

---

27. Future Expansion

The Authoring Toolkit supports future capabilities including:

Visual Story Editor

Timeline Designer

Economic Simulator

AI Balancing Assistant

Collaborative Editing

Cloud Publishing

Version History

Rollback

Community Reviews

Educational Analytics

These additions enhance the authoring experience without affecting the simulation engine.

---

28. Educational Integrity

The toolkit must protect the educational mission.

It should prevent creators from producing worlds that:

- Reward irresponsible financial behavior.
- Promote financial misinformation.
- Encourage gambling as a financial strategy.
- Misrepresent protection planning.
- Distort basic financial principles.

Validation rules and certification help maintain quality across all published Financial Worlds.

---

29. Success Criteria

The Financial World Authoring Toolkit succeeds when:

- A university can build a localized Financial World without writing code.
- A government agency can publish an official financial literacy campaign.
- A financial institution can create an industry-specific scenario pack.
- Independent educators can contribute new content safely.
- Thousands of Financial Worlds can coexist while using the same simulation engine.

The engine should evolve through **content**, not code.

---

30. Chapter Summary

The Financial World Authoring Toolkit is the bridge between the SPIKE LIFE engine and the global community of educators, financial professionals, and content creators.

It transforms SPIKE LIFE from a single educational game into a platform capable of supporting thousands of localized learning experiences.

By making every Financial World data-driven, validated, digitally signed, and independently versioned, the toolkit ensures that innovation can happen rapidly without compromising simulation integrity or educational quality.

This chapter fulfills one of the original architectural goals established in the Architecture Bible:

Write the engine once. Empower the world to build upon it.

The Financial World Authoring Toolkit ensures that SPIKE LIFE can continue growing for decades through community-driven content while preserving a stable, secure, and deterministic simulation core.

End of Chapter 24

This chapter completes the Content Framework by defining the tools, workflows, validation systems, packaging standards, and publishing process required to create and distribute Financial Worlds. Together with the Financial Worlds System, Situation Library, and Career Engine, it establishes SPIKE LIFE as a scalable platform where content creation is democratized while the integrity of the simulation remains protected.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume VII – Operations & Platform

Chapter 25 – AI Architecture & Intelligence Framework

Document Version: 1.0

Status: Architecture Approved

Priority: Critical

---

1. Purpose

This chapter defines how Artificial Intelligence is integrated into SPIKE LIFE.

AI is **not** the game.

AI is **not** the Financial Engine.

AI is **not** the source of financial truth.

AI exists to make SPIKE LIFE more personal, engaging, conversational, and adaptive—while the deterministic simulation engine remains authoritative.

The governing principle is:

AI creates experiences. The Simulation Engine creates reality.

---

2. Design Philosophy

Every AI component must enhance the player experience without changing the financial integrity of the simulation.

AI may generate:

- Conversations

- Narratives
- Coaching
- Explanations
- Storytelling
- Personalization

AI must never generate:

- Financial formulas
- Interest calculations
- Probability tables
- Life Score
- Random outcomes
- Financial results

The Simulation Engine always has the final authority.

---

3. AI Architecture

PLAYER

|

▼

AI EXPERIENCE LAYER

Advisor AI

Story AI

Mentor AI

NPC AI

Reflection AI

Tutorial AI

Content AI

|

▼

SIMULATION ENGINE (Authority)

Financial Engine

Goal Engine

Life Score Engine

Risk Engine

Planning Cycle Engine

Randomization Engine

____

|

▼

GAME STATE

AI reads.

The Simulation decides.

---

4. AI Design Principles

Every AI service follows seven principles.

Principle 1

Never change financial calculations.

Principle 2

Never invent financial results.

Principle 3

Always explain—not calculate.

Principle 4

Always respect player agency.

Principle 5

Remain educational.

Principle 6

Maintain consistent personality.

Principle 7

Be replaceable.

Every AI module should be interchangeable without affecting gameplay.

---

5. AI Modules

The platform consists of seven AI modules.

- Advisor AI
- Mentor AI
- Story AI
- NPC AI
- Reflection AI
- Tutorial AI
- Content AI

Each module has a distinct responsibility.

---

6. Advisor AI

Advisor AI is the primary player companion.

Responsibilities:

- Explain trade-offs
- Explain opportunity cost
- Explain protection gaps
- Encourage long-term thinking
- Celebrate achievements
- Suggest areas for improvement

It never says:

"You should choose Option A."

Instead it says:

"If you choose this option, here are the likely trade-offs."

---

7. Mentor AI

Mentor AI acts like an experienced financial coach.

It observes:

- recurring mistakes
- decision patterns
- financial habits
- planning discipline

After several Planning Cycles it provides coaching such as:

"You've consistently prioritized lifestyle spending over long-term goals. Would you like to see how that affects retirement over time?"

Mentor AI teaches through patterns.

---

8. Story AI

Story AI converts simulation data into meaningful life stories.

Example:

Instead of

"Income increased by ₱12,000"

Story AI generates:

"Your hard work paid off. After several years of consistent performance, your employer recognized your contribution with a promotion."

Story AI adds emotional engagement.

---

9. NPC AI

Future versions include intelligent characters.

Examples:

Employer

Bank Manager

Business Partner

Friend

Spouse

Mentor

Client

Government Officer

NPCs remember previous conversations.

They respond based on Game State.

---

10. Reflection AI

At the end of every year,

Reflection AI helps players think about their journey.

Examples

"What decision are you most proud of this year?"

"What surprised you?"

"What would you do differently?"

Reflection remains optional.

---

11. Tutorial AI

Instead of static tutorials,

Tutorial AI explains mechanics only when needed.

Example

Player opens Investments for the first time.

Tutorial AI explains:

"What would you like to know about investing?"

Learning becomes contextual.

---

12. Content AI

Content AI assists creators,

not players.

Functions:

Generate situations

Draft careers

Suggest illustrations

Translate content

Write advisor dialogue

Generate educational scenarios

Human approval remains mandatory.

---

13. AI Memory

Each AI module maintains contextual memory.

Examples:

Dreams

Career

Previous advice

Major mistakes

Achievements

Business ownership

Protection history

AI remembers the player's journey,

not every conversation.

---

14. AI Personality

The AI personality should always be:

Professional

Encouraging

Calm

Respectful

Objective

Optimistic

Never:

Judgmental

Fear-based

Aggressive

Sales-oriented

Manipulative

---

15. Conversation Style

Responses should be:

Short

Actionable

Educational

Natural

Average length:

60–120 words.

Avoid lectures.

---

16. AI Safety Rules

AI must never:

Guarantee financial outcomes.

Predict future events.

Reveal hidden probabilities.

Reveal future random events.

Modify financial calculations.

Provide regulated financial advice.

Instead,

AI provides educational guidance.

---

17. AI Prompt Architecture

Every AI request contains:

Game State

+

Current Situation

+

Player History

+

Dream Board

+

Financial Health

+

Conversation Context

=

Prompt

Raw financial formulas are never included.

---

18. AI Knowledge Boundaries

AI knows:

Current simulation.

Player history.

Educational framework.

It does not know:

Hidden probabilities.

Future random events.

Simulation internals.

Upcoming situations.

This preserves fairness.

---

19. Offline Fallback

If AI services are unavailable,

SPIKE LIFE continues normally.

Fallback:

Template Advisor

Static Narratives

Rule-based Reflection

Gameplay never depends on AI availability.

---

20. AI Performance Targets

Advisor Response

<2 seconds

Reflection

<3 seconds

Story Generation

<2 seconds

NPC Conversation

Streaming preferred

The game must remain responsive.

---

21. Privacy

AI should receive only the minimum required context.

Never send:

Authentication credentials

Payment information

Sensitive personal identifiers

Internal security tokens

Player data should be anonymized whenever possible.

---

22. Cost Management

AI usage should be optimized.

Strategies:

Prompt caching

Conversation summarization

Reusable templates

Context trimming

Streaming responses

Only invoke AI when it adds value.

23. Future Multi-Agent System

Future architecture may include specialized agents.

Examples:

Financial Planner

Career Coach

Investment Mentor

Entrepreneur Coach

Retirement Specialist

Behavioral Coach

Each agent specializes in one domain while sharing the same Game State.

---

24. Cursor Implementation Rules

Mandatory implementation requirements:

- AI isolated behind an AI Service Layer.
- Simulation Engine never depends on AI.
- Prompt builders separate from UI.
- AI modules replaceable by provider.
- Fallback templates mandatory.
- AI context generated from Game State only.
- Never expose hidden simulation values.
- AI responses cached where appropriate.
- Streaming supported for long conversations.
- AI disabled without affecting gameplay.

---

25. Future AI Features

Planned capabilities include:

Voice Advisor

Voice NPCs

Personalized coaching

Speech recognition

Natural language decisions

Family discussions

Business negotiations

Career interviews

Financial planning workshops

All future features plug into the same AI Service Layer.

---

26. Educational Integrity

AI should always reinforce the learning philosophy.

It should encourage players to:

Think critically.

Compare alternatives.

Understand trade-offs.

Reflect on consequences.

Develop confidence.

The objective is not to give answers.

The objective is to build judgment.

---

27. Success Criteria

The AI framework succeeds when:

- Players feel guided rather than instructed.

- Conversations feel personal and contextual.
- AI never contradicts the Simulation Engine.
- Gameplay continues even when AI is offline.
- AI enhances replayability without affecting fairness.
- Every AI module can be replaced without changing the engine.

---

28. Chapter Summary

The AI Architecture transforms SPIKE LIFE from a sophisticated simulation into an intelligent learning companion.

Rather than replacing the deterministic simulation, AI sits above it as an experience layer that interprets, explains, coaches, and personalizes the journey.

This separation preserves the integrity of the financial model while unlocking richer storytelling, adaptive education, conversational guidance, and future voice-based interactions.

The guiding principle of the architecture is:

The Simulation Engine determines what happens. AI helps players understand why it matters.

By maintaining this boundary, SPIKE LIFE remains both technically trustworthy and emotionally engaging, ensuring that future advances in artificial intelligence can be adopted without compromising educational quality or financial accuracy.

End of Chapter 25

This chapter establishes the AI Architecture & Intelligence Framework as the experiential layer of SPIKE LIFE. Together with the Simulation Engine, Financial Worlds, and UX framework, it provides a future-proof foundation for AI-powered coaching, storytelling, adaptive learning, and conversational gameplay while preserving deterministic simulation, security, and educational integrity.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume VII – Operations & Platform

Chapter 26 – Analytics, Learning Intelligence & Educational Assessment

Document Version: 1.0

Status: Architecture Approved

Priority: Critical

---

1. Purpose

This chapter defines how SPIKE LIFE measures learning, player behavior, financial decision-making, engagement, and educational outcomes.

Unlike traditional analytics that focus on clicks and usage, SPIKE LIFE measures:

How players think.

The objective is not to collect data.

The objective is to improve financial literacy through measurable learning.

---

2. Design Philosophy

Analytics exists to answer four questions:

Did the player learn?

Did the player improve?

Which financial concepts remain weak?

How can the game adapt to help?

Everything else is secondary.

---

3. Analytics Architecture

Player Actions

↓

Event Stream

↓

Analytics Engine

↓

Learning Intelligence

↓

Reports

↓

Instructor Dashboard

↓

Player Reflection

Analytics never changes gameplay.

It observes.

It measures.

It recommends.

4. Analytics Categories

SPIKE LIFE collects five categories of analytics.

Gameplay Analytics

Planning Cycles

Decision Speed

Completion Rate

Play Time

Game Outcomes

Financial Analytics

Savings Behavior

Protection Coverage

Investment Patterns

Debt Management

Goal Completion

Learning Analytics

Concept Mastery

Repeated Mistakes

Improvement Trends

Advisor Usage

Reflection Quality

Behavioral Analytics

Risk Appetite

Spending Habits

Consistency

Long-Term Thinking

Financial Discipline

Educational Analytics

Knowledge Retention

Decision Quality

Learning Progress

Competency Growth

Overall Financial Literacy

---

5. Event Tracking

Every meaningful interaction generates an event.

Examples

PLAYER_JOINED

GAME_STARTED

GOAL_CREATED

DOMAIN_SELECTED

SITUATION_REVEALED

ADVISOR_OPENED

DECISION_SUBMITTED

AUTO_DECISION_USED

GOAL_COMPLETED

ACHIEVEMENT_UNLOCKED

GAME_FINISHED

Events are immutable.

---

6. Learning Competencies

SPIKE LIFE evaluates financial competencies rather than memorization.

Core competencies include:

Emergency Fund Planning

Insurance & Protection

Cash Flow Management

Goal Planning

Debt Management

Investment Basics

Opportunity Cost

Risk Management

Retirement Planning

Business Thinking

Financial Discipline

Decision Making

Each competency develops independently.

---

7. Competency Scores

Each competency is scored from

0–100.

Example

Competency	Score
Cash Flow Management	88
Emergency Fund	92
Retirement Planning	61
Insurance Planning	84
Investing	70

These are educational indicators,

not game scores.

---

8. Decision Pattern Analysis

The Analytics Engine identifies recurring behaviors.

Examples

Frequently delays saving.

Consistently ignores insurance.

Invests aggressively.

Builds emergency funds early.

Overspends on lifestyle.

Avoids debt.

These patterns drive coaching recommendations.

---

9. Financial Behavior Profiles

Players are classified into evolving behavioral profiles.

Examples

Disciplined Planner

Opportunity Seeker

Risk Taker

Lifestyle Maximizer

Balanced Builder

Entrepreneurial Thinker

Protection First

Growth Investor

The profile changes as behavior changes.

---

10. Learning Timeline

Every Planning Cycle contributes to a learning timeline.

Example

Year 1

Poor budgeting.

Year 3

Emergency Fund completed.

Year 5

Improved investment discipline.

Year 8

Retirement planning strengthened.

Players can visualize their educational growth.

---

11. Reflection Engine

At the end of each year,

players receive guided reflection.

Examples

"What financial decision helped you most this year?"

"What would you change?"

"What surprised you?"

Reflection responses are optional but contribute to learning insights.

---

12. Advisor Effectiveness

The Analytics Engine measures:

Advisor consultations.

Advice acceptance.

Behavior improvement after advice.

Repeated ignored recommendations.

This improves future coaching.

---

13. Facilitator Dashboard

Educators receive aggregated insights.

Examples

Average Life Score

Emergency Fund Completion

Most Common Mistake

Most Improved Competency

Average Decision Time

Advisor Usage

No private financial information is exposed without permission.

---

14. Classroom Analytics

For SPIKE workshops,

facilitators may view:

Class Competency Heat Map

Dream Completion Rates

Protection Readiness

Financial Health Distribution

Decision Confidence

Completion Statistics

Analytics support teaching,

not grading.

---

15. Multiplayer Analytics

For multiplayer sessions,

the system records:

Average Decision Time

Collaboration Frequency

Advisor Requests

Life Score Trends

Goal Completion Comparison

No confidential financial details are shared between players.

---

16. Replay Analytics

Every replay includes:

Decision Timeline

Missed Opportunities

Alternative Outcomes

Advisor Commentary

Financial Turning Points

Replay becomes a learning experience,

not just entertainment.

---

17. Learning Recommendations

Based on analytics,

the system recommends:

Practice scenarios

Articles

Mini lessons

Financial concepts

Replay opportunities

Future Financial Worlds

Recommendations remain educational,

never promotional.

---

18. Progress Dashboard

The player's long-term dashboard displays:

Competency Radar

Life Score History

Financial Health Trend

Goal Achievement

Achievements

Career Timeline

Behavior Profile

Improvement Areas

The dashboard emphasizes growth over comparison.

---

19. Certification Support

Future versions may issue competency certificates.

Examples

Emergency Fund Mastery

Investment Foundations

Financial Planning Essentials

Entrepreneurship Readiness

Retirement Planning

Certification is based on demonstrated decision quality,

not quiz scores.

---

20. Institutional Reporting

Universities and organizations may generate:

Course Completion Reports

Competency Distribution

Learning Outcomes

Engagement Statistics

Participation Metrics

Reports remain anonymized unless explicit identification is required.

---

21. Privacy Principles

Educational analytics must respect player privacy.

Rules:

Collect minimum necessary data.

Allow data export.

Allow account deletion.

Support anonymization.

Separate educational analytics from authentication data.

No analytics may be sold to third parties.

---

22. AI Learning Insights

Future AI modules may analyze:

Decision patterns

Behavioral changes

Reflection responses

Competency growth

AI may suggest personalized learning plans,

but never alter scores or simulation outcomes.

---

23. Performance Targets

Analytics processing must not interrupt gameplay.

Targets:

Event logging

<10 ms

Dashboard generation

<500 ms

Learning report generation

<2 seconds

Replay analytics

Background process

---

24. Cursor Implementation Rules

Mandatory implementation requirements:

- Event-driven analytics architecture.
- Immutable event logs.
- Competency engine independent from Life Score.
- Analytics generated asynchronously.
- Personally identifiable information separated from gameplay analytics.
- Aggregated classroom analytics supported.
- Reflection responses stored separately from simulation state.
- Export analytics in JSON and CSV formats.
- Future Learning Record Store (LRS/xAPI) compatibility.
- Analytics must never influence deterministic simulation.

---

25. Future Expansion

The Analytics Framework supports:

SCORM integration

xAPI integration

Learning Management Systems

University accreditation

Corporate wellness reporting

Government financial literacy programs

Research datasets

Adaptive learning pathways

Predictive learning models

Longitudinal financial literacy studies

---

26. Educational Integrity

Analytics should encourage improvement,

not competition.

Players should compare themselves to:

Yesterday's version of themselves,

not other players.

Learning progress should always be presented positively and constructively.

---

27. Success Criteria

The Analytics Framework succeeds when:

- Players clearly understand how they have improved.
- Facilitators identify class-wide learning gaps.
- Institutions measure educational impact.
- Analytics remain privacy-conscious.
- Reports support coaching rather than judgment.
- The simulation remains unaffected by analytics collection.

---

28. Chapter Summary

The Analytics, Learning Intelligence & Educational Assessment Framework transforms SPIKE LIFE from a game into a measurable learning platform.

By capturing decision patterns, competency growth, behavioral trends, and educational outcomes, the platform enables players, facilitators, schools, employers, and researchers to understand not only **what happened** during gameplay, but **what was learned**.

Unlike conventional game analytics, SPIKE LIFE measures financial judgment, resilience, planning discipline, and long-term thinking.

This fulfills one of the founding educational goals of the platform:

The purpose of the simulation is not simply to play better. It is to live wiser.

The Analytics Framework ensures that every decision contributes not only to the player's virtual life, but also to their real-world financial capability.

End of Chapter 26

This chapter establishes the Analytics, Learning Intelligence & Educational Assessment framework as the measurement layer of SPIKE LIFE. Together with the AI Architecture, Simulation Engine, Financial Worlds, and UX framework, it completes the educational feedback loop—allowing players to learn through experience, facilitators to teach with evidence, and institutions to evaluate financial literacy outcomes through authentic decision-making rather than traditional examinations.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume VII – Operations & Platform

Chapter 27 – Deployment, DevOps & Live Operations

Document Version: 1.0

Status: Architecture Approved

Priority: Critical

---

1. Purpose

This chapter defines how SPIKE LIFE is built, tested, deployed, monitored, maintained, and continuously improved throughout its lifecycle.

Unlike traditional educational software that is released occasionally, SPIKE LIFE is designed as a **continuously evolving platform**.

The goal is to ensure that every deployment is:

- Safe
- Repeatable
- Observable
- Reversible
- Scalable

without interrupting player experience.

---

2. Deployment Philosophy

SPIKE LIFE follows one fundamental principle:

Small, safe, frequent releases are better than large, risky releases.

Every deployment should improve the platform without compromising simulation integrity.

---

3. Platform Architecture

Developer

↓

GitHub

↓

Continuous Integration

↓

Automated Testing

↓

Build Pipeline

↓

Preview Environment

↓

Quality Verification

↓

Production Deployment

↓

Monitoring

↓

Analytics

↓

Continuous Improvement

Deployment is fully automated.

---

4. Environments

SPIKE LIFE maintains separate environments.

Local Development

Developer workstation.

Development

Shared internal environment.

Preview

Generated automatically for every Pull Request.

Staging

Production mirror.

Used for acceptance testing.

Production

Public release.

Only approved builds may reach Production.

---

5. Source Control

Git is the single source of truth.

Main branches

main

develop

feature/*

release/*

hotfix/*

Direct commits to main are prohibited.

---

6. Continuous Integration

Every commit automatically triggers:

- Install dependencies
- Type checking
- Linting
- Unit tests
- Integration tests
- Simulation tests
- Build verification
- Security scanning

- Package validation

No manual intervention required.

---

7. Continuous Delivery

If CI succeeds:

Preview deployments are created automatically.

Staging deployment requires approval.

Production deployment requires:

- Successful CI
- Successful QA
- Version tagging
- Release approval

---

8. Infrastructure

Recommended deployment architecture.

Frontend

Cloudflare Pages

Backend APIs

Cloudflare Workers

Database

PostgreSQL

Storage

Cloudflare R2

CDN

Cloudflare Global Network

Future:

Redis

Queues

Object caching

---

9. Infrastructure as Code

Infrastructure must be reproducible.

Configuration stored using:

Terraform (future)

Wrangler

Environment configuration

No manual production configuration.

---

10. Configuration Management

Configuration separated by environment.

Examples

Development

Staging

Production

Secrets never stored in source code.

---

11. Secret Management

Secrets include:

Database credentials

API Keys

AI Provider Keys

JWT Secrets

Encryption Keys

Cloud credentials

Secrets stored only in secure secret managers.

---

12. Release Versioning

Semantic Versioning

Example

v1.2.4

Major

Minor

Patch

Game content versioned independently from engine version.

---

13. Release Process

Feature Complete

↓

Code Review

↓

CI

↓

Preview

↓

QA

↓

Staging

↓

Approval

↓

Production

↓

Monitoring

Every release is documented.

---

14. Feature Flags

New functionality should be controlled through Feature Flags.

Examples

Voice Advisor

Online Multiplayer

Marketplace

Replay System

Financial Worlds

AI Mentor

Flags allow safe rollout without redeployment.

---

15. Blue-Green Deployment

Production deployments should support:

Current Version

↓

New Version

↓

Health Check

↓

Traffic Switch

↓

Rollback if necessary

Players should experience zero downtime.

---

16. Rollback Strategy

Every deployment must be reversible.

Rollback time target:

Less than

5 minutes.

Database migrations must also support rollback.

---

17. Database Migration

Migration rules:

Forward compatible.

Backward compatible where possible.

Idempotent.

Versioned.

Automatically tested.

Never modify production data manually.

---

18. Monitoring

System health monitored continuously.

Metrics include:

API latency

Database latency

Planning Cycle duration

Simulation failures

Memory usage

CPU usage

Storage utilization

Error rate

---

19. Application Monitoring

Track:

Page load time

FPS

Animation performance

Client errors

Network failures

Replay failures

Save failures

Multiplayer synchronization

Monitoring must be proactive.

---

20. Logging

Centralized structured logging.

Levels

Debug

Info

Warning

Error

Critical

Every log entry includes:

Timestamp

Environment

Correlation ID

Version

Request ID

Player Session ID (anonymized)

---

21. Alerting

Critical alerts include:

Deployment failure

Simulation failure

Database outage

High error rate

High latency

Authentication failure

Storage failure

AI service outage

Alerts routed to operations teams.

---

22. Backup Operations

Automatic backups:

Database

Cloud saves

Configuration

Media assets

Retention policy configurable.

Recovery tested regularly.

---

23. Disaster Recovery

Recovery objectives.

Recovery Time Objective (RTO)

<1 hour

Recovery Point Objective (RPO)

<5 minutes

Documented recovery playbooks mandatory.

---

24. Operational Dashboard

Administrators view:

System Health

Deployments

Traffic

Errors

Simulation Performance

Active Games

Storage

AI Usage

Background Jobs

Everything observable from one dashboard.

---

25. Performance Budgets

Targets

Initial Load

<2 seconds

Planning Cycle

<500 ms

API

<200 ms

FPS

60

Bundle Size

Monitored continuously.

Performance regressions block releases.

---

26. Security Operations

Continuous security includes:

Dependency scanning

Vulnerability alerts

Container scanning (future)

Penetration testing

Secret rotation

Audit logging

Security reviews

Security is continuous,

not periodic.

---

27. Quality Gates

Production deployment blocked if:

Unit tests fail.

Coverage below threshold.

Security vulnerabilities exceed policy.

Simulation determinism fails.

Performance budget exceeded.

Architecture validation fails.

---

28. Operational Analytics

Platform metrics include:

Daily Active Players

Completion Rate

Average Session Length

Planning Cycle Duration

Replay Usage

Financial World Usage

Advisor Usage

Crash-Free Sessions

These metrics improve operations,

not gameplay balance.

---

29. Live Operations

Future Live Operations support:

Seasonal Events

Holiday Themes

Educational Campaigns

Government Financial Literacy Programs

Corporate Wellness Challenges

University Competitions

Content updates through Financial World Packs.

No engine redeployment required.

---

30. Maintenance Windows

Scheduled maintenance should:

Be announced in advance.

Minimize downtime.

Avoid peak usage periods.

Allow graceful game completion before shutdown.

Future online multiplayer should support rolling maintenance without disconnecting active players.

---

31. Cursor Implementation Rules

Mandatory implementation requirements:

- GitHub Actions for CI/CD.
- Preview deployment for every Pull Request.
- Cloudflare Pages + Workers deployment.
- PostgreSQL migrations automated.
- Feature Flag system implemented.
- Structured logging throughout the platform.
- OpenTelemetry-compatible instrumentation.
- Health check endpoints for all services.
- Zero-downtime deployment strategy.
- Automated rollback support.
- Environment variables validated at startup.
- Monitoring dashboards configurable.

---

32. Future Platform Services

Future infrastructure may include:

Dedicated Simulation Cluster

Real-time Multiplayer Gateway

Replay Service

Analytics Warehouse

Recommendation Engine

Content Marketplace

Financial World CDN

AI Orchestration Layer

Regional Data Centers

Global Tournament Platform

The current architecture supports these expansions without redesign.

---

33. Operational Excellence

Operational success is measured by:

99.9% uptime.

Fast deployments.

Rapid rollback.

Low defect rate.

Stable performance.

Predictable releases.

Reliable backups.

Confident development.

Operations should become routine,

not stressful.

---

34. Chapter Summary

Deployment, DevOps, and Live Operations ensure that SPIKE LIFE remains reliable long after development is complete.

By combining automated testing, continuous deployment, infrastructure as code, structured monitoring, and robust rollback strategies, the platform becomes resilient enough for schools, universities, enterprises, government agencies, and eventually millions of players.

This operational architecture reflects the same philosophy as the simulation itself:

Success comes from preparation, discipline, and continuous improvement—not from reacting to crises.

Every deployment should therefore be as carefully planned as the financial lives the game teaches players to build.

End of Chapter 27

This chapter establishes the operational foundation of SPIKE LIFE, completing the production architecture alongside the Software Architecture, Security, Analytics, and AI frameworks. It provides the DevOps blueprint required to build, deploy, monitor, scale, and continuously evolve the platform while preserving reliability, determinism, educational integrity, and a world-class player experience.

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Volume VIII – Governance & Product Roadmap

Chapter 28 – Product Roadmap, Governance & Future Vision

Document Version: 1.0

Status: Master Blueprint

Priority: Strategic

---

1. Purpose

This final chapter defines the long-term evolution of SPIKE LIFE.

It establishes how the platform will grow over the next decade while preserving its educational philosophy, architectural integrity, and product vision.

It answers the question:

"Where is SPIKE LIFE ultimately going?"

This chapter serves as the constitutional document for future product decisions.

Every major enhancement should align with the vision defined here.

---

2. Vision Statement

SPIKE LIFE exists to become

the world's most trusted life and financial decision simulator.

It is not merely a financial literacy game.

It is a platform where people can safely experience life before living it.

---

3. Mission

SPIKE LIFE empowers people to make better life decisions through immersive simulation, experiential learning, intelligent coaching, and realistic financial planning.

Players learn by experiencing the consequences of decisions rather than memorizing financial concepts.

---

4. Product Philosophy

SPIKE LIFE is built upon five enduring principles.

Learn by Living

Players learn through realistic life experiences.

Decisions Have Consequences

Every decision creates short-term and long-term effects.

There Are No Perfect Lives

Every player experiences both opportunities and setbacks.

Balance Beats Extremes

Financial success comes from disciplined balance rather than maximizing one metric.

Build Better Humans

The ultimate objective is not better players—

it is wiser people.

---

5. Long-Term Platform Vision

SPIKE LIFE evolves through four stages.

Educational Game

↓

Financial Simulation Platform

↓

Global Financial Worlds Ecosystem

↓

Universal Life Decision Platform

Every development milestone supports this evolution.

---

6. Product Evolution Roadmap

Phase 1 — MVP (Current)

Objective:

Deliver a fully playable financial life simulation.

Includes:

- Philippine Financial World
- 2–6 Players
- 10-Year Campaign
- 20 Planning Cycles
- Dream Board
- Financial Planning Engine
- Life Score
- Advisor
- Replay
- Analytics

Estimated Release:

Version 1.x

Phase 2 — Enhanced Platform

Adds:

- Cloud Saves
- Online Multiplayer
- Voice Advisor
- Mobile Companion
- Replay Theatre

- More Careers
- More Situations
- Additional Financial Worlds

Estimated Release:

Version 2.x

Phase 3 — Creator Platform

Adds:

- Financial World Authoring Toolkit
- Marketplace
- Community Worlds
- University Content
- Corporate Training Packs
- Government Campaign Packs
- AI Content Authoring

Estimated Release:

Version 3.x

Phase 4 — Global Platform

Adds:

- Global Leaderboards
- Certification Programs
- Research Platform
- Educational Accreditation
- AI Multi-Agent Coach
- Financial Digital Twin

- Enterprise Solutions

Estimated Release:

Version 4.x+

---

7. Product Governance

Every proposed feature must satisfy three questions.

Question 1

Does it improve financial learning?

Question 2

Does it preserve architectural integrity?

Question 3

Does it improve player experience?

If the answer to any question is "No,"

the feature should be reconsidered.

---

8. Architectural Governance

The Architecture Bible remains the supreme technical authority.

Priority order:

1. Architecture Bible
2. Domain Model Blueprint
3. Game Design Specification
4. PRD
5. UI Mockups
6. Implementation Code

Code must conform to architecture,

never the reverse.

---

9. Educational Governance

Educational accuracy takes precedence over entertainment.

If a gameplay mechanic encourages harmful financial behavior,

it must be redesigned,

regardless of how entertaining it is.

---

10. Simulation Governance

The Financial Engine remains deterministic.

Future enhancements must never compromise:

Financial accuracy

Simulation fairness

Replay determinism

Life Score integrity

Randomization fairness

These principles are non-negotiable.

---

11. AI Governance

AI remains an assistant.

Never an authority.

AI may:

Explain

Coach

Narrate

Translate

Personalize

AI may never:

Alter calculations

Override simulations

Reveal hidden probabilities

Guarantee outcomes

Generate financial truth

---

12. Content Governance

Every Financial World must undergo review.

Required validation:

Educational accuracy

Cultural authenticity

Simulation balance

Technical validation

Localization review

No content enters production without approval.

---

13. Community Governance

Future community contributions require:

Identity verification

Content validation

Peer review

Automated testing

Digital signatures

Quality certification

The community expands the platform responsibly.

---

14. Open Standards

Where practical,

SPIKE LIFE should support:

xAPI

SCORM

CSV

JSON

REST APIs

Open telemetry

Open educational reporting

Avoid proprietary lock-in.

---

15. Research Platform

Future academic partnerships include:

Financial literacy research

Behavioral economics

Decision science

Educational psychology

Game-based learning

Public policy

Research datasets remain anonymized.

---

16. Enterprise Platform

Future enterprise customers include:

Banks

Insurance companies

Universities

Government agencies

NGOs

Corporations

Financial educators

Enterprise features remain separate from the consumer experience.

---

17. Sustainability Strategy

Revenue should come from:

Premium Financial Worlds

Educational licenses

Enterprise deployments

Certification programs

Marketplace revenue sharing

Institutional subscriptions

Never compromise educational neutrality through product placement or biased financial recommendations.

---

18. Ethical Principles

SPIKE LIFE commits to:

Transparency

Privacy

Fairness

Accessibility

Inclusivity

Educational honesty

Responsible AI

Respect for cultural diversity

These values guide every product decision.

---

19. Accessibility Commitment

Future versions strive for:

WCAG compliance

Multilingual support

Voice interaction

Screen reader compatibility

Low-bandwidth operation

Offline capability

Financial education should be accessible to everyone.

---

20. Internationalization Strategy

Future Financial Worlds will support:

Localized economies

Localized careers

Localized government systems

Localized culture

Localized language

Localized educational objectives

The simulation engine remains universal.

21. Product Success Metrics

Success is measured by:

Financial literacy improvement

Replay rate

Goal completion understanding

Educational adoption

Player retention

Institutional adoption

Content ecosystem growth

Community contributions

Not by microtransactions or advertising.

---

22. Governance Board

Future governance may include:

Product Architect

Educational Director

Financial Experts

Behavioral Economists

Game Designers

AI Ethics Advisor

Technical Architect

Community Representative

Major architectural changes require governance review.

---

23. Backward Compatibility

Future versions should preserve:

Saved games where practical

Financial World compatibility

Replay capability

Content portability

API stability

Breaking changes require migration tools.

---

24. Long-Term Vision

In ten years,

SPIKE LIFE should support:

100+

Financial Worlds

50+

Languages

10,000+

Situations

500+

Careers

Millions of simulated lives

Hundreds of educational institutions

Thousands of facilitators

A global creator ecosystem

All using one deterministic simulation engine.

---

25. Cursor Implementation Rules

Mandatory implementation requirements:

- Preserve architectural boundaries.
- Favor extensibility over shortcuts.
- Never duplicate business logic.
- Maintain deterministic simulation.
- Maintain configuration-driven content.
- Protect backward compatibility.
- Separate platform features from educational content.
- Document architectural decisions.
- Use feature flags for experimental functionality.
- Every major feature accompanied by automated tests and updated documentation.

---

26. SPIKE LIFE Design Manifesto

SPIKE LIFE believes:

People learn best by doing.

Stories teach better than lectures.

Preparation is more valuable than prediction.

Good financial habits create better lives.

Technology should empower judgment—not replace it.

Financial education should be engaging, accessible, and lifelong.

These beliefs shape every line of code, every Planning Cycle, and every Financial World.

---

27. Final Product Statement

SPIKE LIFE is more than a board game.

More than educational software.

More than a financial simulator.

It is a platform for practicing life.

Players make decisions.

The simulation reveals consequences.

Reflection creates wisdom.

Wisdom changes real-world behavior.

That is the purpose of SPIKE LIFE.

---

28. Master Architecture Declaration

The following principles are permanently established:

- The Simulation Engine is the single source of truth.
- Financial Worlds contain localized content, never simulation logic.
- AI enhances understanding but never determines outcomes.
- Learning is measured through authentic decision-making.
- Architecture takes precedence over implementation.
- Content evolves faster than code.
- Every expansion preserves deterministic simulation.
- Every feature must reinforce the educational mission.

These principles define the identity of SPIKE LIFE and must remain unchanged unless a formal architecture revision is approved.

---

29. Chapter Summary

This chapter completes the SPIKE LIFE Game Design Specification by defining not only what the platform is today, but what it is intended to become.

It establishes the governance model, product philosophy, architectural priorities, ethical commitments, educational mission, and long-term roadmap that will guide development for years to come.

More importantly, it preserves the central vision that has emerged throughout this design process:

SPIKE LIFE is not designed to teach people how to become wealthy. It is designed to help people make wiser life decisions.

If the platform succeeds in doing that, then every simulation, every Planning Cycle, every Financial World, and every line of code will have fulfilled its purpose.

---

30. End of the Game Design Specification

This concludes the **SPIKE LIFE™ Game Design Specification (GDS) Version 1.0**.

Together, the 28 chapters define a complete blueprint for the platform, including:

- Gameplay systems
- Financial simulation
- UX philosophy
- AI architecture
- Software architecture
- Security
- Analytics
- Financial Worlds
- Career system
- Situation authoring
- Content creation
- DevOps
- Governance
- Product vision

The document is intentionally designed to serve as the authoritative reference for product owners, architects, developers, designers, AI coding assistants, educators, facilitators, and future contributors.

From this point forward, implementation should focus on realizing this architecture rather than redefining it.

The vision is complete.

The next step is execution.

End of Chapter 28

SPIKE LIFE™ — Version 1.0 Architecture & Game Design Specification Complete

"Practice Life. Make Better Decisions."

SPIKE LIFE™

Game Design Specification (GDS) v1.0

Final Chapter – The SPIKE LIFE Constitution

Document Version: 1.0

Status: Founding Constitution

Authority Level: Supreme

Priority: Non-Negotiable

---

1. Purpose

This document is the permanent Constitution of SPIKE LIFE.

It exists to preserve the vision, philosophy, architecture, educational mission, and design principles of the platform for every future contributor.

This Constitution is intentionally separate from implementation.

Technology will evolve.

Programming languages will change.

Frameworks will come and go.

Artificial Intelligence will improve.

Financial Worlds will multiply.

The Constitution remains.

Whenever uncertainty exists, this document provides the final direction.

---

2. Declaration

SPIKE LIFE is not merely a game.

It is not merely educational software.

It is not merely a financial simulator.

SPIKE LIFE is a **Life Decision Simulator.**

Its purpose is to help people practice life before living it.

The simulation exists so that mistakes can be made safely.

The lessons learned inside the simulation should improve decisions made outside the simulation.

Everything else is secondary.

---

3. The Mission

SPIKE LIFE exists to help people become:

- Better decision makers
- Better financial planners
- Better entrepreneurs
- Better professionals
- Better parents
- Better citizens
- Better human beings

Financial literacy is the beginning.

Life wisdom is the destination.

---

4. The Vision

To become the world's most trusted simulation platform for learning life, finance, and decision-making through experience.

One Engine.

Unlimited Worlds.

Millions of lives improved.

---

5. The Founding Principles

Every feature, every design decision, every Financial World, and every line of code must reinforce these principles.

Learn Through Experience

People remember experiences more than lectures.

Simulation is more powerful than instruction.

Decisions Have Consequences

Every decision creates trade-offs.

Nothing is free.

Every gain has a cost.

Every delay has an opportunity cost.

Balance Wins

The objective is never to maximize one variable.

Not income.

Not investments.

Not protection.

Not business.

Balance creates resilience.

Preparation Beats Prediction

Life is uncertain.

Preparation matters more than forecasting.

Emergency funds.

Protection.

Planning.

Discipline.

These outperform luck.

Reflection Creates Wisdom

Learning occurs after consequences are experienced.

Reflection transforms experience into wisdom.

---

6. The Ten Commandments of SPIKE LIFE

The following commandments are permanent.

I

The Simulation Engine is the only source of truth.

II

Financial calculations shall never be generated by Artificial Intelligence.

III

The Game State shall remain the single authoritative state.

IV

Players shall learn through consequences, never through fear.

V

The interface shall always remain simpler than the underlying simulation.

VI

Every Planning Cycle shall contain one meaningful decision.

VII

Every feature shall improve learning before improving entertainment.

VIII

Content shall evolve faster than software.

IX

Architecture shall always take precedence over implementation convenience.

X

The purpose of SPIKE LIFE is to improve real lives—not merely create enjoyable gameplay.

---

7. Educational Constitution

SPIKE LIFE teaches principles rather than products.

The platform shall never become:

An insurance sales simulator.

An investment recommendation engine.

A budgeting worksheet.

A product marketing tool.

A compliance examination.

Instead,

it teaches timeless financial principles including:

Cash Flow

Protection

Savings

Investing

Entrepreneurship

Opportunity Cost

Delayed Gratification

Risk Management

Long-Term Thinking

Financial Discipline

These principles remain valid regardless of country.

---

8. AI Constitution

Artificial Intelligence exists to support learning.

AI may:

Explain.

Encourage.

Coach.

Reflect.

Narrate.

Translate.

Personalize.

AI shall never:

Override the Simulation Engine.

Generate financial truth.

Predict guaranteed outcomes.

Reveal hidden probabilities.

Manipulate player decisions.

Replace human judgment.

AI serves the player.

It never controls the player.

---

9. Architecture Constitution

The following architectural rules are permanent.

The Simulation Engine remains deterministic.

Financial Worlds contain content only.

The UI never owns business logic.

Configuration replaces hardcoding.

The Event Bus coordinates systems.

Pure simulation remains framework-independent.

The architecture must remain modular.

No expansion may violate these principles.

---

10. Design Constitution

SPIKE LIFE shall always feel like:

A premium strategy board game.

It shall never resemble:

Accounting software.

Insurance software.

Banking software.

Enterprise dashboards.

Players should remember stories,

not spreadsheets.

Every screen should answer:

"What should I do next?"

within three seconds.

---

11. Development Constitution

Every contributor should follow these principles.

Write clean code.

Write readable code.

Write deterministic code.

Write testable code.

Prefer composition.

Prefer configuration.

Prefer simplicity.

Prefer maintainability.

Never optimize prematurely.

Never sacrifice clarity for cleverness.

---

12. Financial Integrity Constitution

Financial realism is sacred.

Inflation always exists.

Risk always exists.

Opportunity cost always exists.

Protection always has value.

Emergency funds always matter.

Debt always has consequences.

There are no shortcuts to sustainable wealth.

The simulation should reward discipline,

not exploitation.

---

13. Product Evolution Constitution

SPIKE LIFE may grow through:

New Financial Worlds.

New Careers.

New Situations.

New AI capabilities.

New platforms.

New educational programs.

It shall never abandon:

The Simulation Engine.

The Financial Engine.

The Life Score Engine.

The Planning Cycle.

The One Workspace philosophy.

Deterministic replay.

Educational integrity.

---

14. Governance Constitution

Major changes require architectural review.

Every proposal should answer:

Does this improve learning?

Does this preserve architecture?

Does this improve player experience?

If the answer is "No"

the proposal should not proceed.

---

15. Ethical Constitution

SPIKE LIFE commits to:

Educational honesty.

Transparency.

Privacy.

Accessibility.

Inclusivity.

Responsible AI.

Cultural respect.

Financial neutrality.

The platform shall never manipulate players for commercial gain.

Trust is more valuable than revenue.

---

16. Creator Constitution

Future creators are guardians of the platform.

Every new Financial World should:

Respect local culture.

Respect financial reality.

Respect educational objectives.

Respect architectural boundaries.

Creativity is encouraged.

Compromising the educational mission is not.

---

17. Success Definition

Success is not measured by:

Downloads.

Revenue.

Advertising.

Microtransactions.

Social media popularity.

Success is measured by:

Better financial decisions.

Greater confidence.

Improved financial resilience.

Healthier financial habits.

Stronger families.

More prepared communities.

Better futures.

---

18. The Promise

SPIKE LIFE promises every player:

You will not always win.

You will make mistakes.

You will experience setbacks.

You will face uncertainty.

You will sacrifice some dreams to achieve others.

But if you learn,

every simulation becomes worthwhile.

---

19. The Legacy

The purpose of SPIKE LIFE is larger than software.

It is to help create a generation that understands:

Money.

Time.

Risk.

Responsibility.

Family.

Opportunity.

Purpose.

A financially literate society creates stronger families.

Stronger families create stronger communities.

Stronger communities create stronger nations.

That is the legacy SPIKE LIFE seeks to leave.

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20. Final Declaration

Every contributor who builds SPIKE LIFE becomes a steward of this vision.

Every educator who teaches through SPIKE LIFE extends this mission.

Every developer who writes code contributes to a platform that may influence millions of financial decisions.

Every player who completes the journey carries its lessons into real life.

Whenever uncertainty exists,

choose simplicity over complexity.

Choose wisdom over cleverness.

Choose education over entertainment.

Choose long-term impact over short-term gain.

Because SPIKE LIFE is not ultimately about wealth.

It is about helping people build lives they can be proud of.

The Founder's Oath

We will build technology that teaches rather than manipulates.

We will create simulations that prepare rather than frighten.

We will value truth over convenience, architecture over shortcuts, and education over monetization.

We will respect every culture while teaching universal financial principles.

We will ensure that every new feature leaves the platform better than we found it.

And we will never forget that behind every player is a real person whose future may be changed by the decisions they practice here.

Final Constitutional Statement

SPIKE LIFE™ is hereby established as:

A deterministic financial life simulation platform.

A global educational ecosystem.

A creator-powered Financial Worlds platform.

A lifelong learning companion.

A place where people safely practice the decisions that shape their future.

From this day forward,

every design,

every architecture,

every simulation,

every Financial World,

every Planning Cycle,

and every line of code

shall honor this Constitution.

End of the SPIKE LIFE™ Constitution

End of the Game Design Specification v1.0

Closing Motto

Practice Life. Make Better Decisions. Build a Better Future.